

Chapter 7: Functional MRI

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Structural vs. Functional MRI



Structural MRI studies
brain anatomy



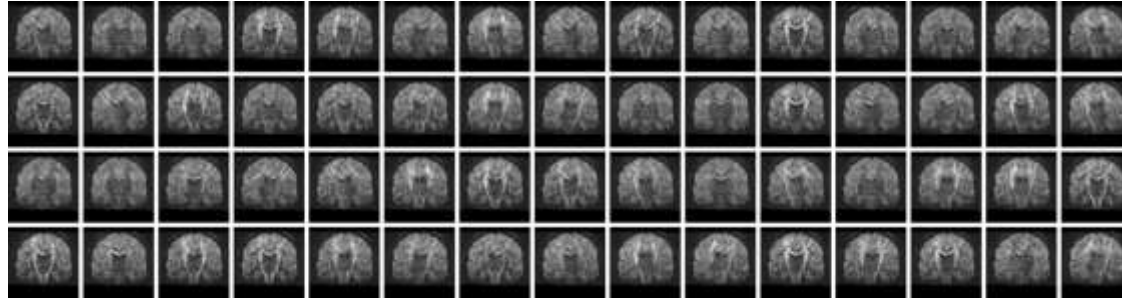
Functional MRI studies
brain function

Image Acquisition and Analysis in Neuroscience

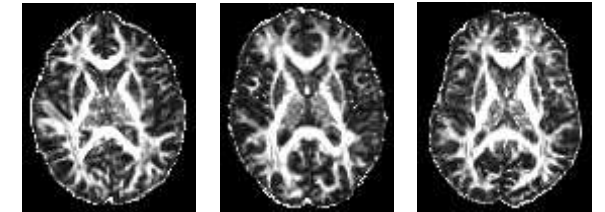
Acquisition



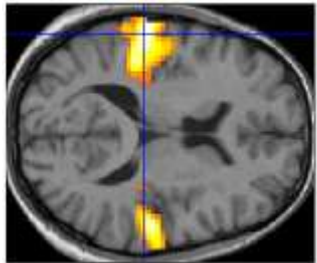
Reconstruction & Artifact Correction



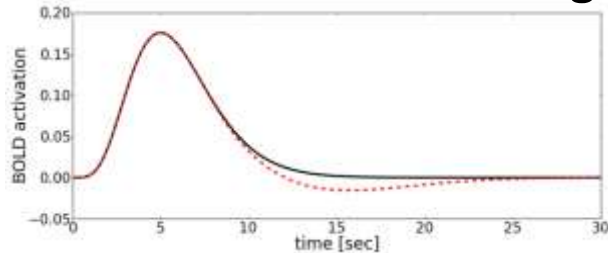
Registration



Hypothesis Testing



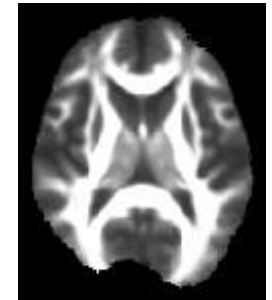
Mathematical Modeling



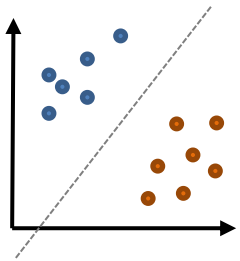
Segmentation



Atlas Construction



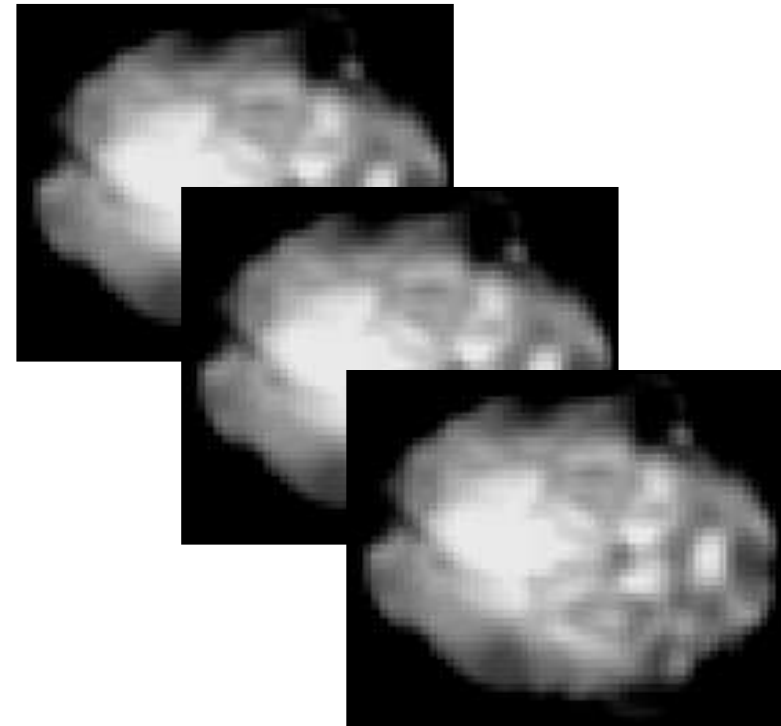
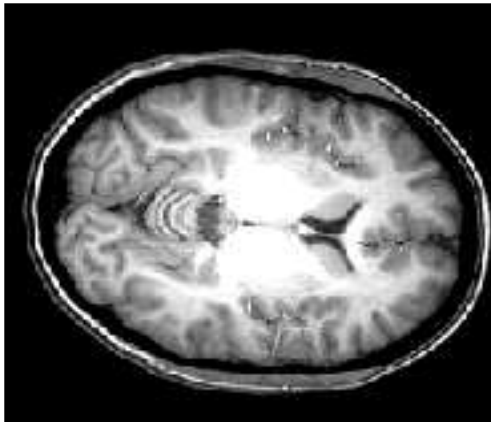
Machine Learning



7.1 The BOLD Effect

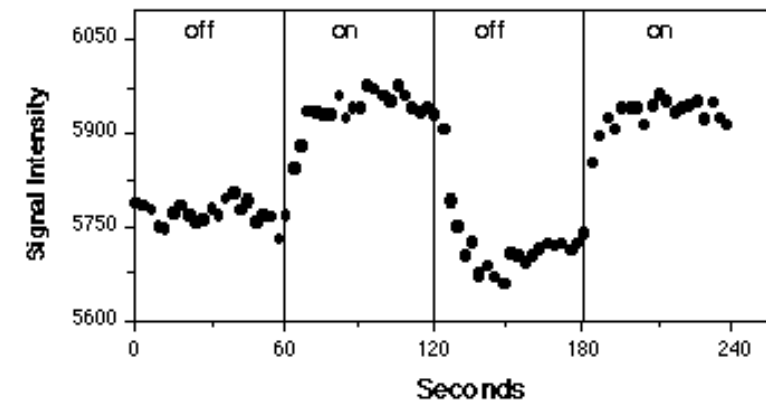
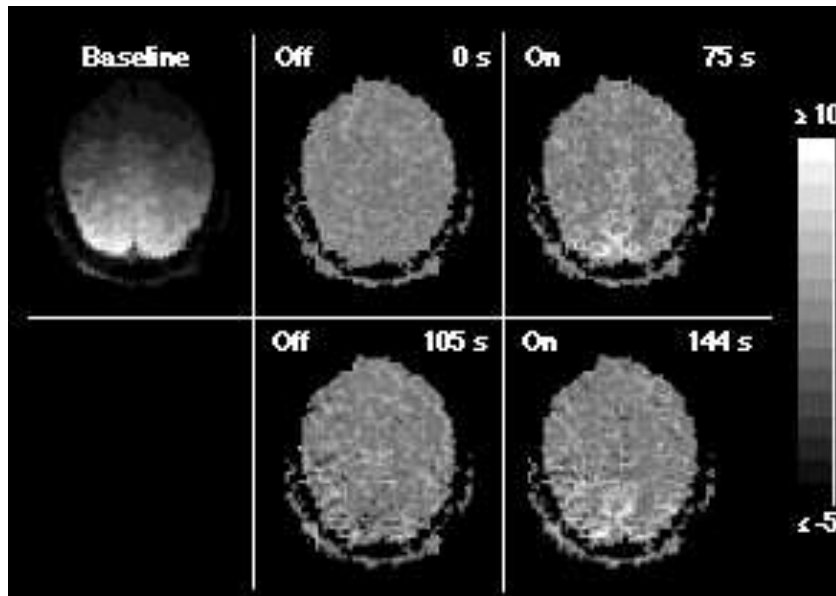
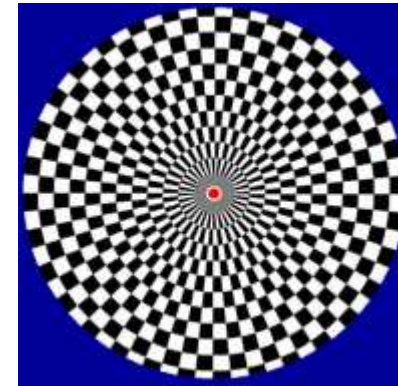
fMRI: The Raw Data

- In **structural MRI**, we take one high-resolution image (e.g., $1 \times 1 \times 1 \text{ mm}^3$)
- In **functional MRI**, we repeatedly (e.g., every 2 sec) take low-resolution images (e.g., $3 \times 3 \times 5 \text{ mm}^3$)



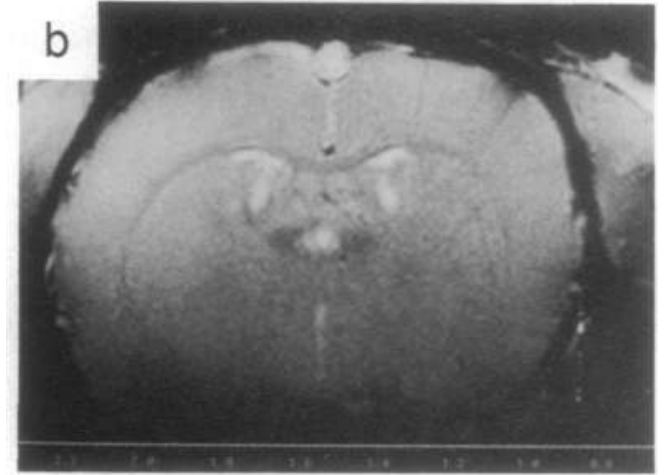
fMRI: Activations

- In fMRI, an **activation** is measured by taking the difference between the signal for different tasks
- **Example stimulus:** Checkerboard
OFF (60 sec) – ON (60 sec) – OFF ...
- **Example MR response:**

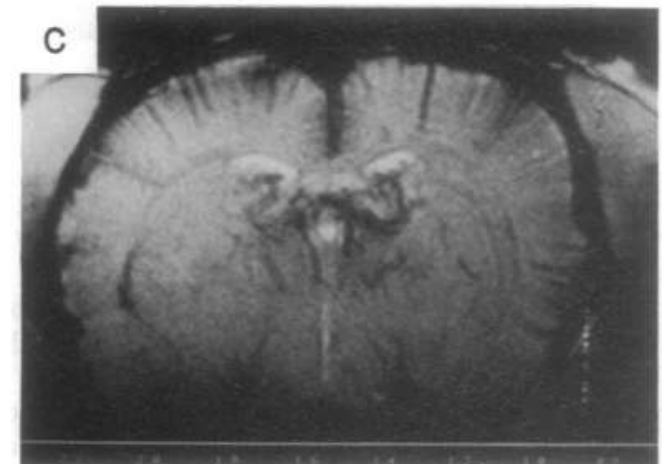


Where Does the Signal Come From?

- **BOLD effect:** MR intensity is **B**lood **O**xygenation **L**evel **D**ependent
- **Oxygenated Blood**
 - **Diamagnetic:** (weakly) counteracts the local magnetic field
 - Effect similar to water
 - (Almost) no change in signal
- **Deoxygenated Blood**
 - **Paramagnetic:** (slightly) enhances the field
 - Decreases T_2^*
 - Attenuates the MR signal



Rat breathing 3% halothane in O₂



Rat breathing 100% N₂

Temporal Complexity of BOLD Response



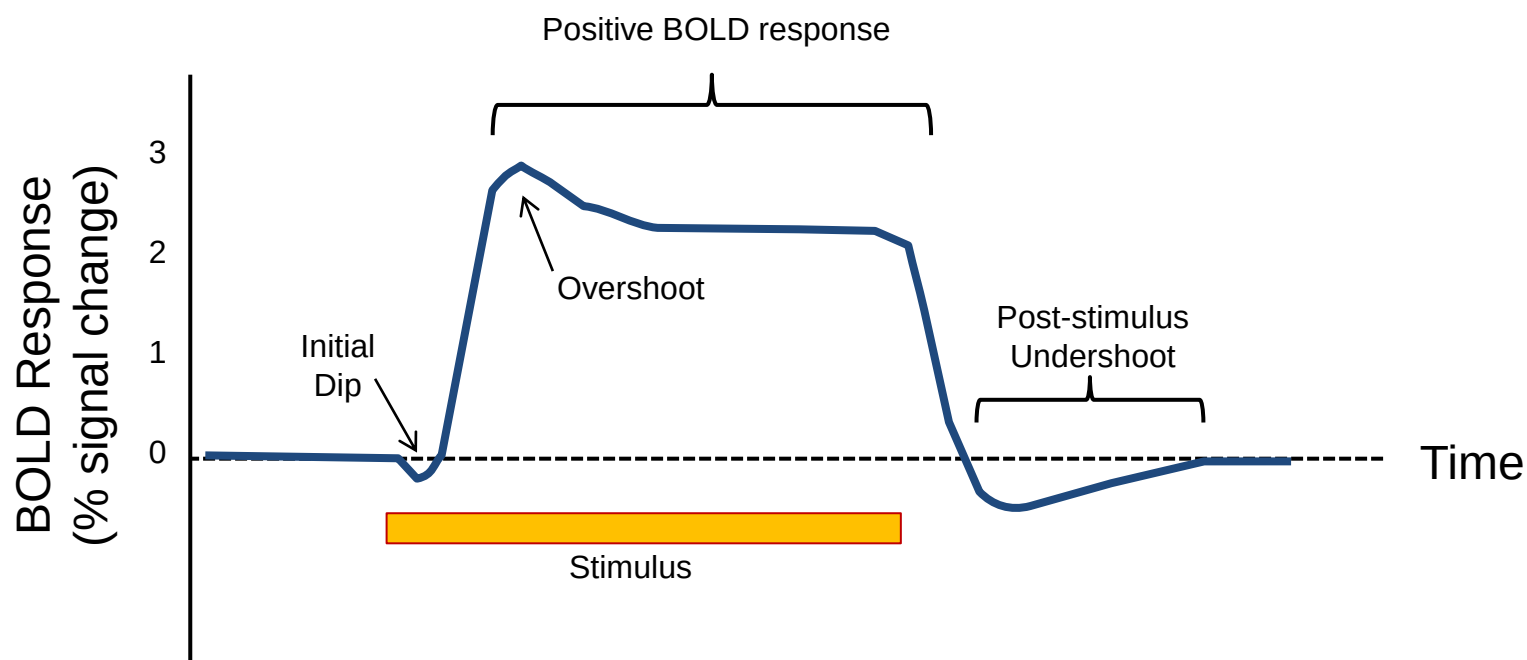
MR response to
flashing visual stimulus

Red: Increased MR
Signal

Blue: Decreased MR
Signal

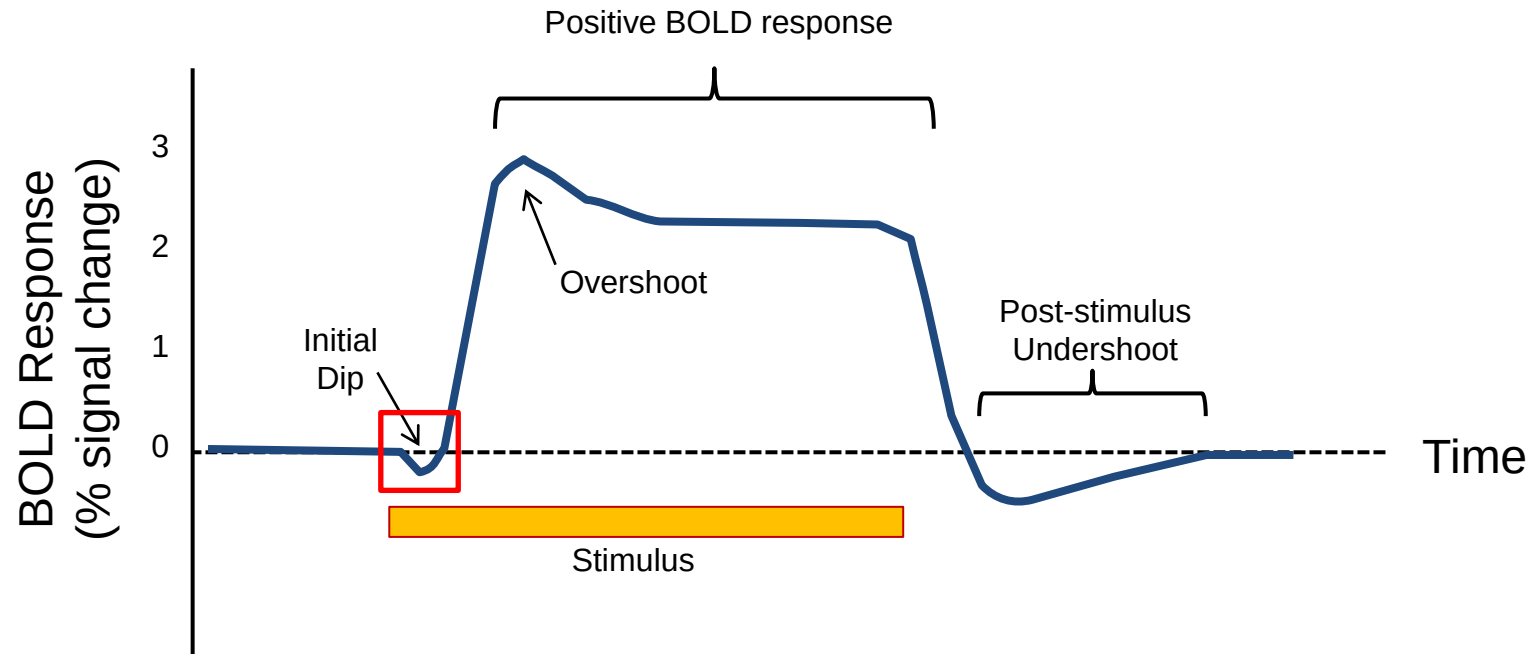
How Does BOLD Relate to Neuronal Activity?

- Typical example of the time course of the MR signal change in response to an extended stimulus:



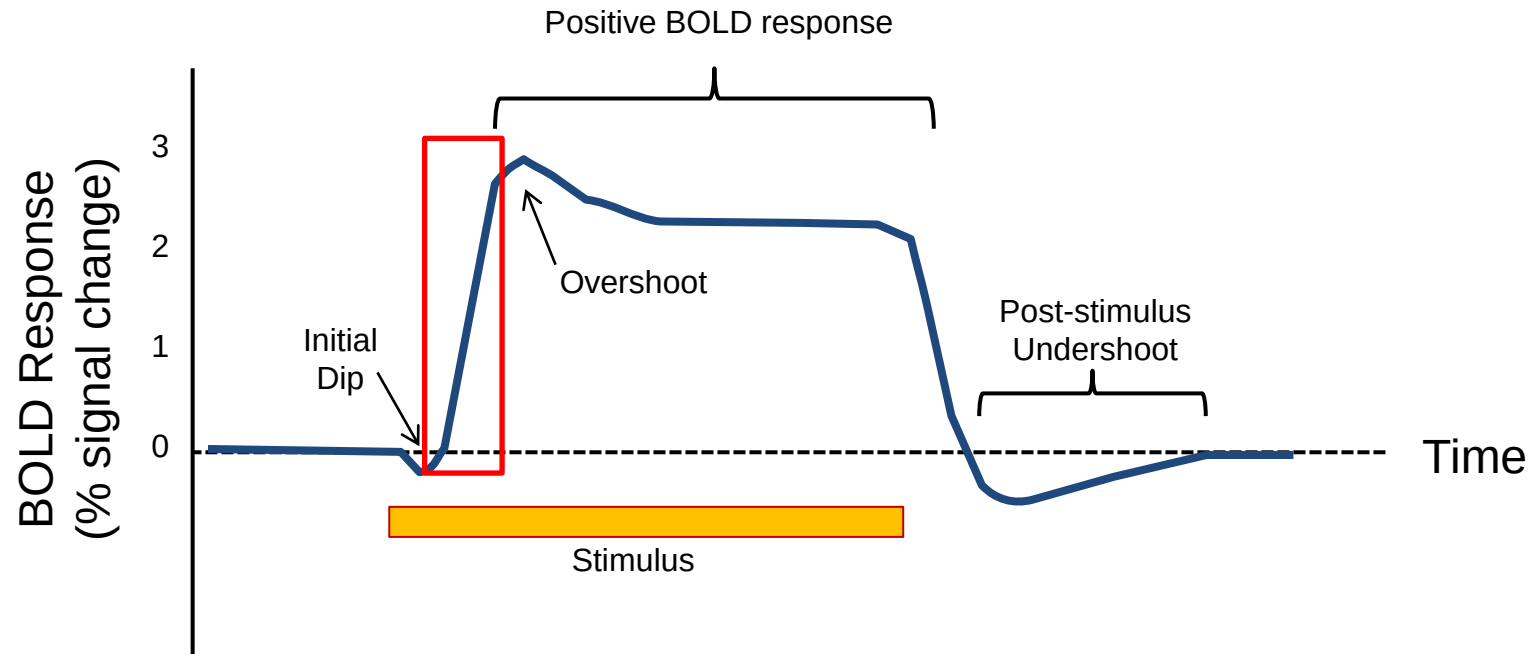
Mechanism: Initial Dip

- Stimulus **increases neuronal activity**, which requires more oxygen and increases the amount of deoxygenated blood
 - Leads to a decrease in MR signal intensity
- **“Initial dip”** rather weak and not observed in all studies
 - If observed, often spatially more restricted than main peak



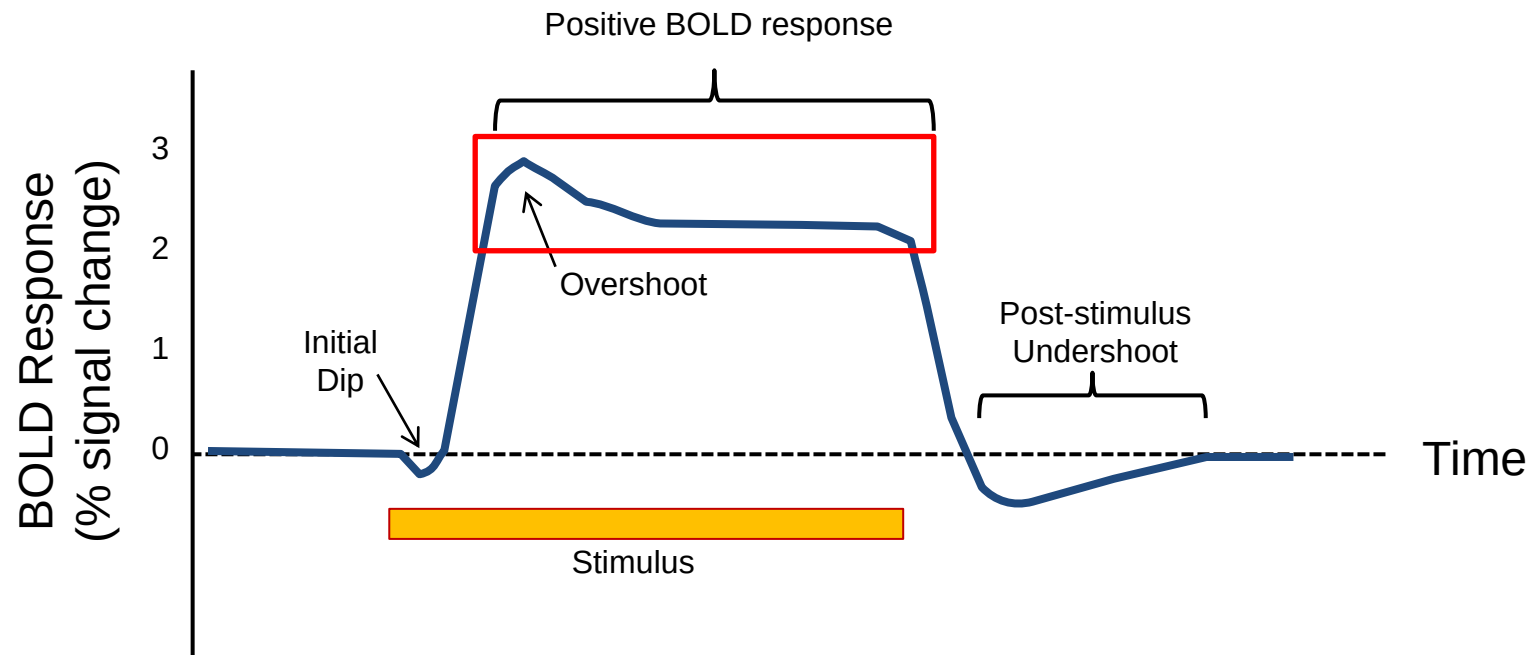
Mechanism: Rise

- **Local blood flow increases**, providing more oxygenated blood and **overcompensating** the additional need from neural activity
 - Change in signal strength between 5% (primary sensory stimulation) and 0.1-0.5% (cognitive tasks)
 - Basis of most fMRI studies



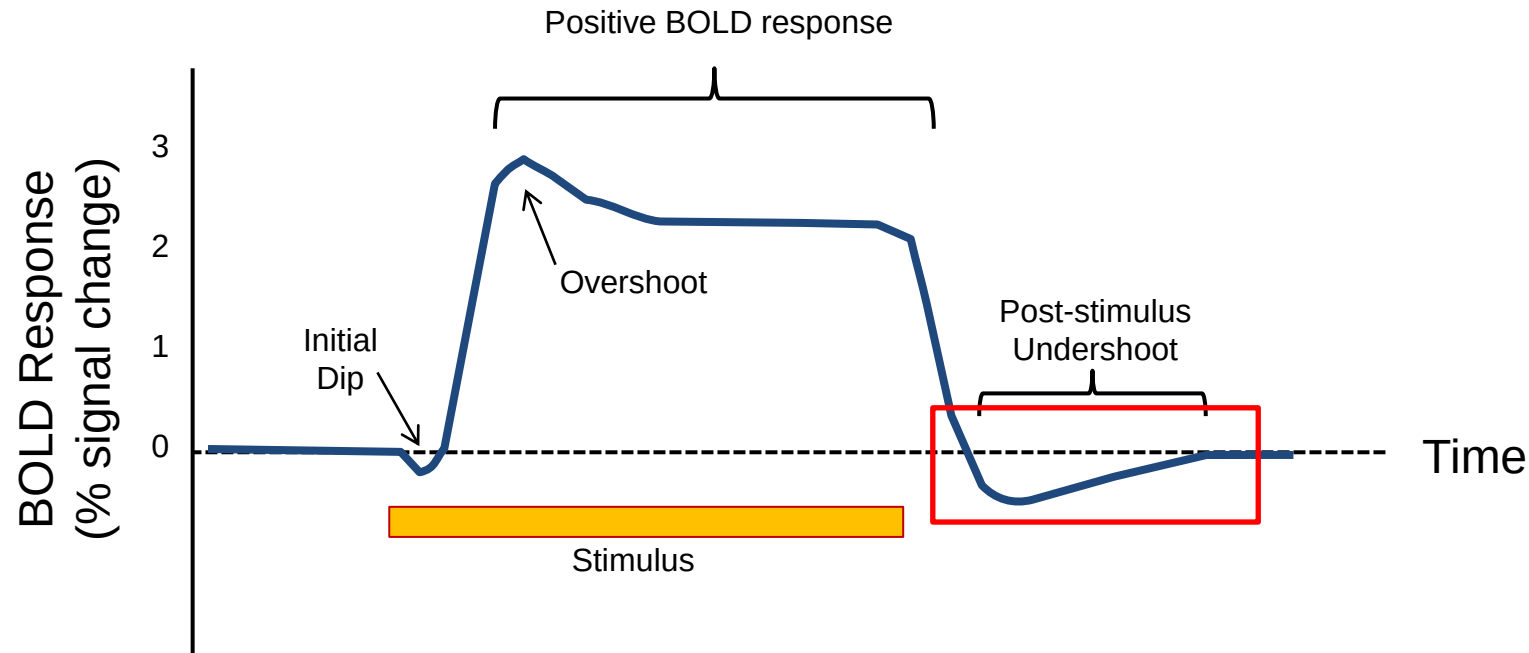
Mechanism: Overshoot / Plateau

- While neuronal activity persists, increased oxygen uptake continues to be overcompensated
 - In “blocked” fMRI design with extended phases of stimulation, a plateau follows an initial overshoot



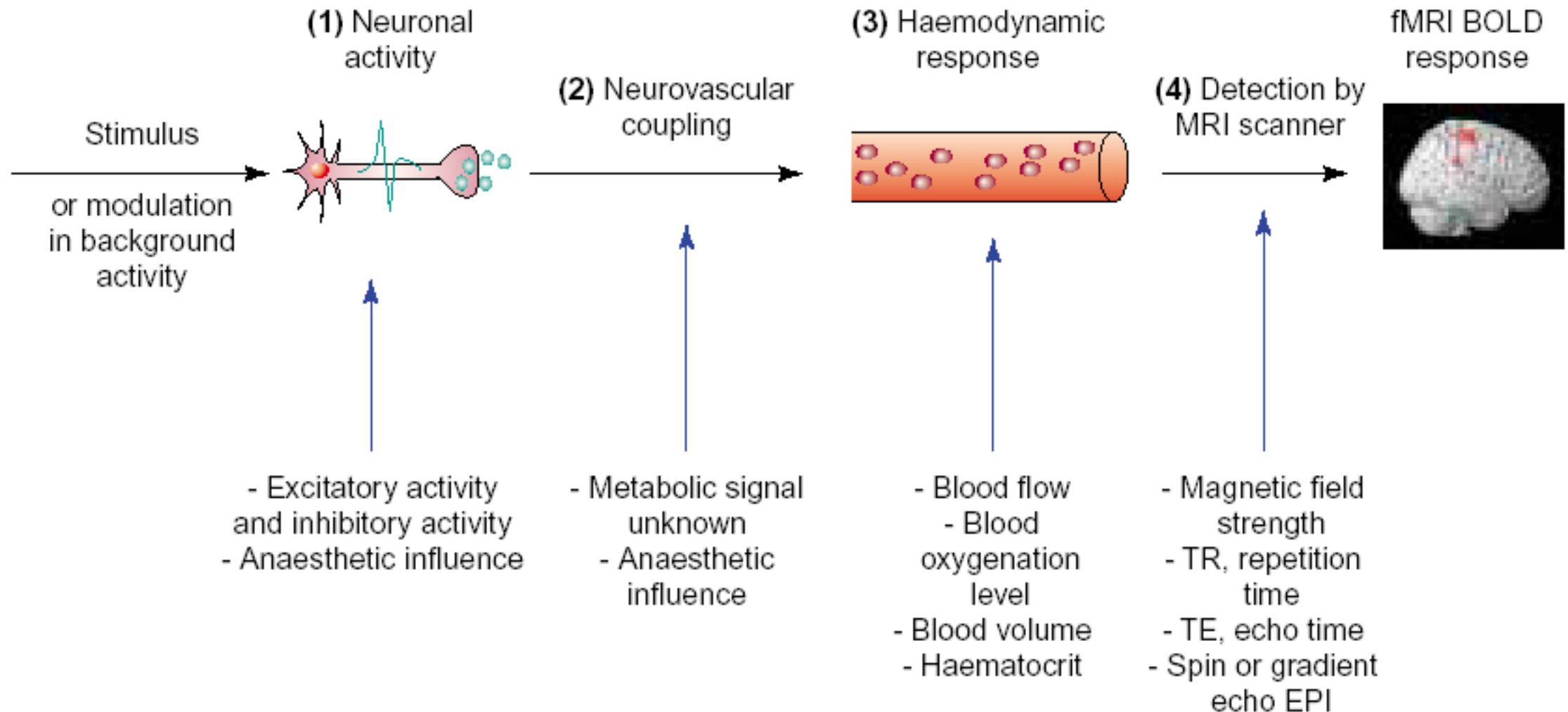
Mechanism: Post-Stimulus Undershoot

- Many studies observe a **post-stimulus undershoot** in MR intensity that often persists for tens of seconds
 - Mechanism not fully agreed upon [van Zijl et al., 2012]
 - May indicate uncoupling of metabolic and blood flow response:
 - Blood flow returns to baseline earlier than need for oxygen
 - Much oxygen used while “cleaning up” after activity



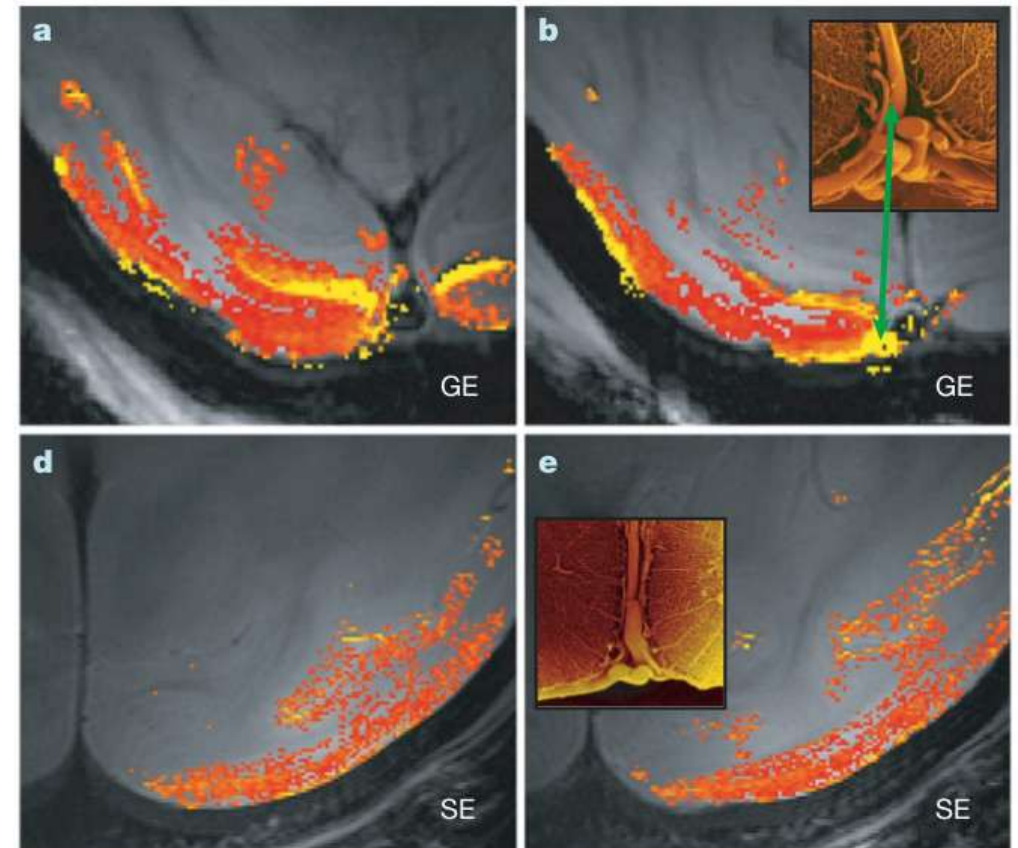
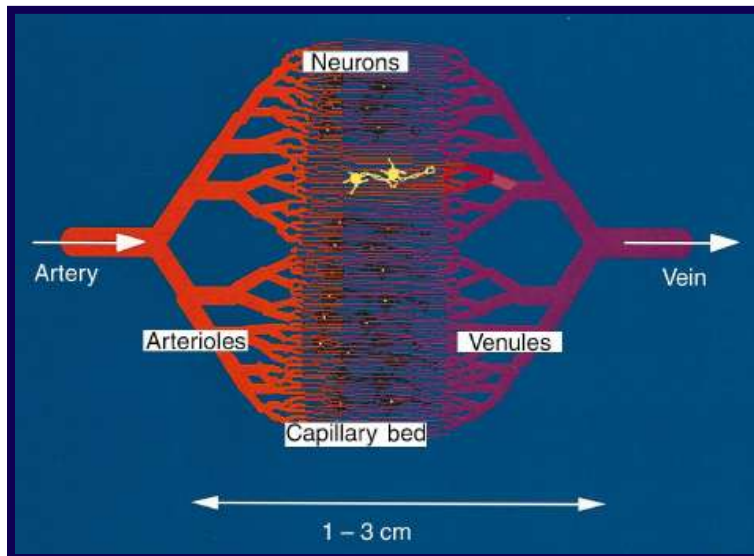
Mechanism: Overview

- fMRI provides a rather **indirect measure** of neuronal activity



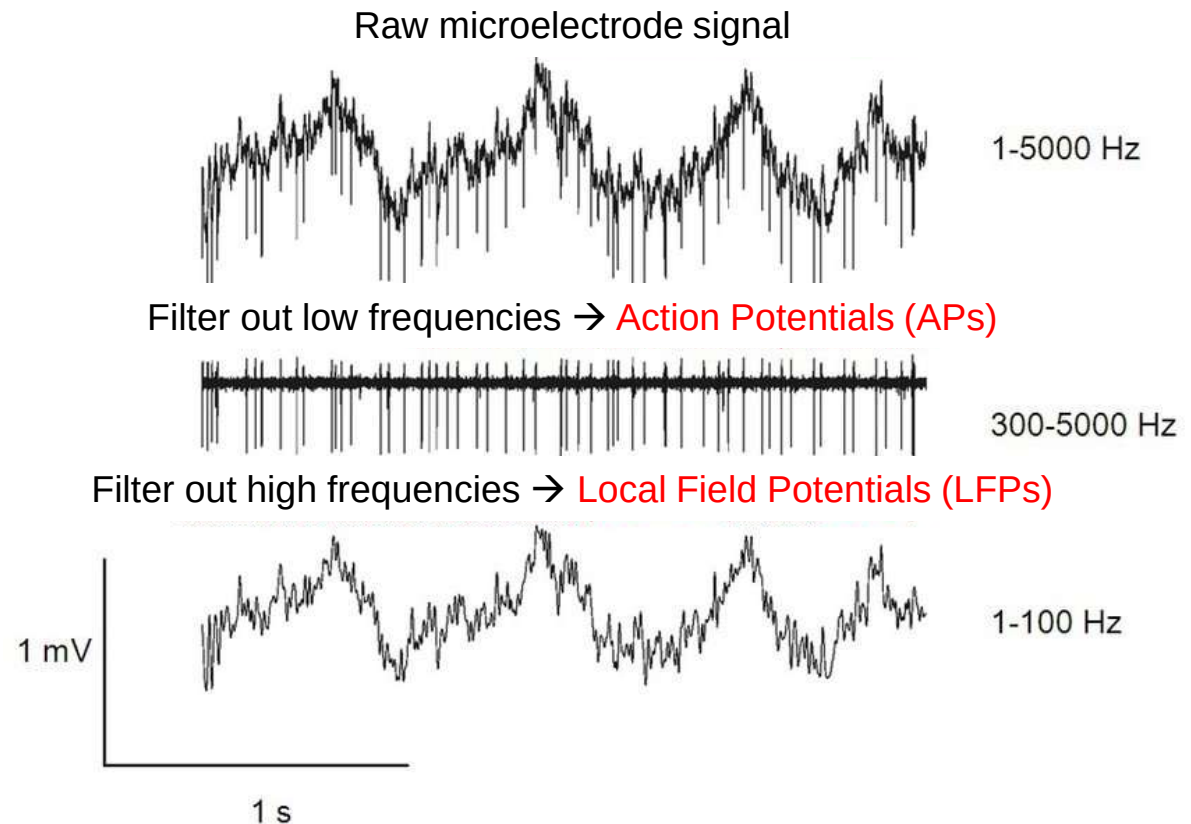
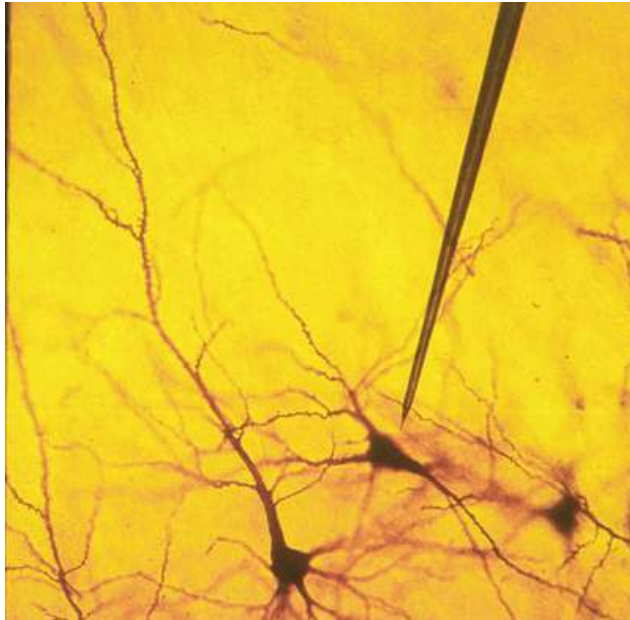
Caveat: “Brain vs. Vein”

- A BOLD response may be seen not only in the active brain tissue, but also in **larger draining veins**, which can be a few centimeters away
 - Reduced in spin echo images
 - But: Also reduces signal-to-noise



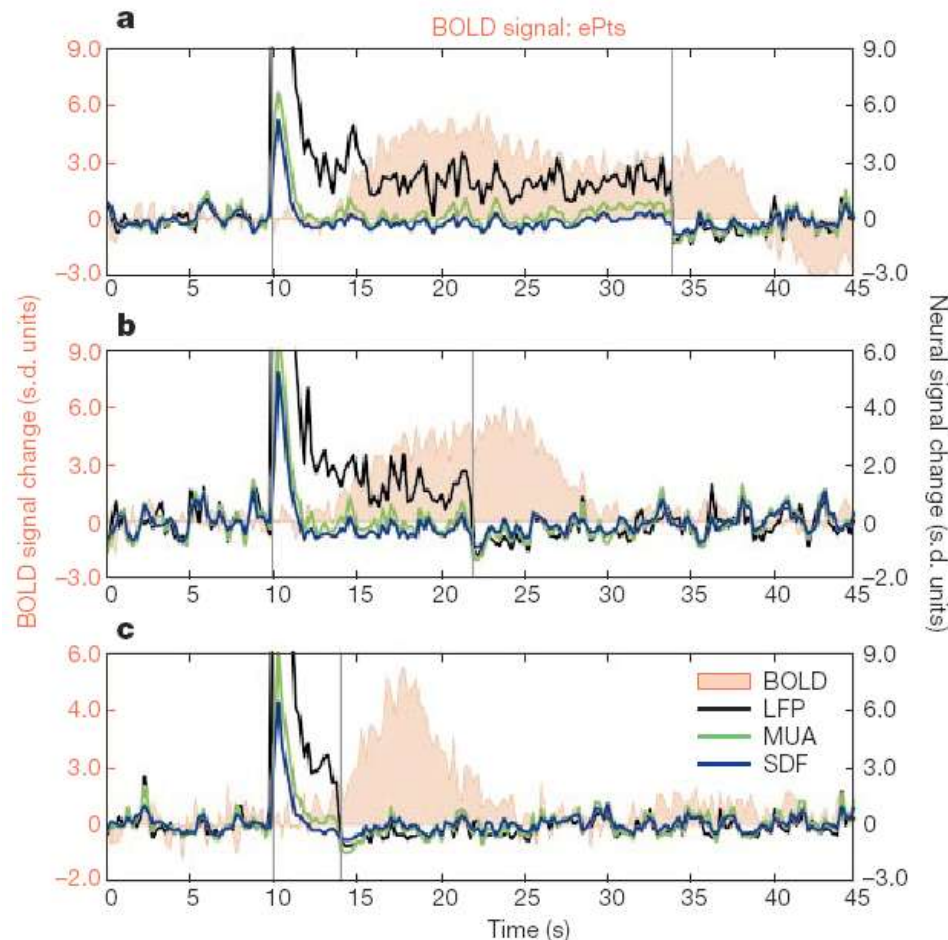
Electrophysiology

- Taking **electrical measurements** in the cortex allows us to monitor the neuronal activity:



What Does BOLD Measure?

- **Logothetis et al. 2001** performed simultaneous electrophysiology and fMRI in monkeys



Conclusion:

- BOLD correlates more clearly with
 - **local field potentials** (post-synaptic potentials; input) than with
 - **multi-unit activity** (action potentials, output)

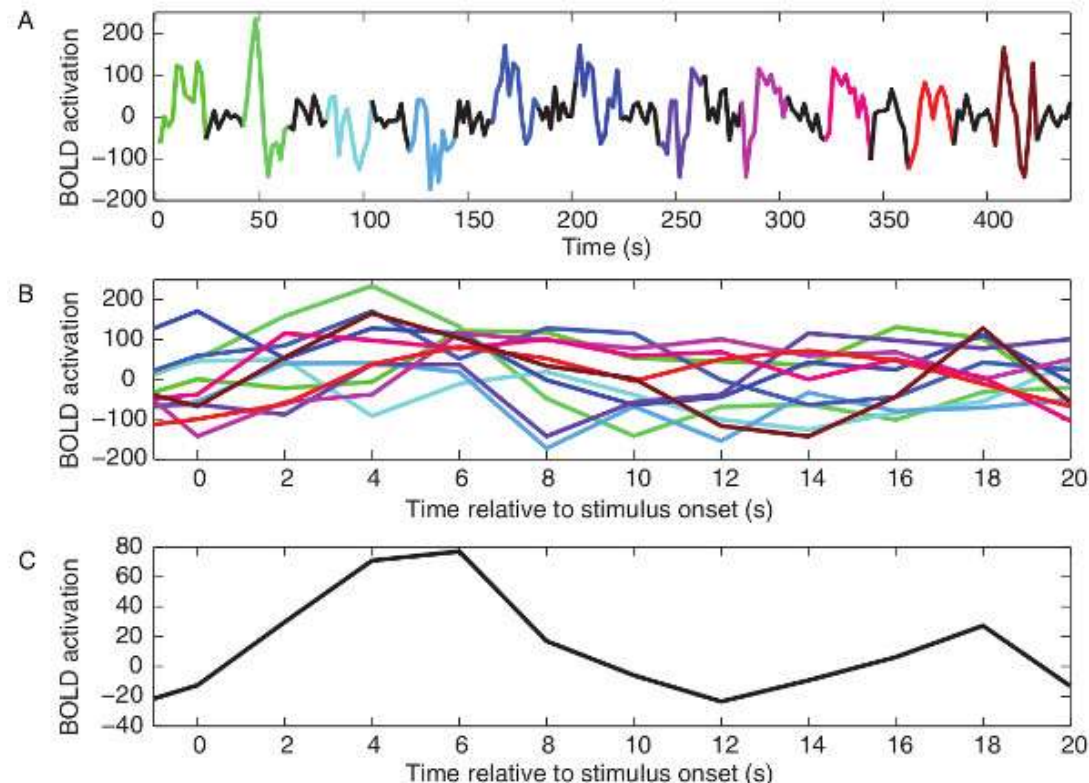
Summary: The BOLD Effect

- **BOLD effect:** MR intensity reduced by presence of deoxygenated blood
- Amount of deoxygenated blood changes during **neuronal activity**
 - After initial dip, additional need for oxygen is overcompensated, leading to a stronger signal
 - Mechanism still not known in full detail, but found to correlate mostly with local field potentials
- **Observing signal change over time** allows us to draw conclusions about brain activity
 - Tradeoff between temporal and spatial resolution

7.2 The Hemodynamic Response Function (HRF)

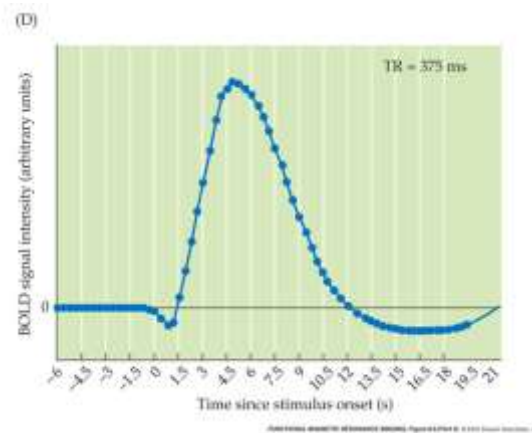
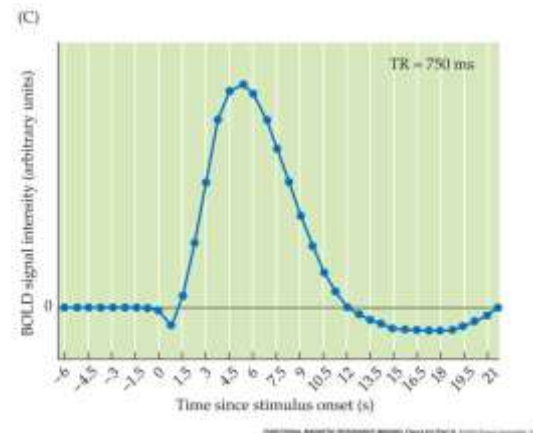
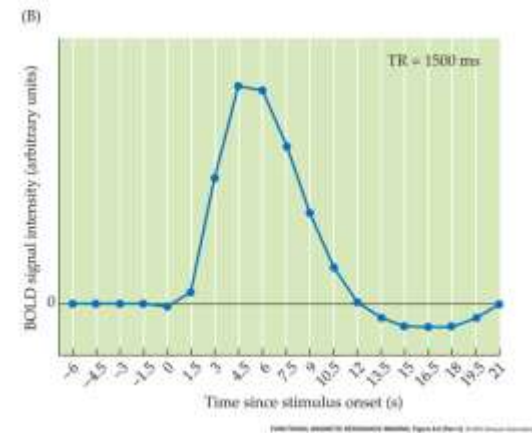
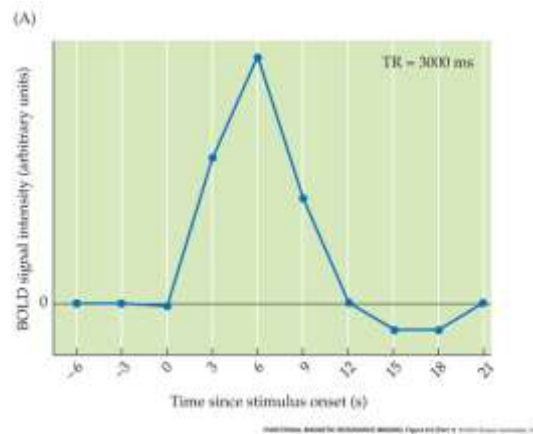
Hemodynamic Response Function

- The **Hemodynamic Response Function (HRF)** describes the time course of MR signal change in response to a short stimulus
 - Can be estimated by repeatedly presenting a short stimulus and averaging the measurements

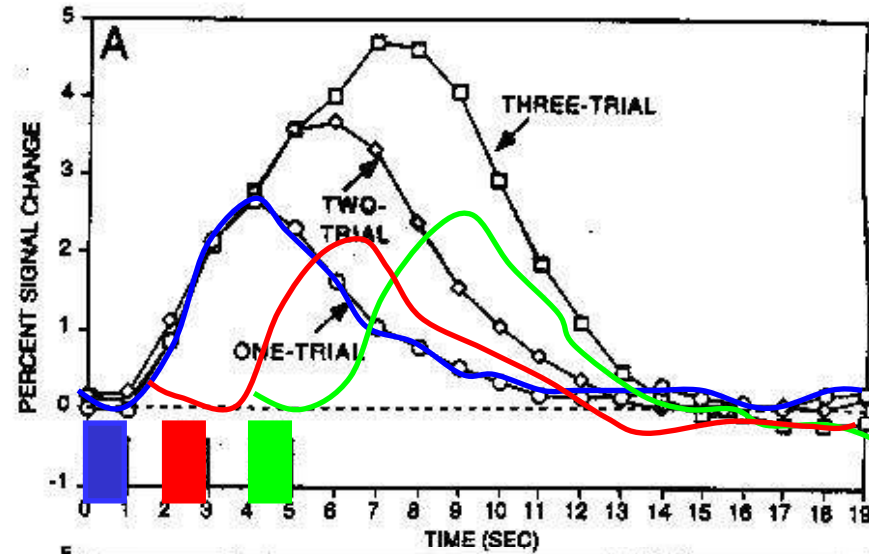


Impact of Sampling Rate

- Reconstructing subtle features of the HRF requires sufficiently high sampling rate
 - Not seeing initial dip can be a matter of resolution

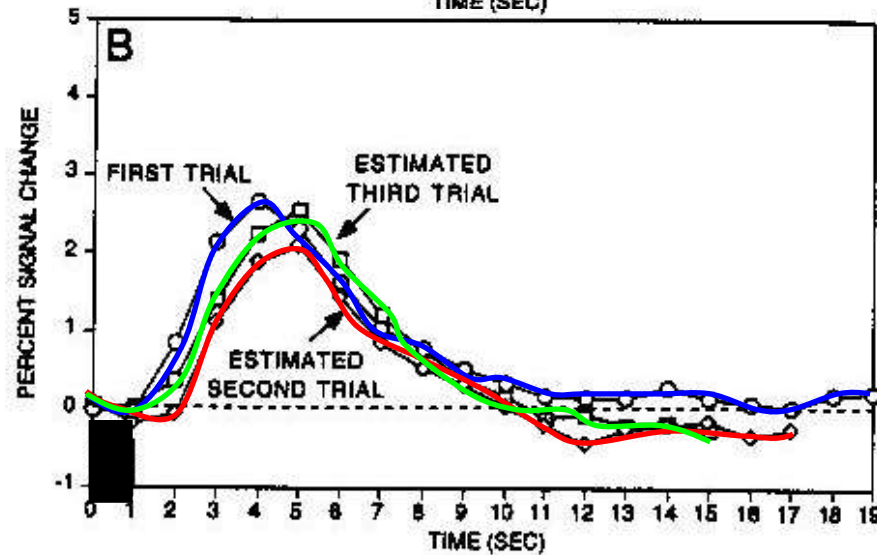


Linearity of Hemodynamic Response



red = 2 - 1

green = 3 - 2



Sync each trial response to start of trial

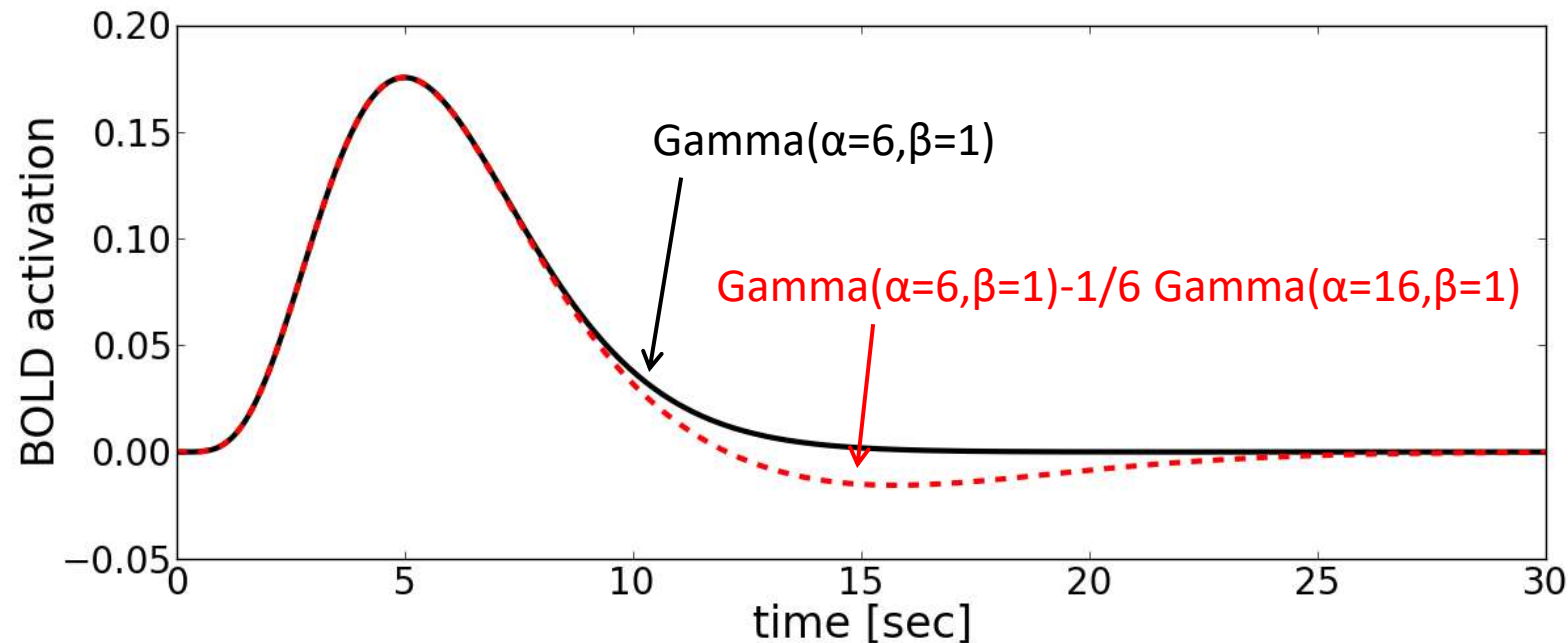
Not quite linear
but good enough!

Limits to Linearity of Hemodynamic Response

- Linearity is widely assumed in fMRI data analysis. Caveats of this assumption include:
 - **Spacing:** Responses to stimuli less than two seconds apart are slightly smaller than expected
 - **Duration:** Very short stimuli have a much larger response than would be expected from longer stimuli
 - Yesilyurt et al. 2008: Response to 5 ms visual stimulus half as large as to 1000 ms stimulus
- **Solution:** Quick succession of stimuli and ultra-short stimuli are rarely used in practice

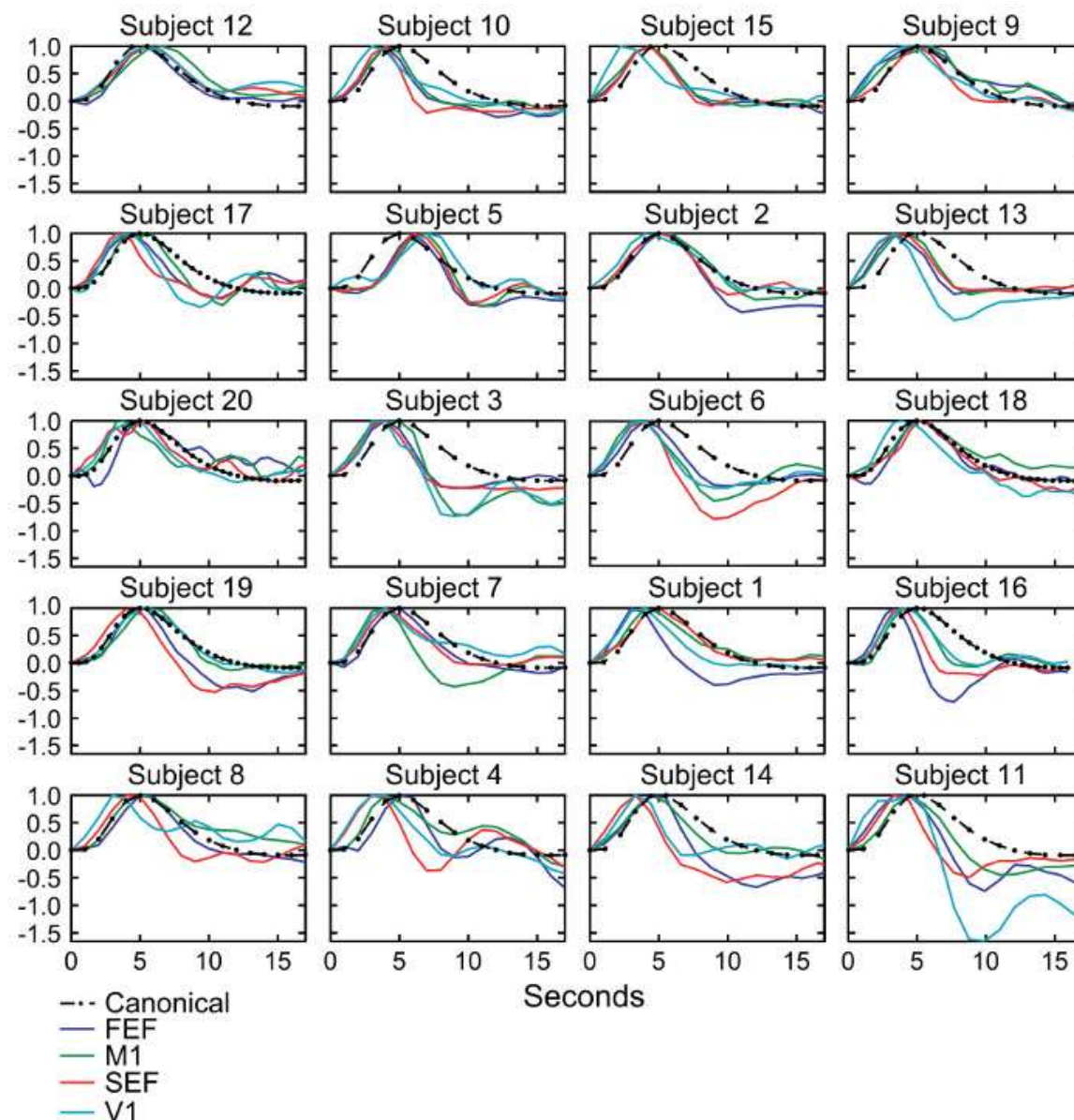
Canonical HRF

- It has been found empirically that the shape of the **Gamma distribution** (for certain parameters) closely resembles the HRF
 - Often used as a “**canonical HRF**” in analysis
 - Usually (imprecisely) called “**Gamma function**”
 - “**Double Gamma**” includes poststimulus undershoot



Variability of HRF

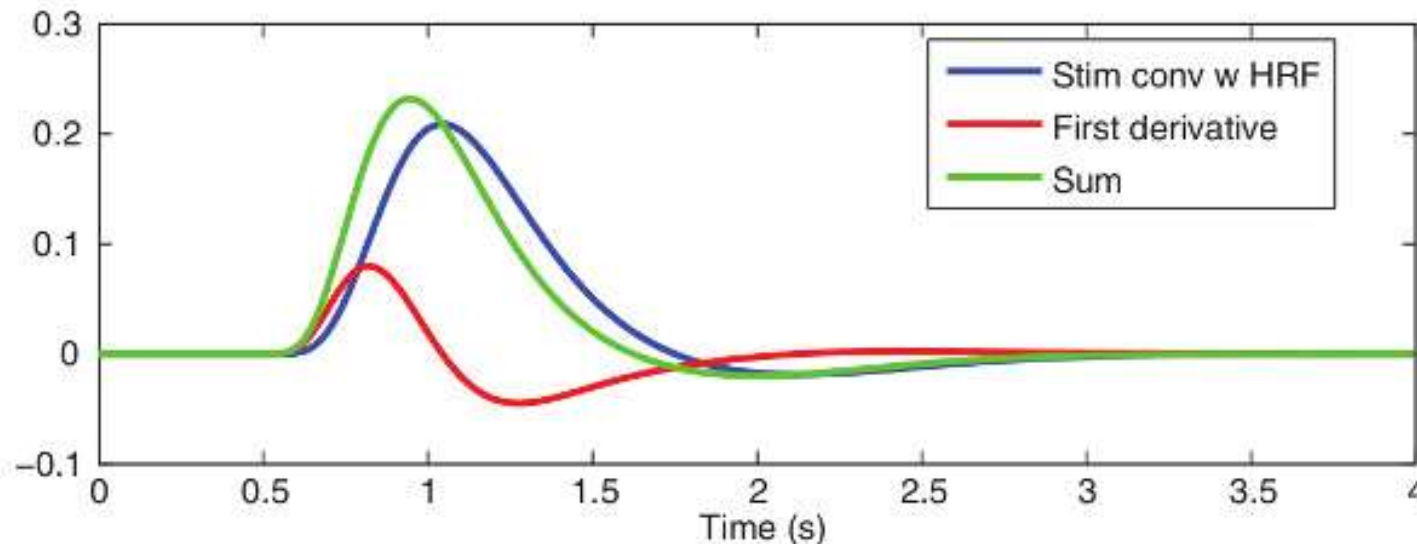
- HRF varies considerably **between subjects**
- HRF varies **between regions**, too, but usually less strongly
- Could perform a **per-subject estimate** of the HRF in a pre-scan, but is relatively rare in practice



Derivative of the Canonical HRF

- Canonical HRF alone is not flexible enough to capture variety in timing found in practice
 - In a linear model (e.g., GLM), small shifts can be captured by including the derivative of the canonical HRF
 - First-order Taylor approximation:

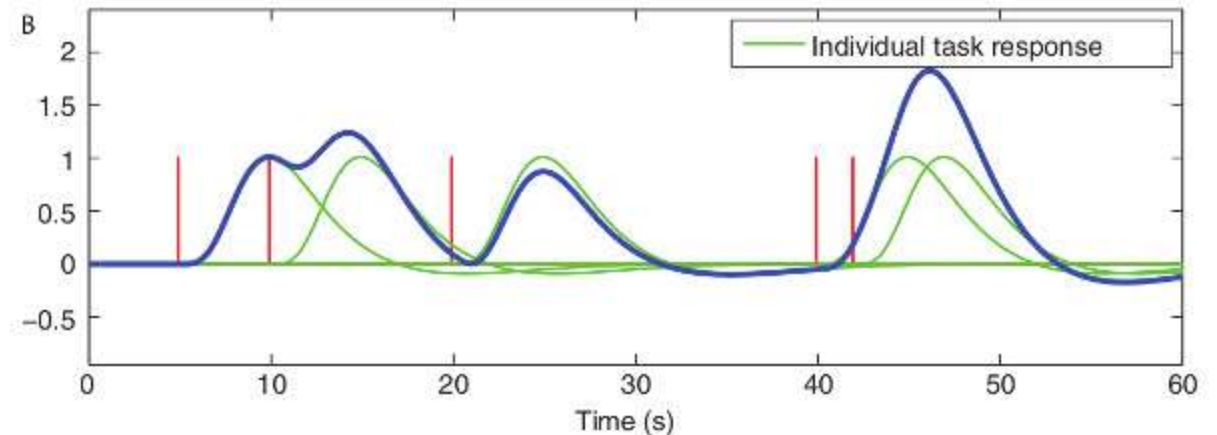
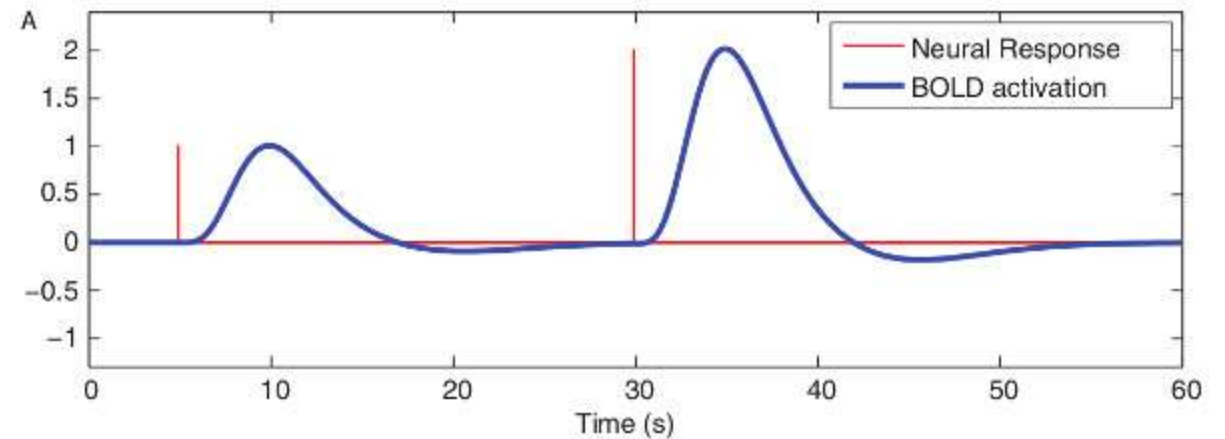
$$f(t_1) \approx f(t) + f'(t) \overbrace{(t_1 - t)}^{\delta}$$
$$f(t + \delta) \approx f(t) + \delta f'(t)$$



Convolution

- If hemodynamic response is linear, the BOLD signal is given by convolving the neural response f with the HRF h

$$(h * f)(t) = \int h(\tau)f(t - \tau)d\tau$$



Summary: Hemodynamic Response

- The MR signal change in response to stimuli is usually assumed to be linear
 - Limitations avoided by experimental design
 - Modeled by convolving stimulus with **Hemodynamic Response Function (HRF)**
- HRF varies between subjects and regions
 - Despite this, a **canonical HRF** is commonly used in fMRI analysis
 - Including first derivative compensates for small deviations in timing

7.3 General Linear Model (GLM)

Motivation: General Linear Model

- **General Linear Model** provides a unified framework for different statistical models
 - Includes linear regression, t tests, ANOVA
 - Provides additional flexibility that we need for fMRI analysis
 - Standard for statistical analysis in neuroimaging
 - Expresses vector $\mathbf{y}$ of N measurements as the product of an $N \times (p+1)$ design matrix $\mathbf{X}$ and a vector $\boldsymbol{\beta}$ of $p+1$ parameters, plus an i.i.d. Gaussian noise vector $\boldsymbol{\varepsilon}$:

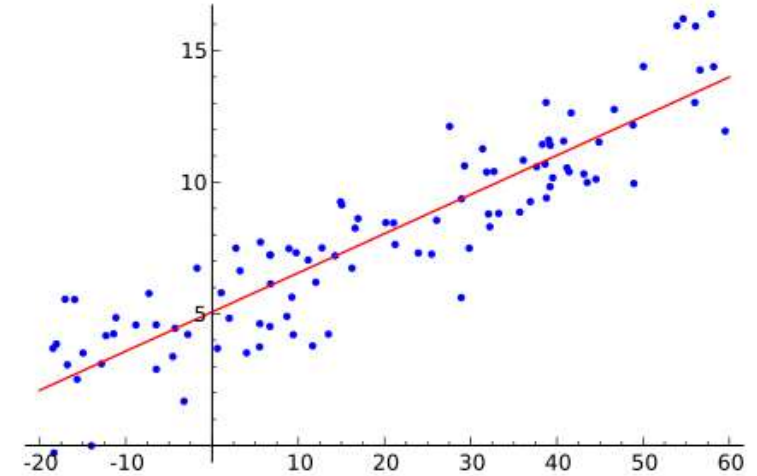
$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Example: Linear Regression

- Ordinary linear regression can be stated as:

$$\mathbf{y} = \beta_0 + \beta_1 \mathbf{x}_1 + \epsilon$$

↑ ↑
intercept slope



- Alternatively, we can write it as a General Linear Model. Given data $\{(x_{1,1}, y_1), \dots, (x_{1,N}, y_N)\}$, we obtain $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with

$$\boldsymbol{\beta} = [\beta_0, \beta_1]^T \text{ and } \mathbf{X} = \begin{bmatrix} 1 & x_{1,1} \\ \vdots & \vdots \\ 1 & x_{1,N} \end{bmatrix}$$

Multiple Linear Regression

- Linear regression with $p > 1$ independent variables is called **multiple regression**:

$$y = \beta_0 + \beta_1 \mathbf{x}_1 + \cdots + \beta_p \mathbf{x}_p + \epsilon$$

- Same GLM $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with longer $\boldsymbol{\beta}$ vector
- Estimation of $\boldsymbol{\beta}$:
 - Since number of measurements typically exceeds $p+1$, we have to solve a linear least squares problem
 - Different numerical algorithms available, including Singular Value Decomposition (SVD)
 - Yields estimates $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\epsilon}}$

Example: Two-Sample t-Test in the GLM

- *Example:* Two-sample t test in the GLM framework: $\boldsymbol{\beta} = [\mu^C, \mu^{NC}]^T$

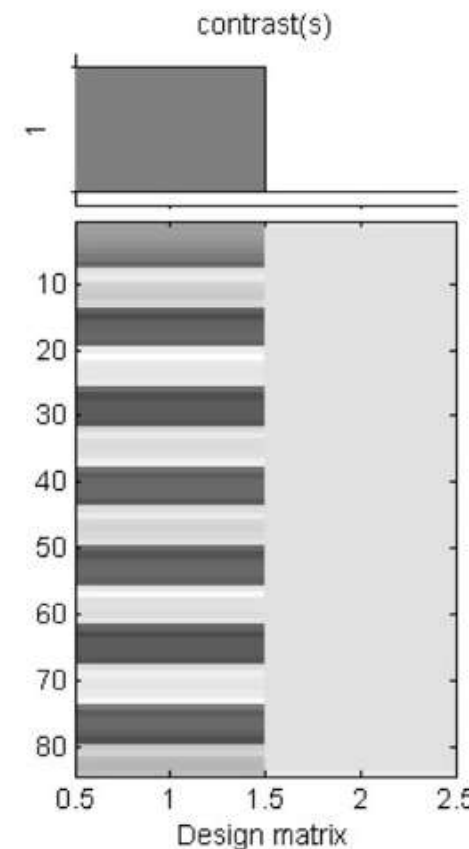
$$\begin{pmatrix} v_1^C \\ \vdots \\ v_n^C \\ v_1^{NC} \\ \vdots \\ v_n^{NC} \end{pmatrix} = \begin{bmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 1 \end{bmatrix} \boldsymbol{\beta} + \boldsymbol{\epsilon}$$

- **Null hypothesis** $\beta_0 = \beta_1$ can be written in the general form $\mathbf{c}^T \boldsymbol{\beta} = 0$ using the contrast vector $\mathbf{c} = [1 \ -1]^T$
- For any $\mathbf{c}$, the t score (with $\nu = N - (p + 1)$) can be shown to be

$$t = \frac{\mathbf{c}^T \hat{\boldsymbol{\beta}}}{\sqrt{\hat{\sigma}^2 \mathbf{c}^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{c}}} \quad \text{with} \quad \hat{\sigma}^2 = \frac{\hat{\boldsymbol{\epsilon}}^T \hat{\boldsymbol{\epsilon}}}{N - (p + 1)}$$

GLM for Hypothesis Tests in fMRI

- A two-sample t -test is not an adequate model to test the null hypothesis in an fMRI experiment
 - “Voxel intensities do not change based on the task or stimulus.”
 - Grouping time steps into “stimulus on/off” groups would not account for the hemodynamic response
 - Instead, compute an expected BOLD timecourse by convolving the stimulus with the HRF (7.2) and use it as a column of design matrix $\mathbf{X}$ in a GLM
 - Null hypothesis: Corresponding $\beta_i = 0$, expressed by a 1 in the corresponding position of contrast $\mathbf{c}$
 - t is computed as on the previous slide



Scaling Issues

$$t = \frac{c^T \hat{\beta}}{\sqrt{\text{var}(c^T \hat{\beta})}} = \frac{\overbrace{c^T \hat{\beta}}}{\sqrt{\hat{\sigma}^2 \underbrace{c^T}_{\text{contrast}} \underbrace{(X^T X)^{-1}}_{\text{regressors}} \underbrace{c}_{\text{contrast}}}}$$

- ❑ The t -statistic does not depend on the scaling of the regressors.
- ❑ The t -statistic does not depend on the scaling of the contrast.
- ❑ Contrast $c^T \hat{\beta}$ depends on scaling.

Regressing Out Nuisance Variables

- **Additional benefit of GLM:** Can “regress out” nuisance variables
 - *Example 1:* Head motion can affect voxel intensities. Inserting a column in **X** that quantifies the motion at each measurement can help absorb related artifacts
 - *Example 2:* If a subject pool is not age-matched, inserting a column of ages in **X** applies a form of “age-correction”
 - Mean-centering co-regressors does not affect p values, but does affect the interpretation of the β_i
 - *Example:* Does average brain volume refer to the average at the mean age in your population, or at the (often extrapolated) age zero?
 - *Limitation:* Assumes that effect of motion/age is linear. It’s still preferable to avoid motion as much as possible / use matched subjects.

Example: ANOVA in the GLM

- GLM representation of ANOVA based on generative interpretation:

$$y_{ij} = \underbrace{\mu + a_i}_{= \mu_i} + \epsilon_{ij}$$

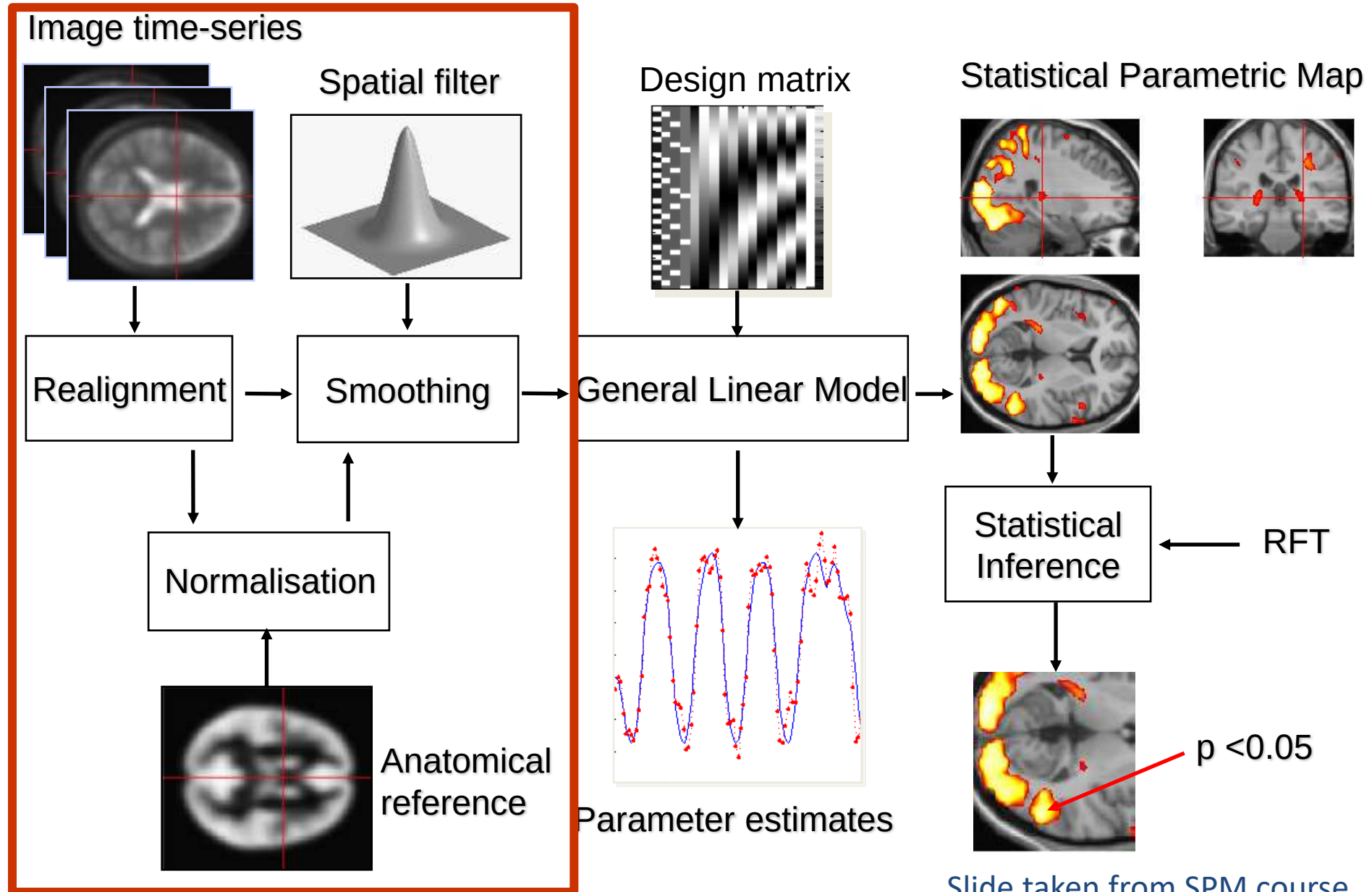
- Assuming three groups, we obtain GLM $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ as
 - $\boldsymbol{\beta} = [\mu_1, \mu_2, \mu_3]^\top$
 - $x_{ij} = 1$ if measurement i belongs to group j ; $x_{ij} = 0$ otherwise
 - To simultaneously test $\mu_1 = \mu_2$ and $\mu_1 = \mu_3$ ($\mu_2 = \mu_3$ follows implicitly), define $\mathbf{C}$ matrix such that $\mathbf{C}\boldsymbol{\beta} = \mathbf{0}$:
$$\mathbf{C} = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$$
 - Similar to t score, $\mathbf{C}$ and $\hat{\boldsymbol{\beta}}$ can be combined to compute F score
 - Formula too long for this slide

Summary: General Linear Model

- The General Linear Model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ unifies the statistical tests that are most widely used in neuroimaging
 - Specification using design matrix $\mathbf{X}$ and contrast vector $\mathbf{c}$ or contrast matrix $\mathbf{C}$
 - Can be used to compute t scores, F scores, or combined with permutation-based testing
 - Additional flexibility allows us to account for hemodynamic response function in fMRI analysis
 - Allows us to “regress out” variables of no interest

7.4 The Task fMRI Processing Pipeline

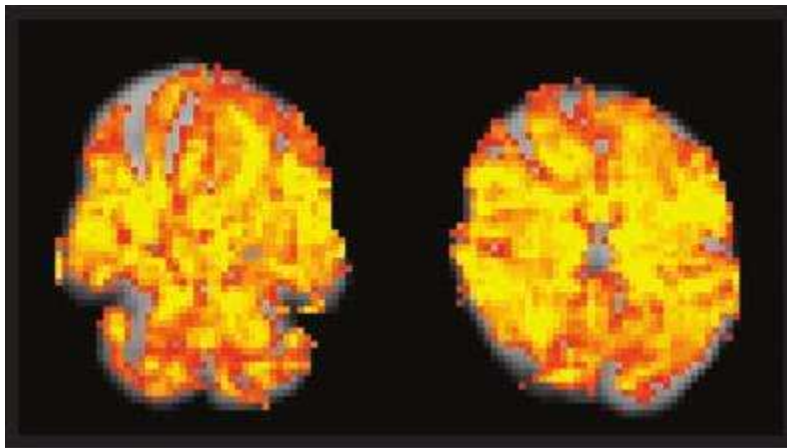
Overview: Task fMRI Processing Pipeline



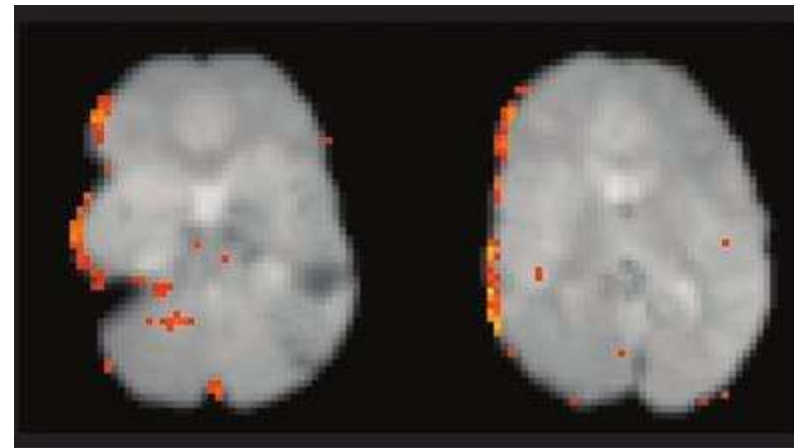
Slide taken from SPM course

Realignment

- Since fMRI experiments typically go on for extended periods of time (at least several minutes), the head is rarely perfectly still
- Realignment uses **rigid registration** to a common reference to compensate this
 - Often, image from the middle is used as a reference



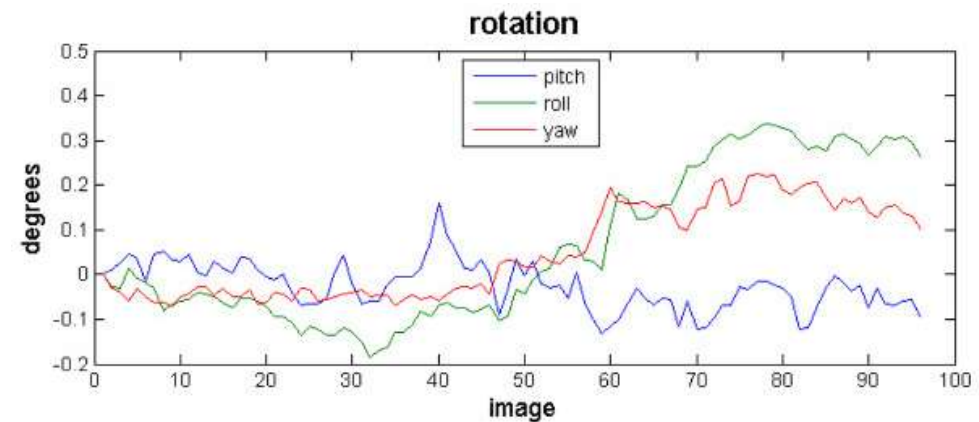
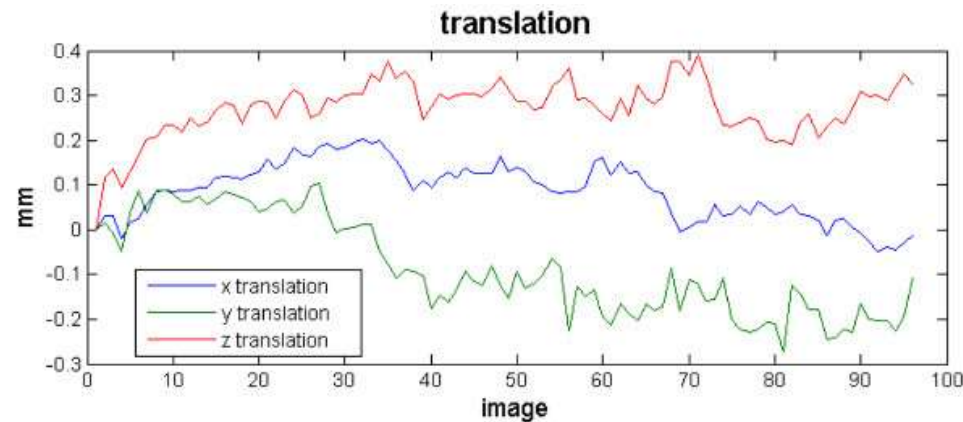
Artifacts from severe head motion
("flaming brain")



Artifacts from slight head motion

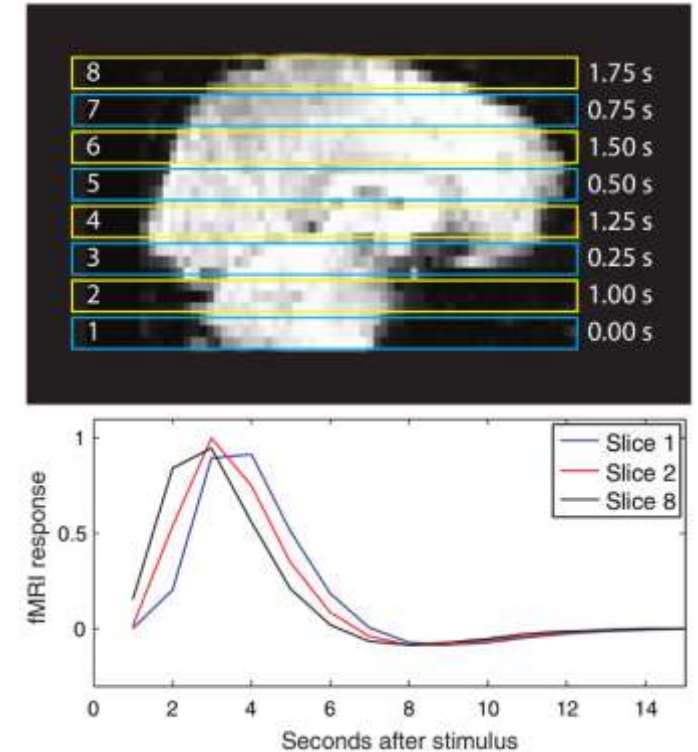
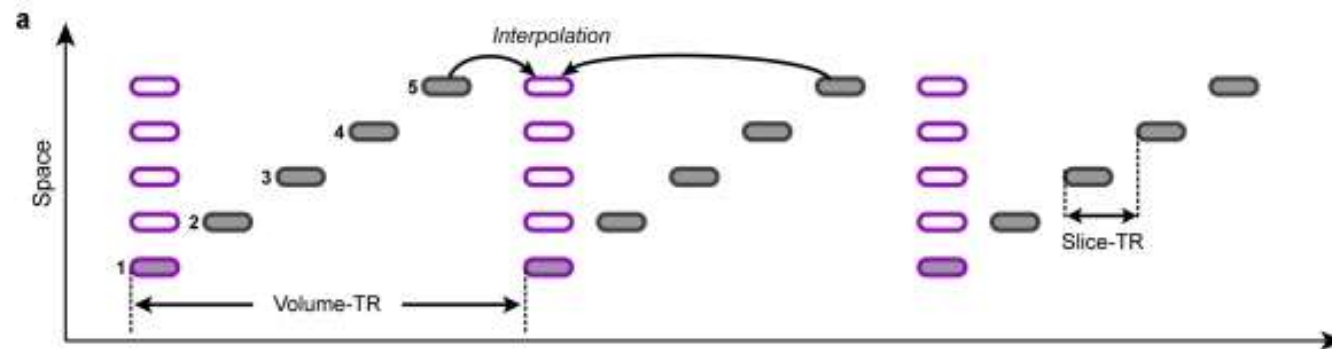
Motion Parameters

- Realignment outputs motion corrected images as well as estimated **motion parameters**
 - Amount of translation/rotation w.r.t. baseline
- Motion parameters and their derivative can be included as coregressors of no interest in statistical analysis to account for impact of motion on MR signal intensities
 - *Caveat*: This impact is rarely linear!



Slice Timing Correction

- **Problem:** 2D slices are measured sequentially, but analysis assumes all voxels are synchronized
- **Solution:** Interpolate (across time) so that all slices correspond to same point in time
 - *Caveats:* Requires exact timing information from scanner, propagates artifacts across time

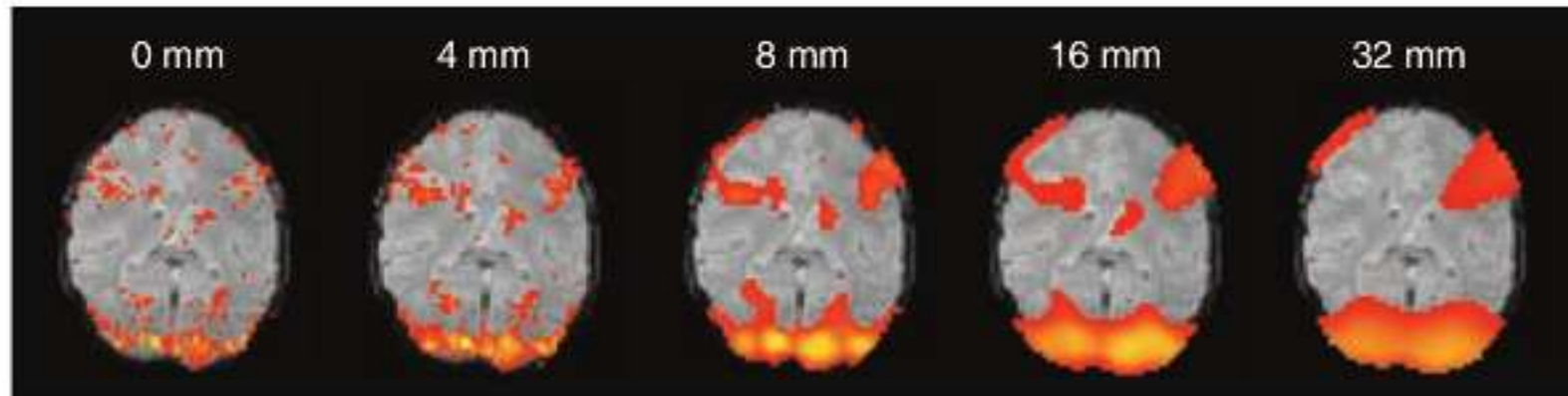


Normalization

- Often, fMRI data is normalized into a **standard space** (e.g., MNI) for analysis
- Due to low resolution of fMRI, commonly done **indirectly**:
 1. **Linearly register** mean fMRI image to a high-resolution structural image of the same subject
 - Use cross-modality similarity measure such as mutual information
 2. **Nonlinearly normalize** structural image to an atlas, apply the same transformation to fMRI images
 - SPM integrates this with tissue segmentation to obtain (and compensate for) **bias fields**

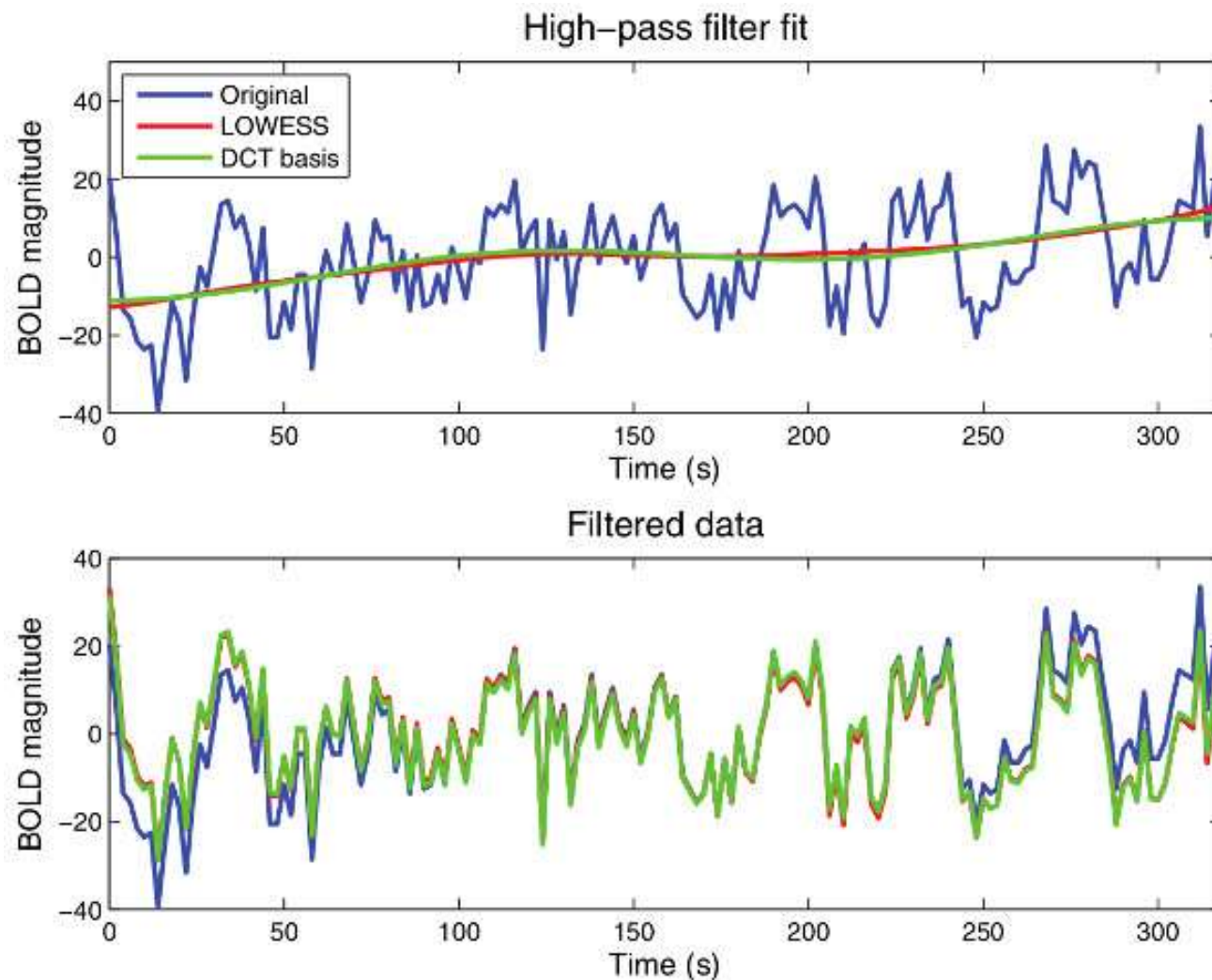
Smoothing

- Some amount of (spatial) **Gaussian smoothing** is typically applied to the data
 - Reduces high (spatial) frequency noise
 - Reduces effective number of comparisons
 - Compensates residual misalignment
 - Enhances activation at scale of smoothing kernel

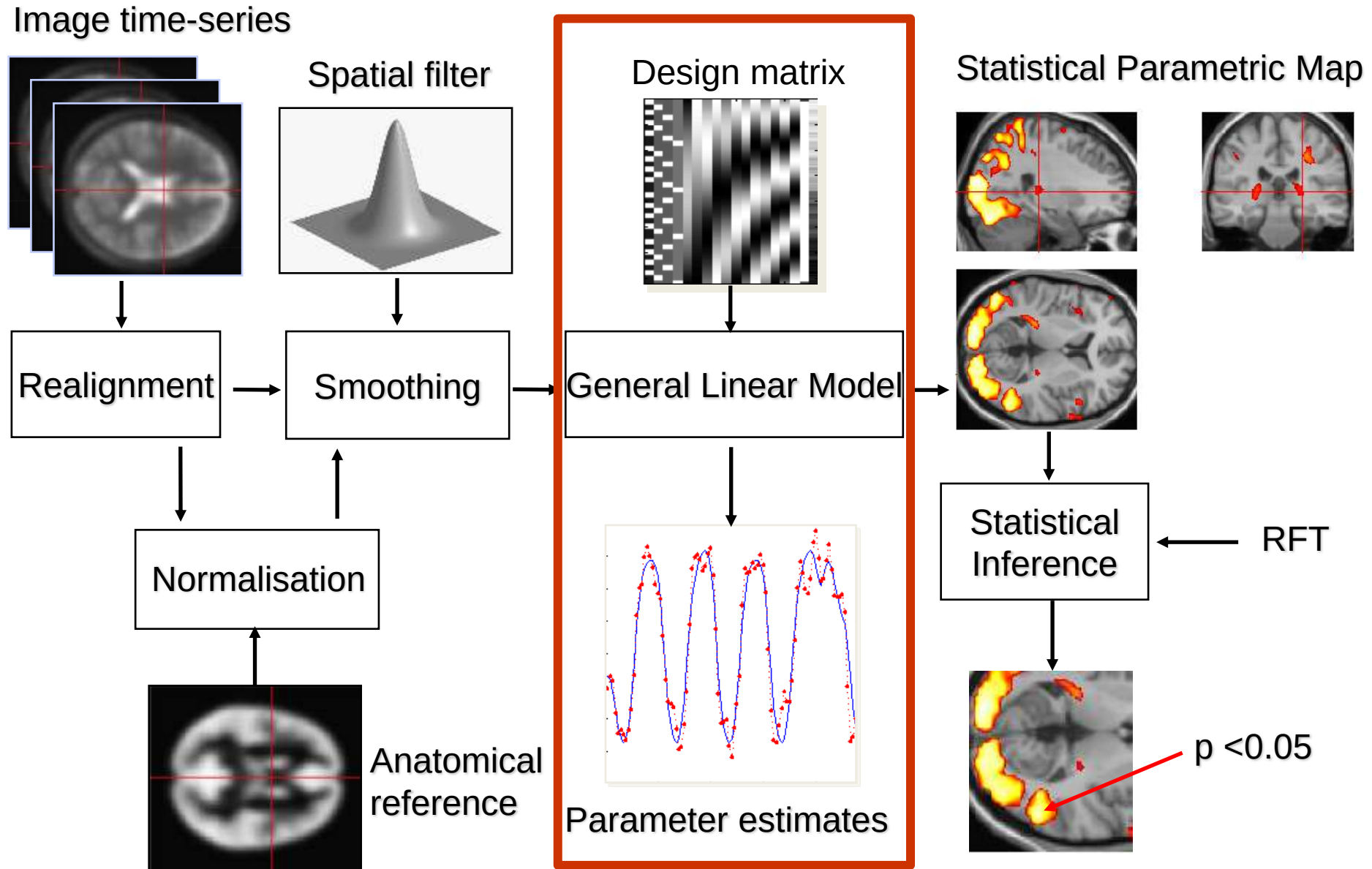


High-Pass Filtering

- The baseline MR intensity often **drifts slowly** during the experiment
- **High-pass filter** removes low (temporal) frequency component of the signal



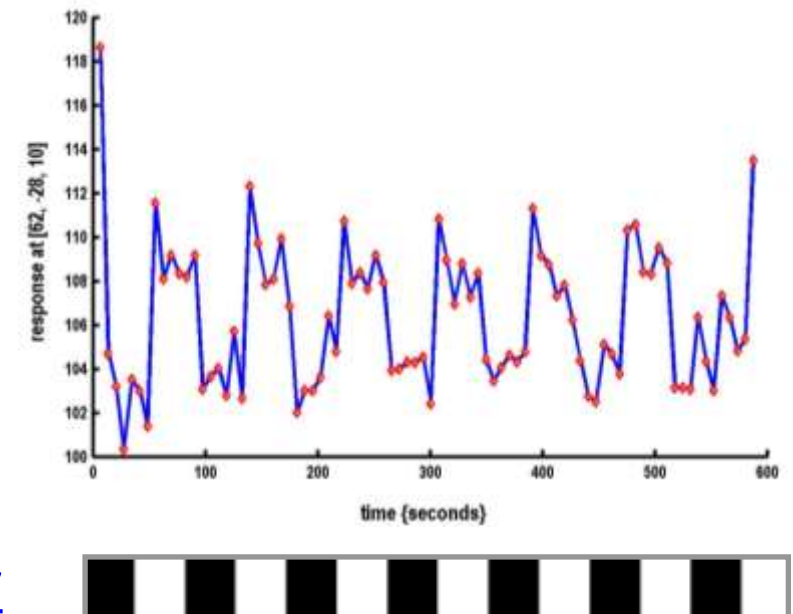
Overview: Task fMRI Processing Pipeline



Slide taken from SPM course

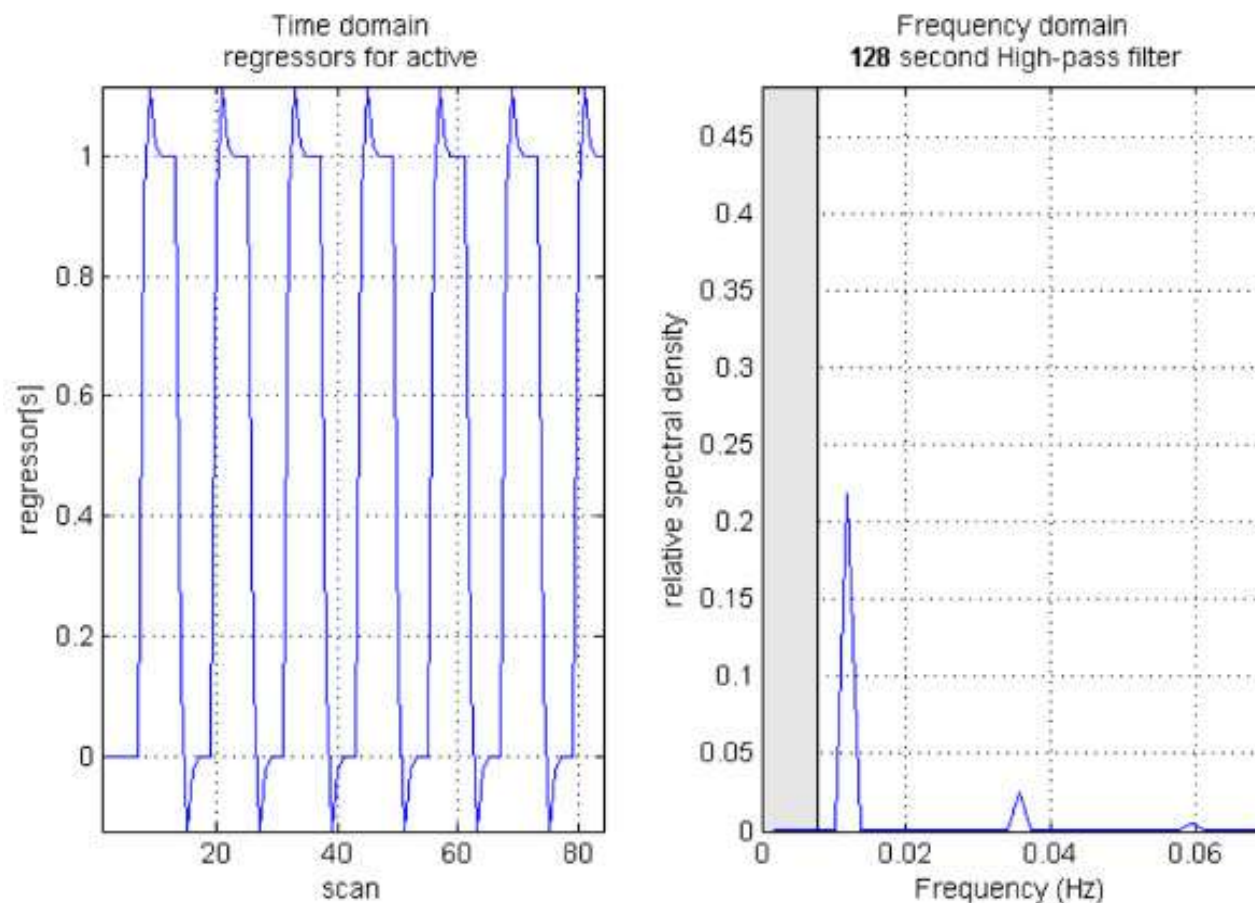
Example: A Basic fMRI Experiment

- Early proof-of-concept experiment with auditory stimulation (spoken words), performed at UCL
 - Full-brain EPI scan with 64x64x64 voxels (3x3x3 mm³ resolution), repeated at TR=7 sec
 - Alternating between 42 sec blocks of rest and auditory stimulation
 - 96 acquisitions overall
 - First few have been discarded in the plot (to reach a T₁ steady state)
 - Data publicly available:
 - <http://www.fil.ion.ucl.ac.uk/spm/data/auditory>



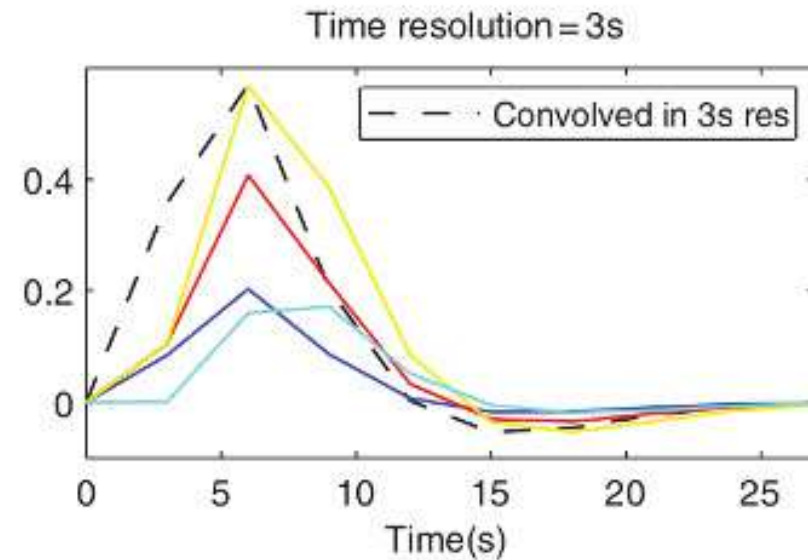
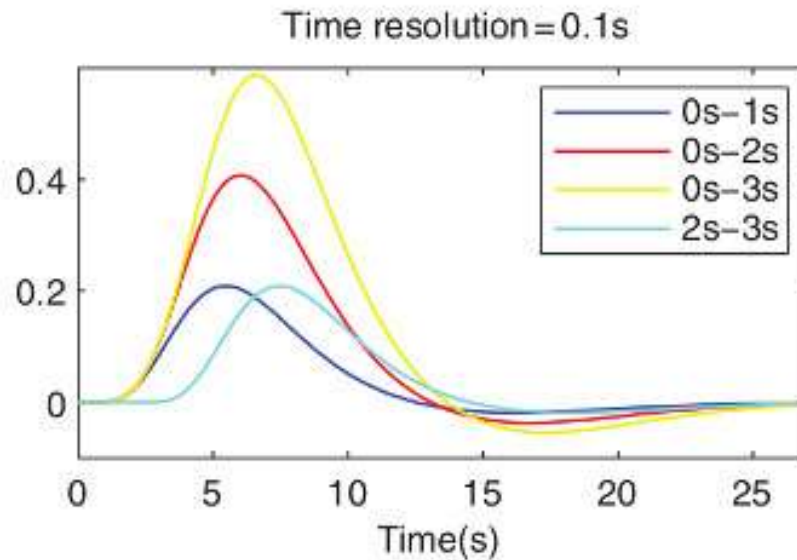
Expected BOLD Time Course

- As discussed in 7.3, the GLM fits the expected BOLD time course to the measured signal
 - Expected time course computed by convolving stimulus with HRF
- Double-check that temporal high pass filter (shaded gray) does not discard relevant part of the signal



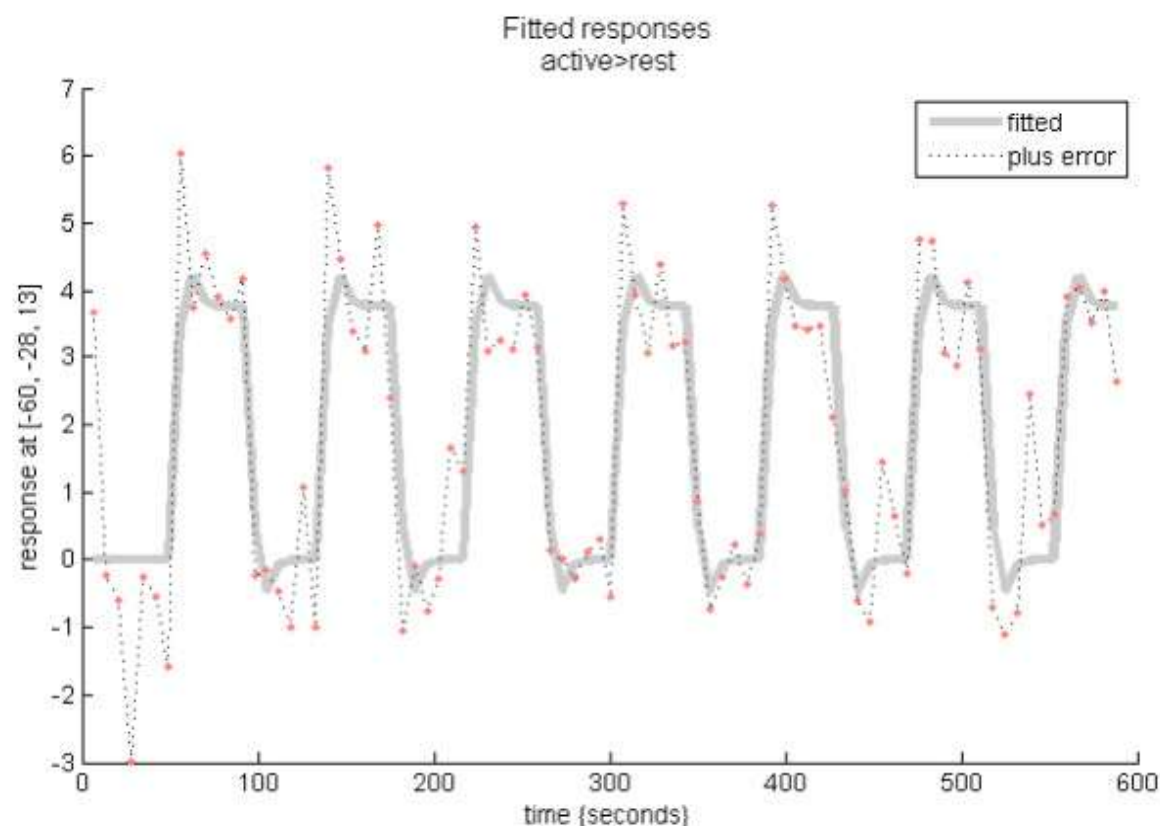
Perform Convolution with High Resolution!

- At low temporal resolution (e.g., $TR=3$ s), short stimuli are undersampled ☹️
- Convolution with HRF smoothes out stimulus 😊
- *But:* Have to perform convolution at high temporal resolution to avoid aliasing

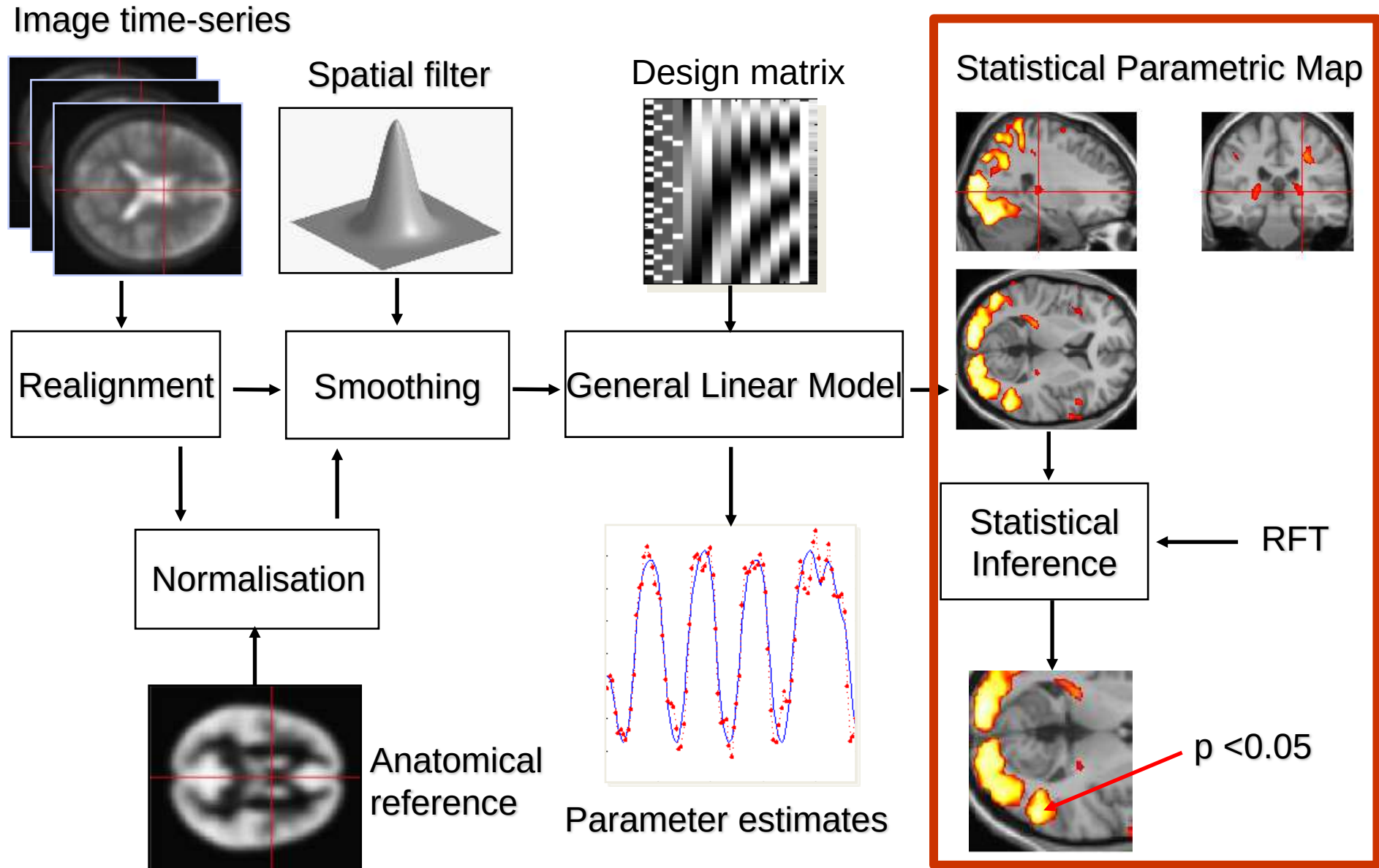


Inspecting Individual Time Series

- To double-check synchronization of stimulation and measurements, it can make sense to compare time series in individual voxels with the model fit



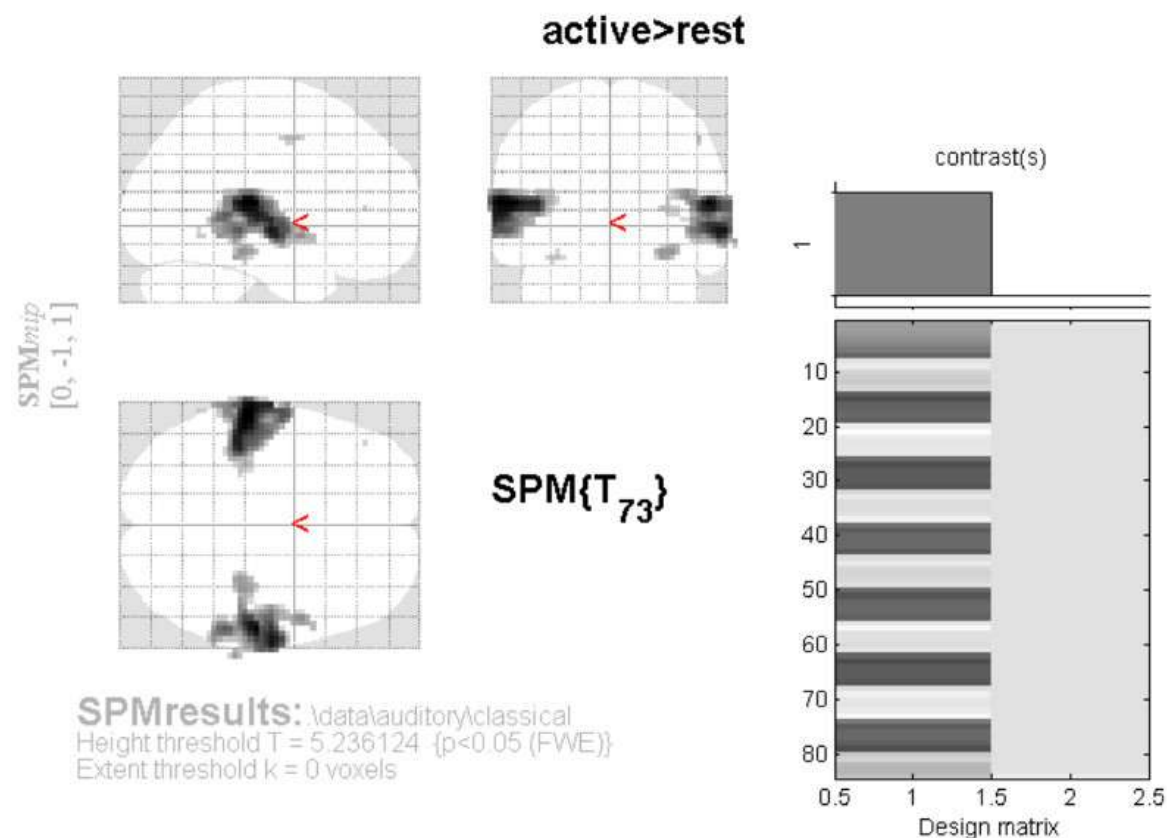
Overview: Task fMRI Processing Pipeline



Slide taken from SPM course

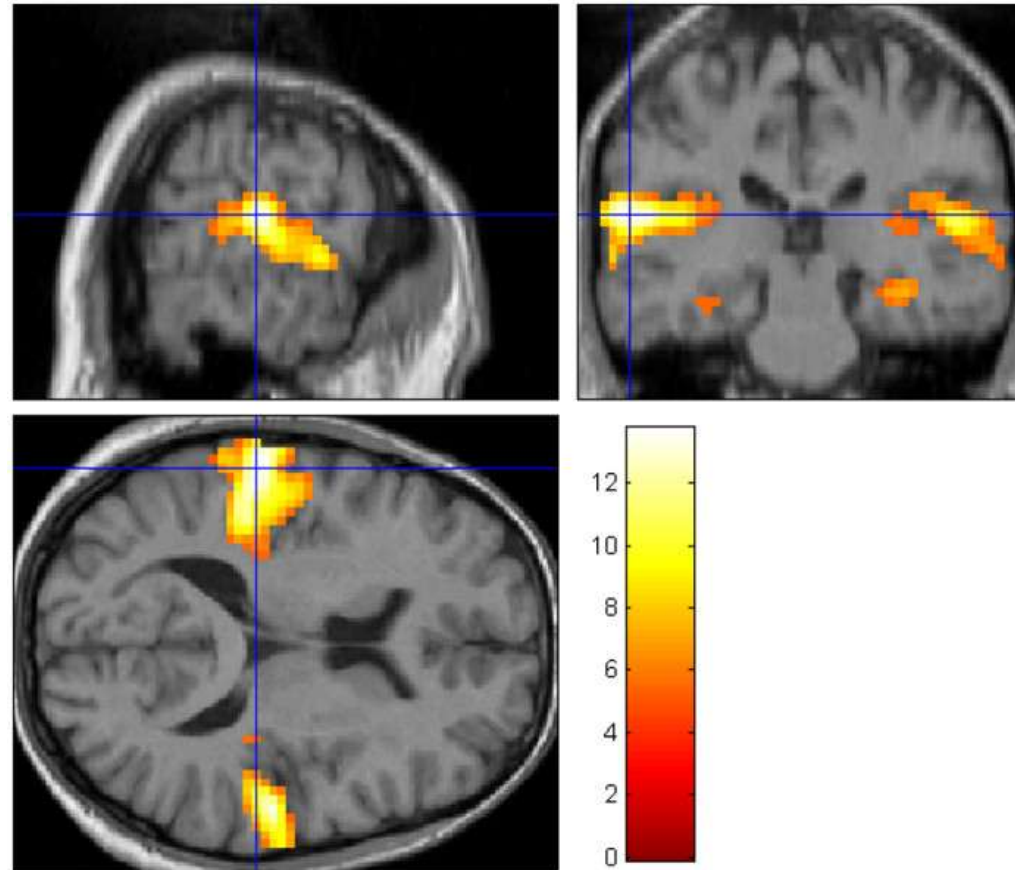
Maximum Intensity Projection

- Perform statistical testing per voxel
 - Random Field Theory most popular, permutation testing does not easily apply due to temporal correlations in the fMRI signal
- **Maximum Intensity Projection** is one way to visualize results
 - 2D map showing maximum across third dimension



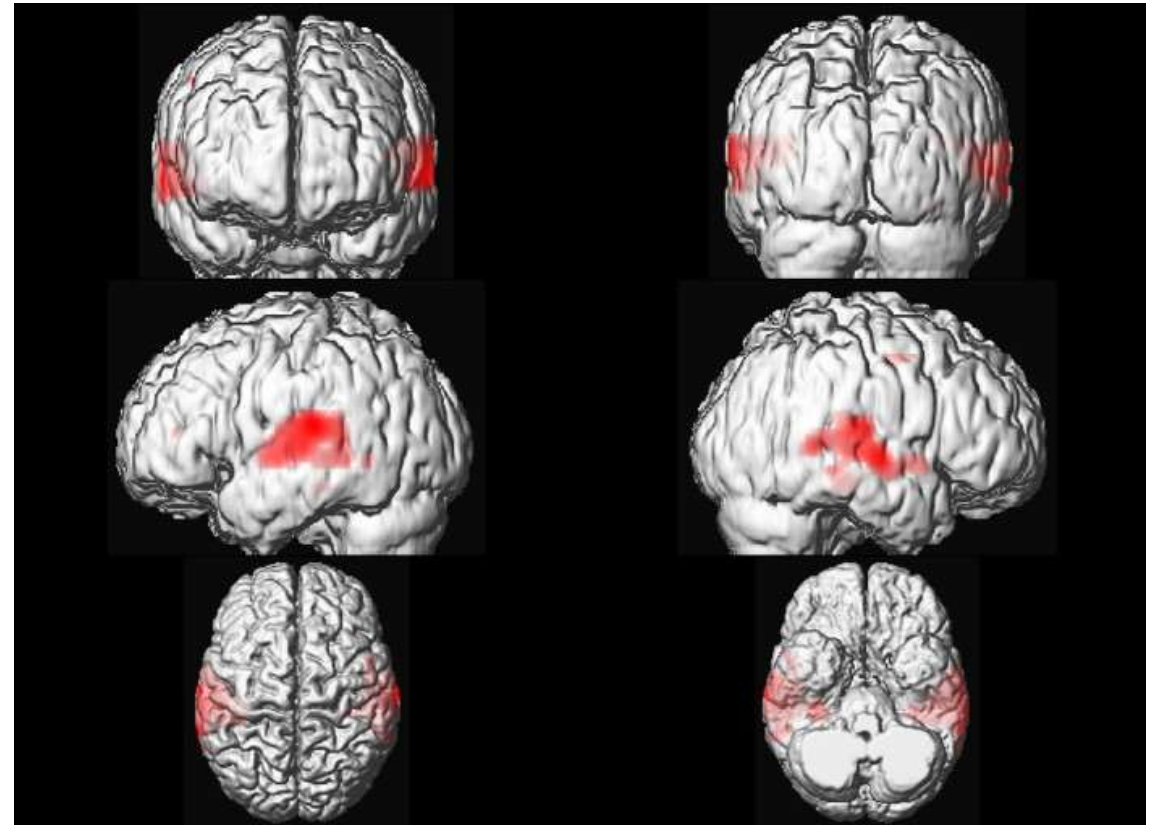
Overlay after Coregistration

- After coregistration of functional and high-resolution structural data, activations can be overlaid



Overlay after Normalization

- To achieve 3D visualization, activations can be overlaid on a geometric model of the cortical surface
 - **Individual surfaces** can be obtained with the methods mentioned at the end of Chapter 5
 - After coregistration to a template, some authors simply use **generic models** that ignore individual variability in brain anatomy

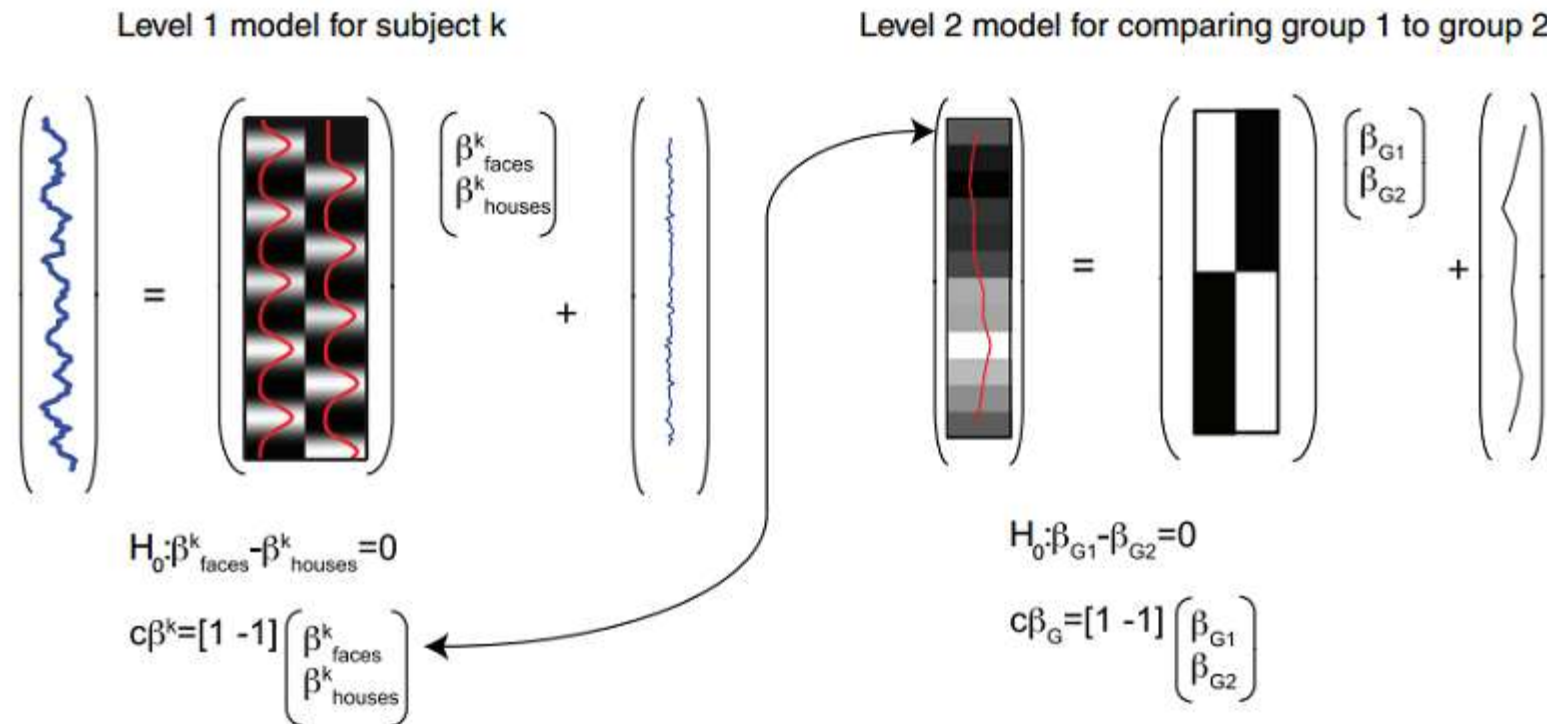


fMRI Group Analysis

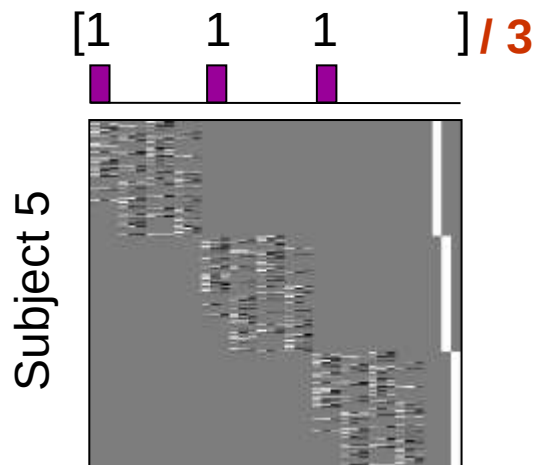
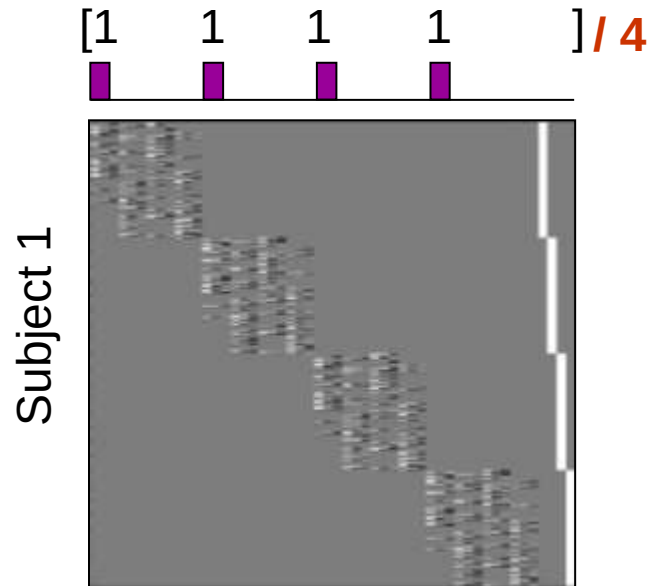
- Group analysis combines fMRI data **from several subjects** to answer questions such as
 - „Where are activations significant *at the population level?*“
 - „Do activation patterns differ between laymen and experts?“
- Simply normalizing and temporally concatenating all scans is called a „fixed effects analysis“ in the fMRI literature
 - It ignores the fact that the subjects themselves are **samples from a larger population**
 - Fails to distinguish variances and degrees of freedom w.r.t. number of fMRI time steps vs. number of subjects
 - Statistical results will **not generalize** to the wider population!

Two-Level Analysis

- „Random effects“ models are commonly estimated in two steps, each of them using a GLM:
 1. Estimate activation per-subject
 2. Test contrasts against zero or compare contrasts between groups



Scaling Issues



$$t = \frac{c^T \hat{\beta}}{\sqrt{\text{var}(c^T \hat{\beta})}} = \frac{c^T \hat{\beta}}{\sqrt{\hat{\sigma}^2 c^T (X^T X)^{-1} c}}$$

- ❑ The t -statistic does not depend on the scaling of the regressors.
- ❑ The t -statistic does not depend on the scaling of the contrast.
- ❑ Contrast $c^T \hat{\beta}$ depends on scaling.
- Be careful of the interpretation of the contrasts $c^T \hat{\beta}$ themselves (eg, for a second level analysis):

sum $\neq$ average

Model Unequal Within-Subject Variance?

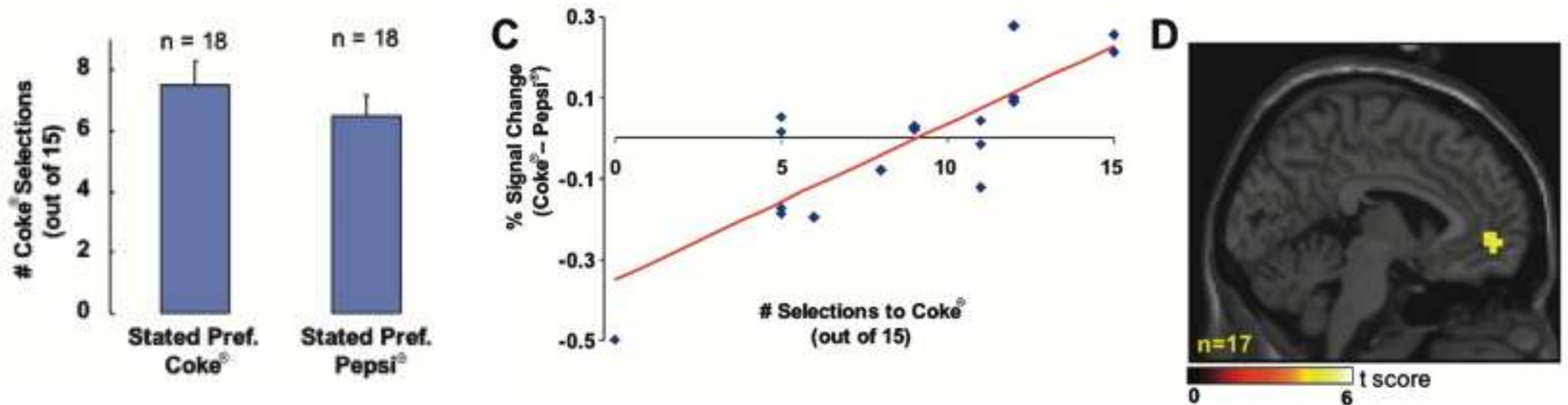
- A **weighted least squares fit** for the second-level GLM can account for variances that differ between subjects:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\Sigma}^{-1} \mathbf{y}$$

- Reduces the impact of highly uncertain subjects
- Some software packages **do not propagate** individual within-subject variances to the second level
 - *Justification*: Estimates of variances that have been obtained from few samples are themselves quite uncertain

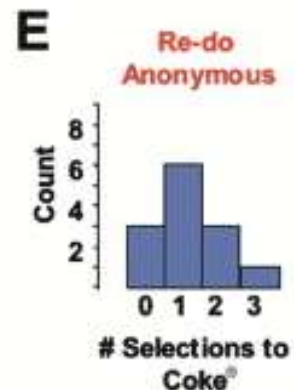
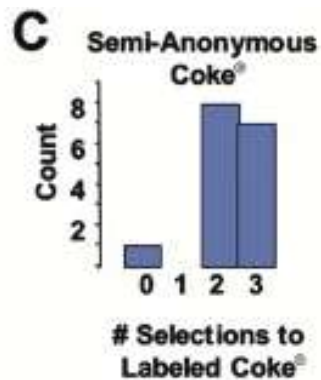
fMRI Headline: Coke vs. Pepsi

- **McClure et al. 2004** report results on behavioral preference and fMRI response to cola with and without knowing the brand
 - Stated preference of Coke vs. Pepsi unreliable in a blind taste test
 - Prefrontal activity correlated with behavioral preference

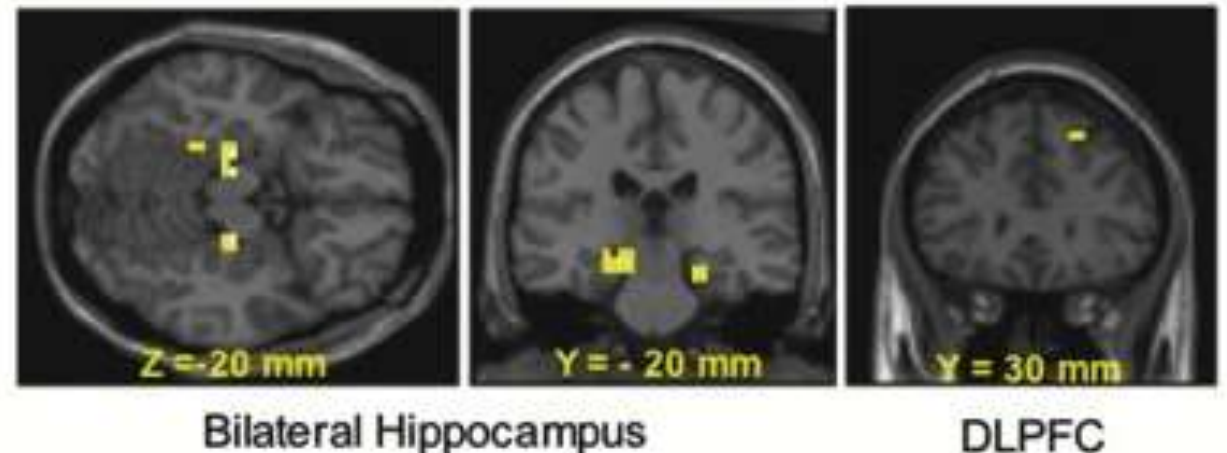


fMRI Headline: Coke vs. Pepsi

- Most subjects preferred labeled cup of coke over unlabeled cup, even if they could not reliably identify coke in a blind taste test
- Brand-cued vs. light-cued delivery of coke led to activations in several other brain regions, which are hypothesized to influence perception based on prior affective biases

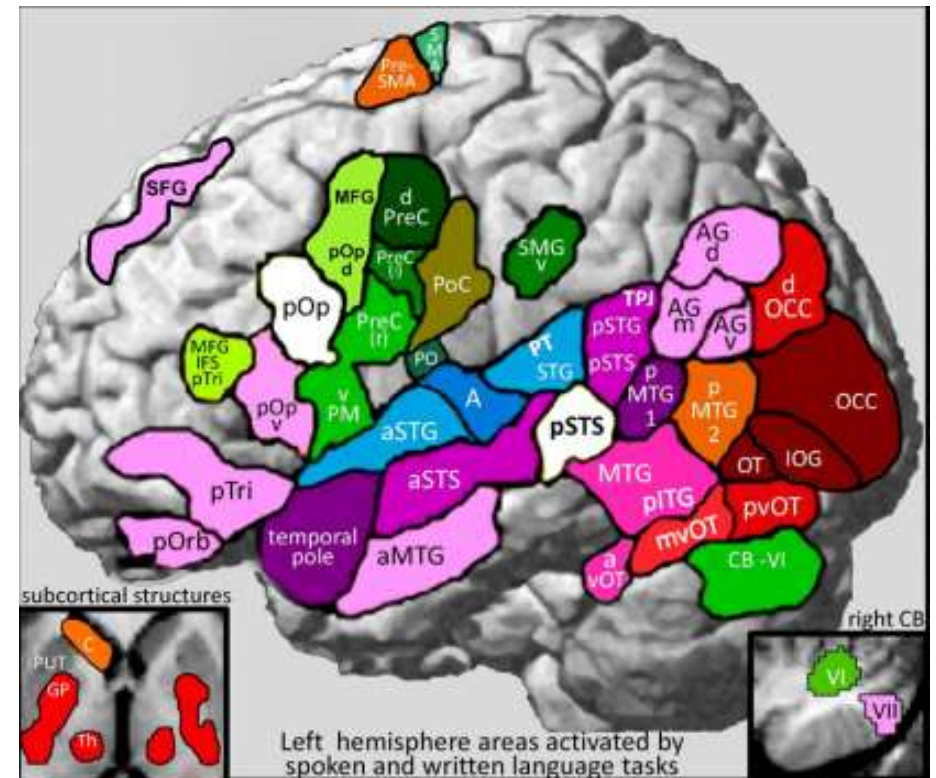


Coke® Delivery (Brand-cued - Light-cued)



fMRI Example: Speech

- **Price 2012** summarizes hundreds of studies that provide converging evidence on which brain regions are involved in hearing speech, producing speech, or reading
 - Reading meaningful vs. meaningless words, generating vs. repeating words, processing simple vs. complex grammar, etc.
 - Concludes that next big challenge is to understand how brain regions interact



Summary: fMRI Analysis Pipeline

- **Pre-processing** fMRI data involves
 - Realignment
 - Smoothing
 - High-pass Filtering
 - *Sometimes*: Normalization, Timing Correction
- **Per-voxel testing** based on GLM
 - Fit the expected BOLD response to the measured signal
- For **visual presentation**
 - slice through data or present MIP
 - overlay on top of anatomical image

7.5 Pitfalls in fMRI Analysis

fMRI Pitfall: The Dead Salmon

- [Bennett et al. 2009]: Neural correlates of interspecies perspective taking in the post-mortem Atlantic Salmon

METHODS

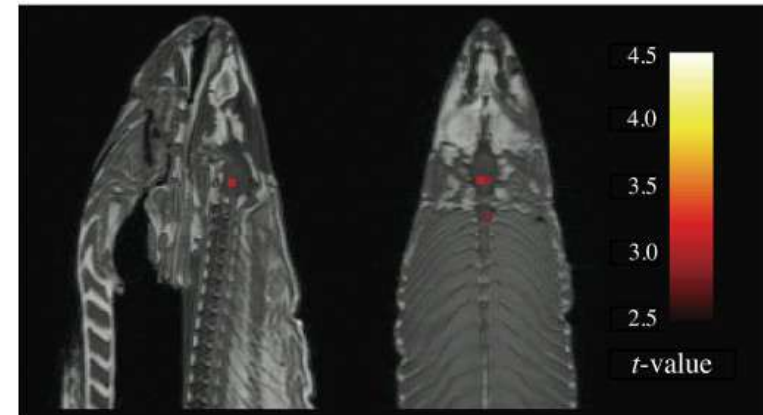
Subject. One mature Atlantic Salmon (*Salmo salar*) participated in the fMRI study. The salmon was approximately 18 inches long, weighed 3.8 lbs, and was not alive at the time of scanning.

Task. The task administered to the salmon involved completing an open-ended mentalizing task. The salmon was shown a series of photographs depicting human individuals in social situations with a specified emotional valence. The salmon was asked to determine what emotion the individual in the photo must have been experiencing.

Design. Stimuli were presented in a block design with each photo presented for 10 seconds followed by 12 seconds of rest. A total of 15 photos were displayed. Total scan time was 5.5 minutes.

Preprocessing. Image processing was completed using SPM2. Preprocessing steps for the functional imaging data included a 6-parameter rigid-body affine realignment of the fMRI timeseries, coregistration of the data to a T_1 -weighted anatomical image, and 8 mm full-width at half-maximum (FWHM) Gaussian smoothing.

Analysis. Voxelwise statistics on the salmon data were calculated through an ordinary least-squares estimation of the general linear model (GLM). Predictors of the hemodynamic response were modeled by a boxcar function convolved with a canonical hemodynamic response. A temporal high pass filter of 128 seconds was include to account for low frequency drift.



A t -contrast was used to test for regions with significant BOLD signal change during the photo condition compared to rest. The parameters for this comparison were $t(131) > 3.15$, $p(\text{uncorrected}) < 0.001$, 3 voxel extent threshold.

Quiz: What went wrong here?

Journalism Pitfall: The Dead Salmon



Scanning Dead Salmon in fMRI Machine Highlights Risk of Red Herrings

By Alexis Madrigal | September 18, 2009 | 5:37 pm | Categories: Brains and Behavior

Neuroskeptic Black and white and blue matter

Wednesday, 16 September 2009
fMRI Gets Slap in the Face with a Dead Fish



Are Brain Scan Findings Fishy?

Wed, Sep 30, 2009

Neuromarketing



A DEAD FISH HAS CAUSED A STINK OVER FALSE POSITIVES
IN FMRI STUDIES

Slide from Jody Culham

fMRI Pitfall: RFT-based Cluster Statistics

- In [Eklund et al., PNAS 2016], **cluster-based testing with Random Field Theory** at nominal FWE=5% produced false positives up to 70% of the time
 - **Reason:** Invalid assumption on autocorrelation
 - **Remedy:** Use voxel-level Random Field Theory or permutation tests where possible



„Cluster Failure“: How Bad Is It Really?

- **Original claim:**
 - “These results question the validity of some 40,000 fMRI studies and may have a large impact on the interpretation of neuroimaging results.”
- **Published correction:**
 - “These results question the validity of a number of fMRI studies and may have a large impact on the interpretation of weakly significant neuroimaging results.”

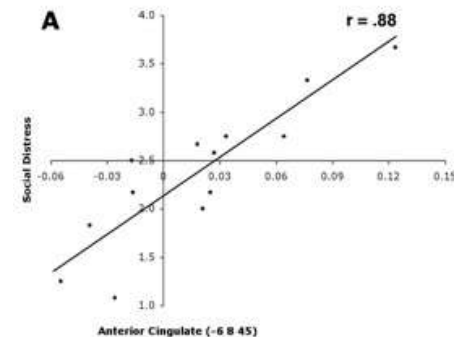
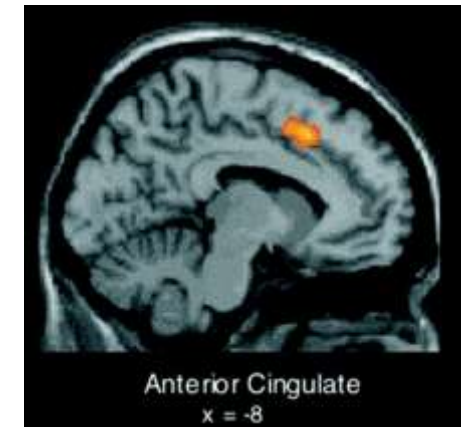
fMRI Pitfall: Correlations in “Active” Areas

- Example of a **widely cited paper** (Science 2003, >6500 citations in Dec 2025) that was later critiqued because of the way correlations were computed:

Does Rejection Hurt? An fMRI Study of Social Exclusion

Naomi I. Eisenberger,^{1*} Matthew D. Lieberman,¹
Kipling D. Williams²

A neuroimaging study examined the neural correlates of social exclusion and tested the hypothesis that the brain bases of social pain are similar to those of physical pain. Participants were scanned while playing a virtual ball-tossing game in which they were ultimately excluded. Paralleling results from physical pain studies, the anterior cingulate cortex (ACC) was more active during exclusion than during inclusion and correlated positively with self-reported distress. Right ventral prefrontal cortex (RVPFC) was active during exclusion and correlated negatively with self-reported distress. ACC changes mediated the RVPFC-distress correlation, suggesting that RVPFC regulates the distress of social exclusion by disrupting ACC activity.



Finding the Obvious



A priori probability of getting JQKA sequence = $(1/13)^4 = 1/28,561$

A posteriori probability of getting JQKA sequence = $1/1 = 100\%$

Non-independence error

- occurs when statistical tests performed are not independent from the means used to select the brain region
- The activations found in such studies are still non-inflated
 - **IF** they corrected for multiple comparisons

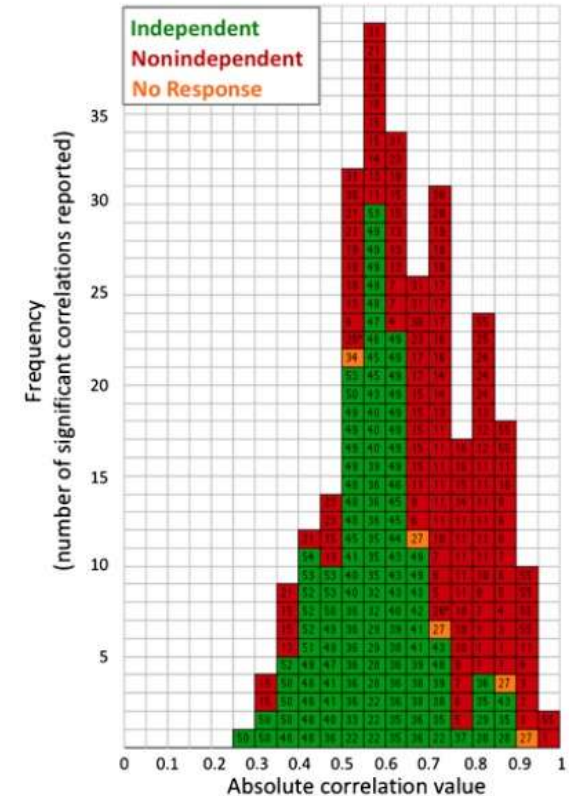
fMRI Pitfall: “Voodoo” Correlations

PERSPECTIVES ON PSYCHOLOGICAL SCIENCE

Voodoo Correlations in fMRI Studies of Emotion, Personality, and Social Cognition¹

Edward Vul,¹ Christine Harris,² Piotr Winkielman,² & Harold Pashler²

¹Massachusetts Institute of Technology and ²University of California, San Diego



"Notably, 53% of the surveyed studies selected voxels based on a correlation with the behavioral individual-differences measure and then used those same data to compute a correlation within that subset of voxels."

Summary: Pitfalls in fMRI Analysis

- Improperly correcting for **multiple comparisons** can substantially increase the probability of false positive findings
 - Salmon example: Use of ad-hoc thresholds
 - RFT-based cluster testing: Incorrect assumptions
- Computing correlations on regions of interest that have been determined in a manner that is **not independent** from the effect leads to inflated correlations

7.6 Resting-State fMRI and ICA

Introduction: Resting State fMRI

- In a **resting state fMRI** (rs-fMRI) experiment, functional images are taken over an extended period of time during which the subject relaxes without falling asleep and is told not to think about anything in particular
- **Motivations:**
 - Brain activity is intrinsic, even in the absence of external stimuli/tasks
 - Task performance has relatively little effect on energy consumption (Huang 2019: “often less than 5%”)
 - Has been found to provide information relevant to clinical and cognitive status
 - Reduced experimental effort
 - Extends the potential pool of subjects

Motivation: Localisationism vs. Globalism



David Ferrier
(1843-1928)

Localisationism

- Functions are localised in anatomic cortical regions
- Damage to circumscribed regions results in loss of specific functions

Functional Segregation

- Functions are carried out by specific areas/cells in the cortex that can be anatomically separated



Globalism

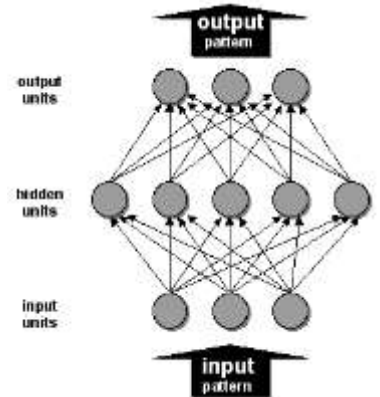
- The brain works as a whole
- Brain can (partially) recover from damage
- Some regions can be removed without apparent impairment



Friedrich Goltz
(1834-1902)

Connectionism

- Networks of simple connected units

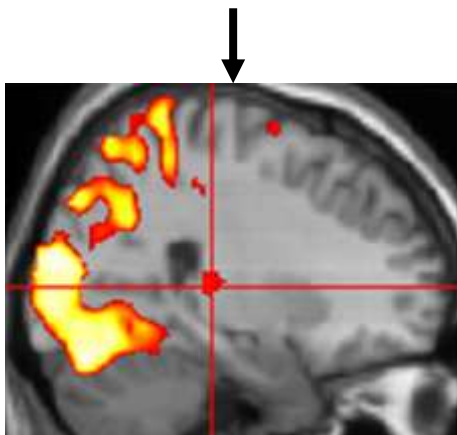


Functional Segregation *and* Integration

Functional Segregation

Specialised areas exist in the cortex

- Analyses of regionally specific effects
- Identifies regions specialized for a particular task



Standard SPM

Functional Integration

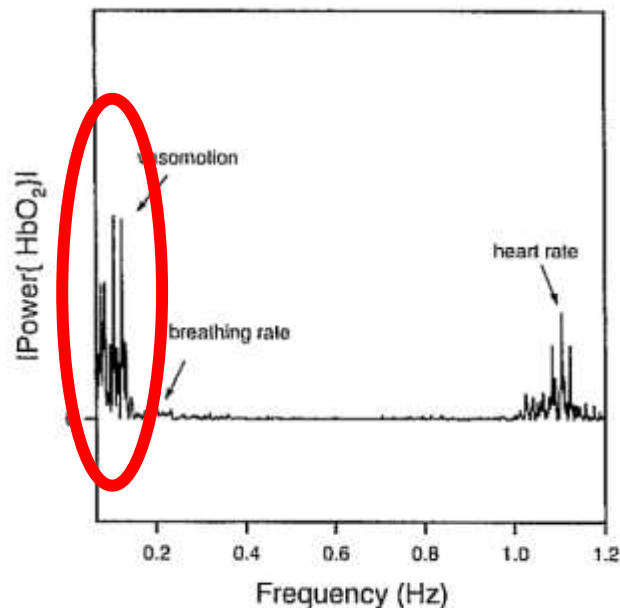
Networks of interactions among specialised areas

- Analysis of how different regions in a neuronal system interact (coupling)
- Determines how group differences or an experimental manipulation affect coupling between regions

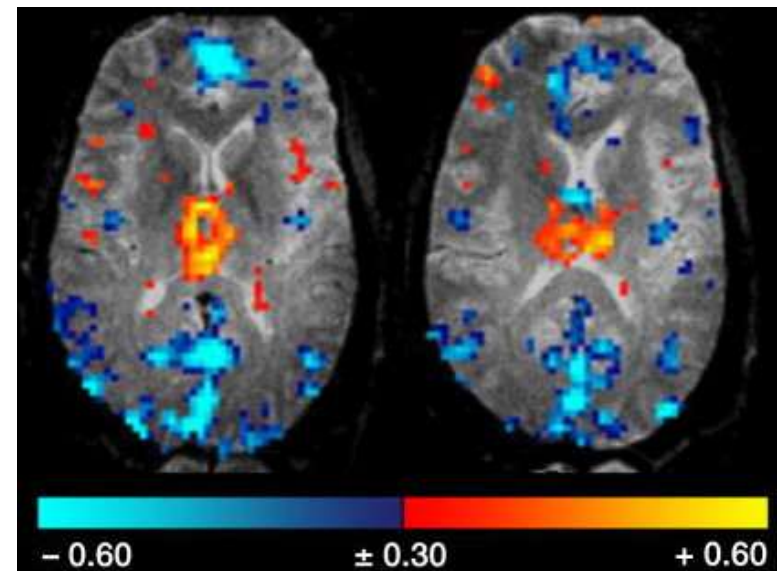
↓
**Analysis of structural /
functional connectivity**

Spontaneous BOLD Activity

- Inspecting the spectrum of rs-fMRI data reveals an interesting peak between 0.01-0.10 Hz
 - Below the frequency of heart beat / respiration
 - Not adequately explained even by aliasing
 - Assumption: Signal is partly due to oscillating neuronal activation
 - Simultaneous EEG recordings indicate correlations with resting state fMRI



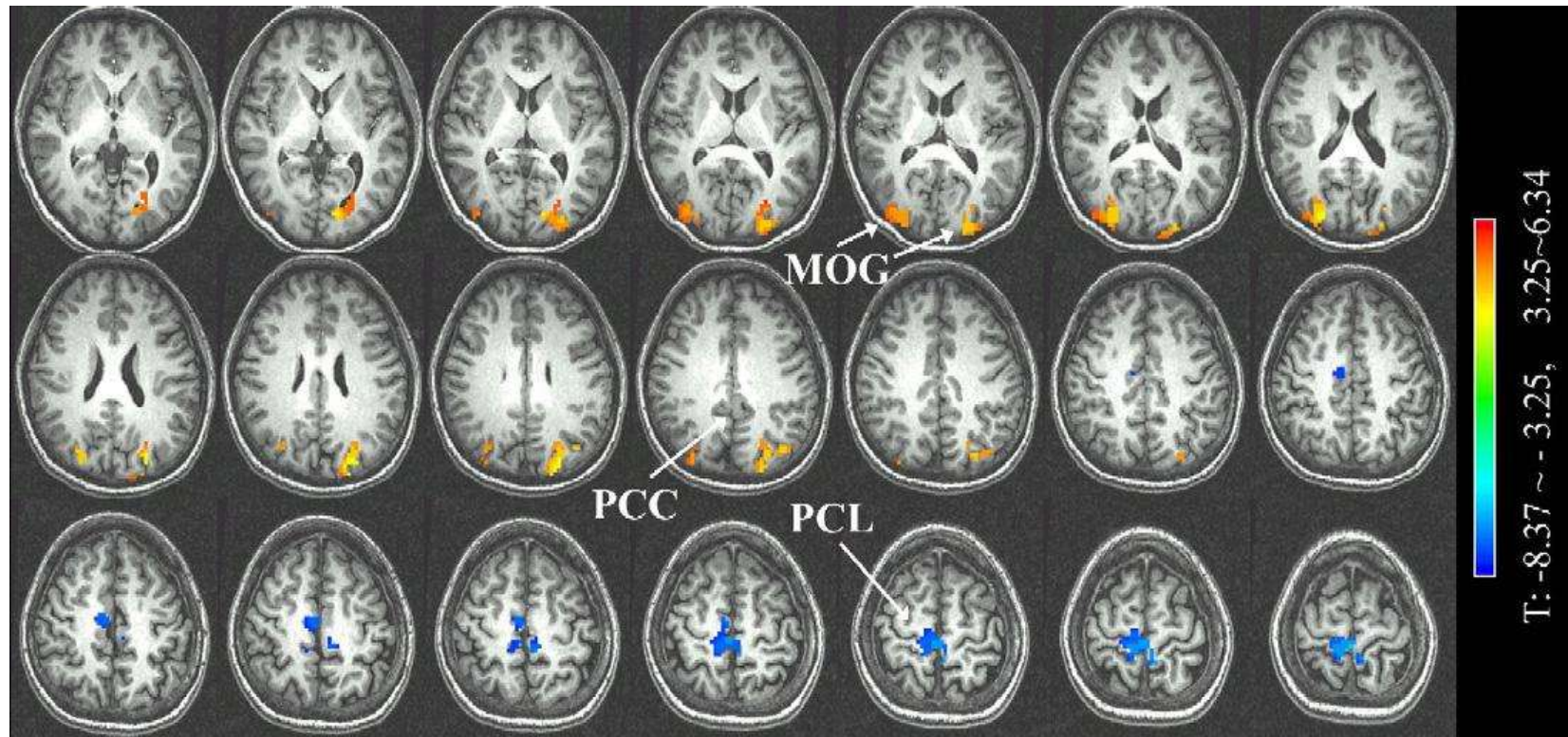
Elwell et al. 1999



Goldman et al. 2002

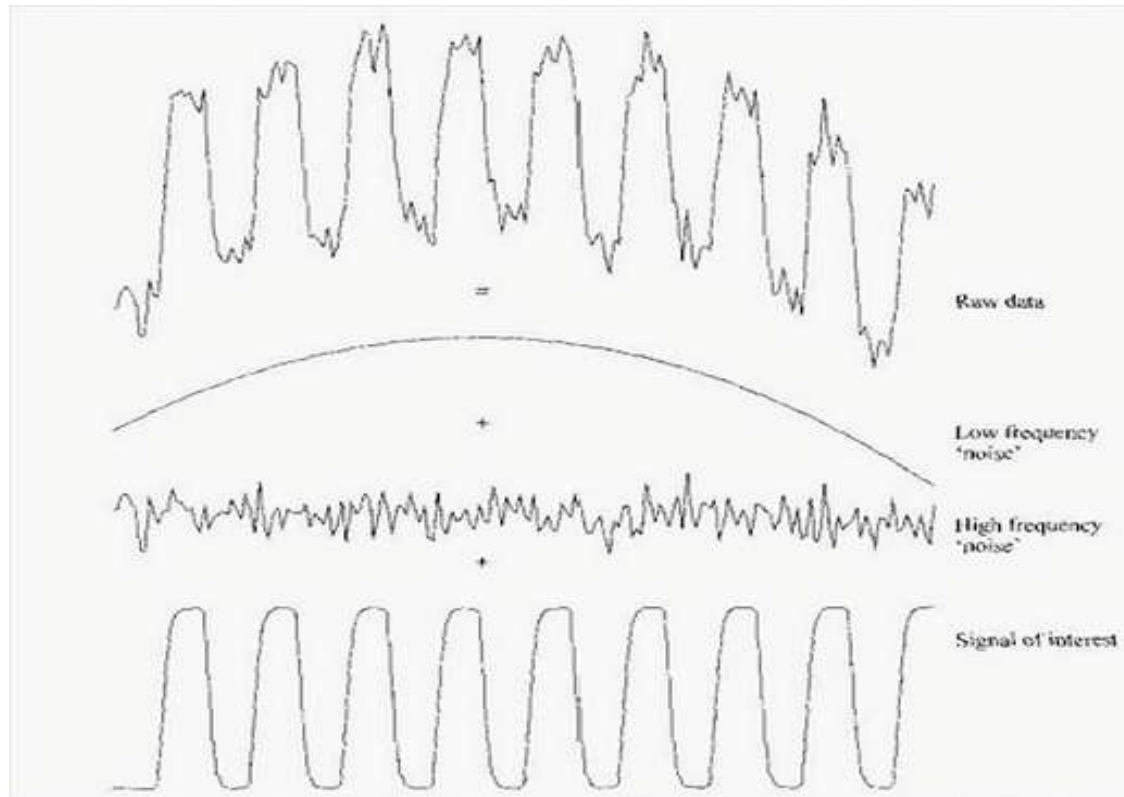
Amplitude of Low Frequency Fluctuation

- Yang et al. 2007 report that **amplitude of low frequency fluctuation** (ALFF; power in range 0.01-0.08 Hz) differs when eyes are open rather than closed



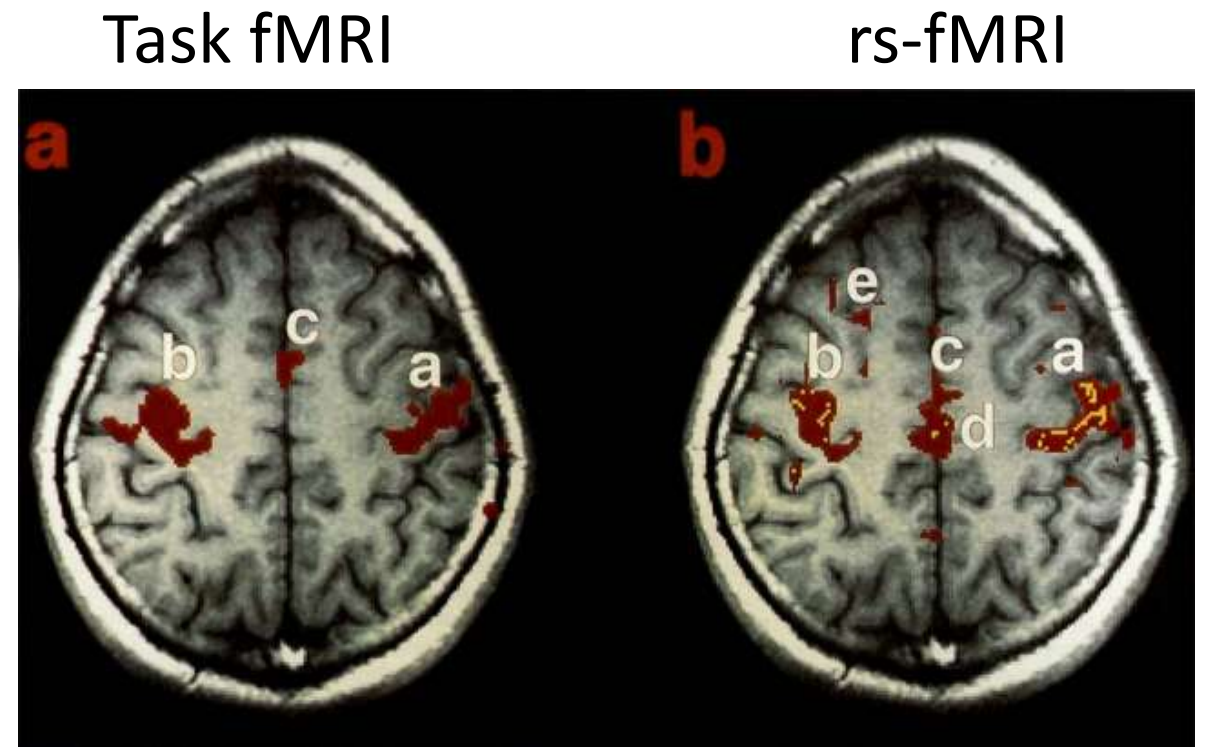
Signal Considered in rs-fMRI

- rs-fMRI undergoes the **same preprocessing** as regular fMRI plus bandpass filtering to around 0.01-0.1 Hz



Correlation to Seed

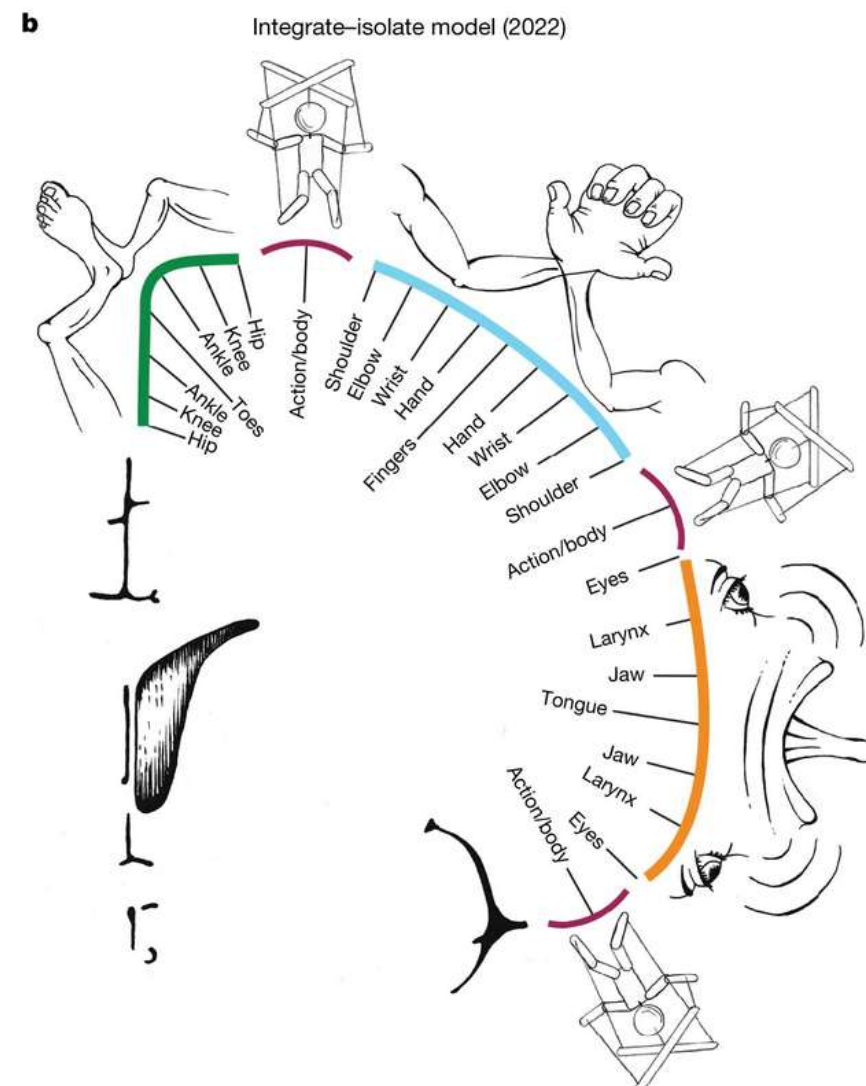
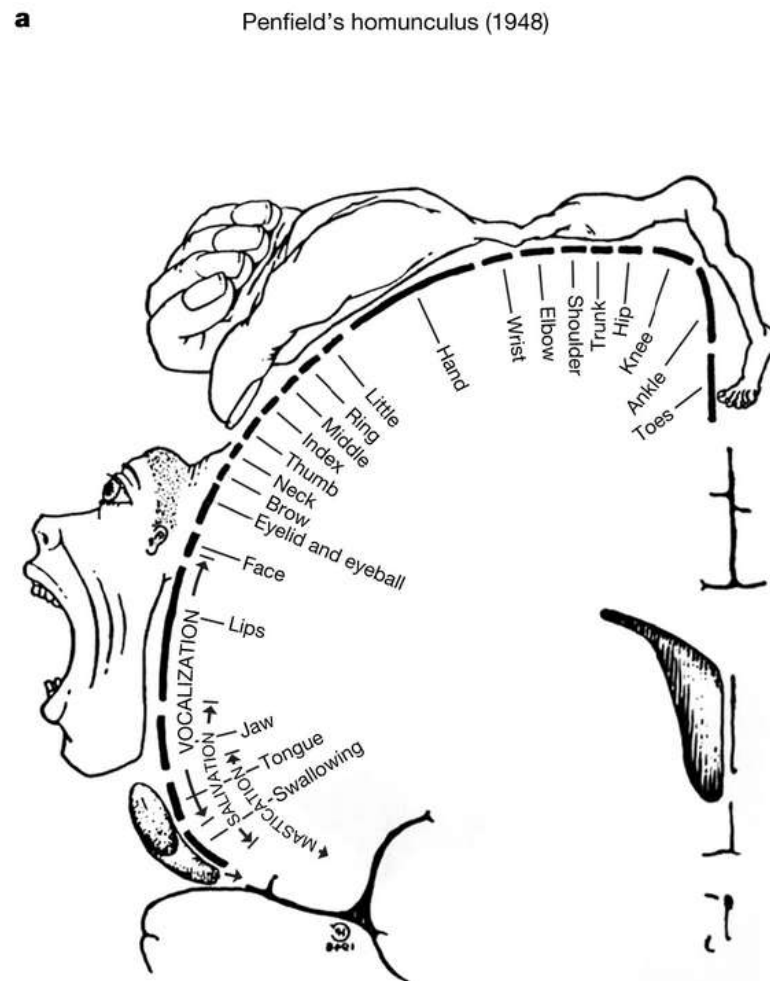
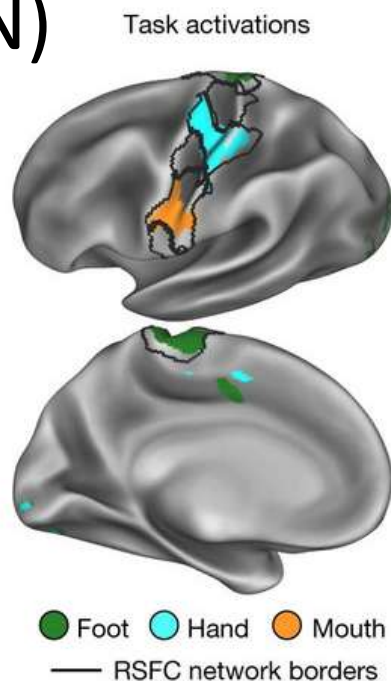
- Correlating time courses with a **given seed voxel** reveals patterns that are consistent with task fMRI, indicate „functional connectivity“
 - *Example*: bilateral finger tapping task vs. rs-fMRI with seed in left motor cortex
 - red = positive correlation
 - yellow = negative correlation



[Biswal et al. 1995]

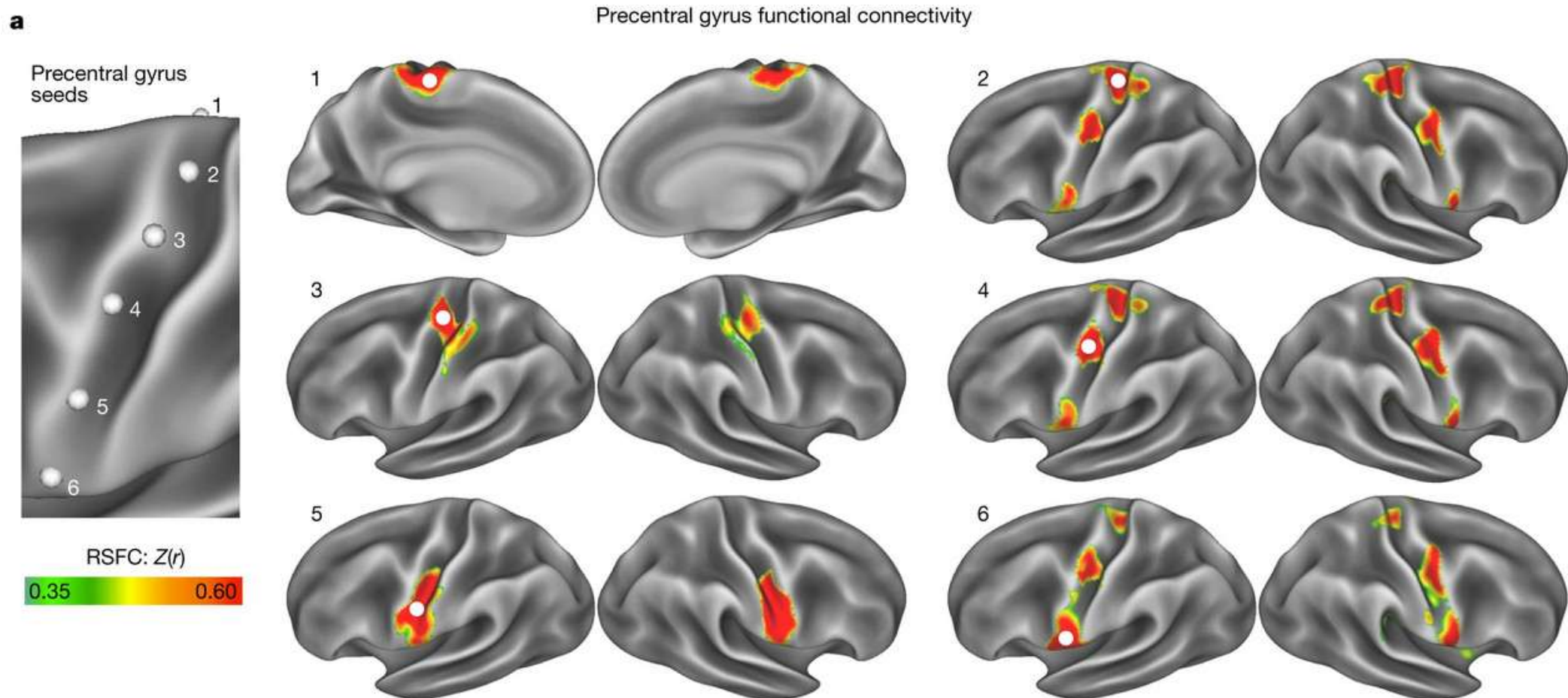
Reminder: The Interrupted Homunculus

- [Gordon et al. 2023] propose that the homunculus is interrupted by a somato-cognitive action network (SCAN)



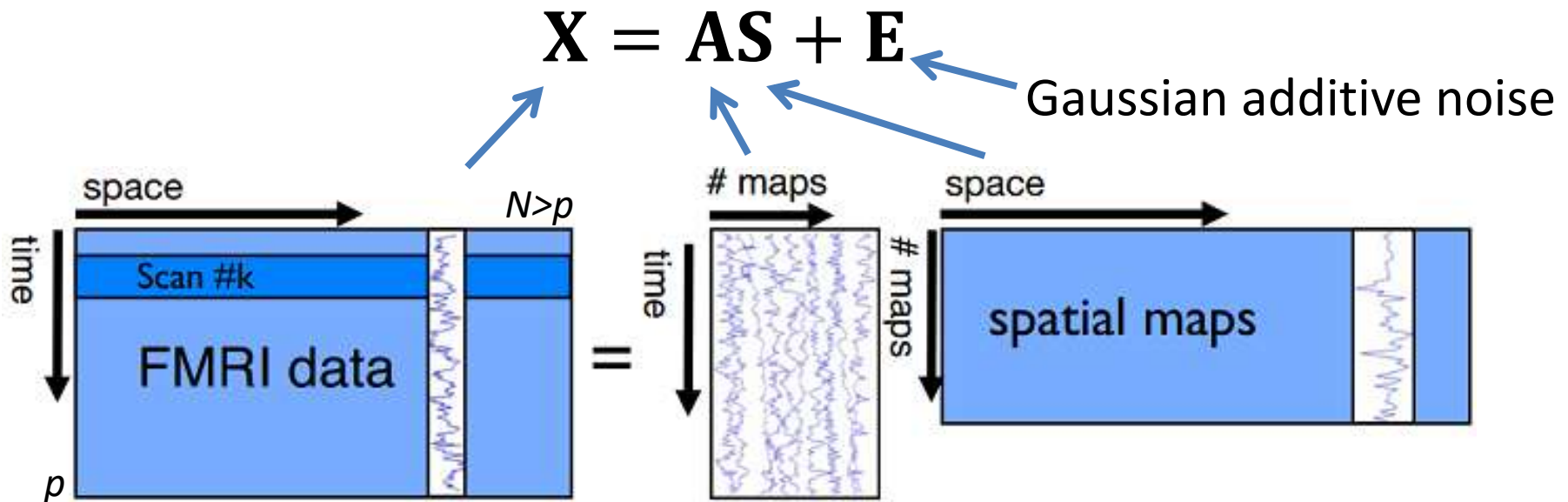
Example: Seed-based Analysis

- Seed-based rs-fMRI analysis indicates functional connectivity within the foot/hand/mouth regions and the proposed SCAN



Independent Component Analysis

- **Independent Component Analysis (ICA)** decomposes the observed spatiotemporal MR signal into *spatially independent components* that are activated according to corresponding time courses



Quiz: How does this relate to GLM analysis of task fMRI?

ICA vs GLM

In the context of fMRI analysis,

- **GLM** is used for **confirmatory** analysis
 - Spatially map the evidence against the null hypothesis that the observed fMRI signal is locally uncorrelated with a (known) expected BOLD timecourse, after accounting for (known) confounders
- **ICA** is used for **exploratory** analysis
 - Jointly estimate (unknown) timecourses and (unknown) spatial maps that together explain the observed signal
 - In resting-state data, we have no prior hypothesis about expected timecourses
 - Results require interpretation, e.g., labeling as “signal” or “artifact”
 - *Next*: Clarify assumptions that make such a decomposition meaningful and an algorithm for computing it

Related Problem: Blind Source Separation

“Cocktail party problem”: Given a mixture of multiple audio signals, isolate the individual sources

- *Example*: Two people are talking simultaneously (independently from each other), and are being recorded by two microphones at different locations. Given the two mixtures, can we recover the two sources?

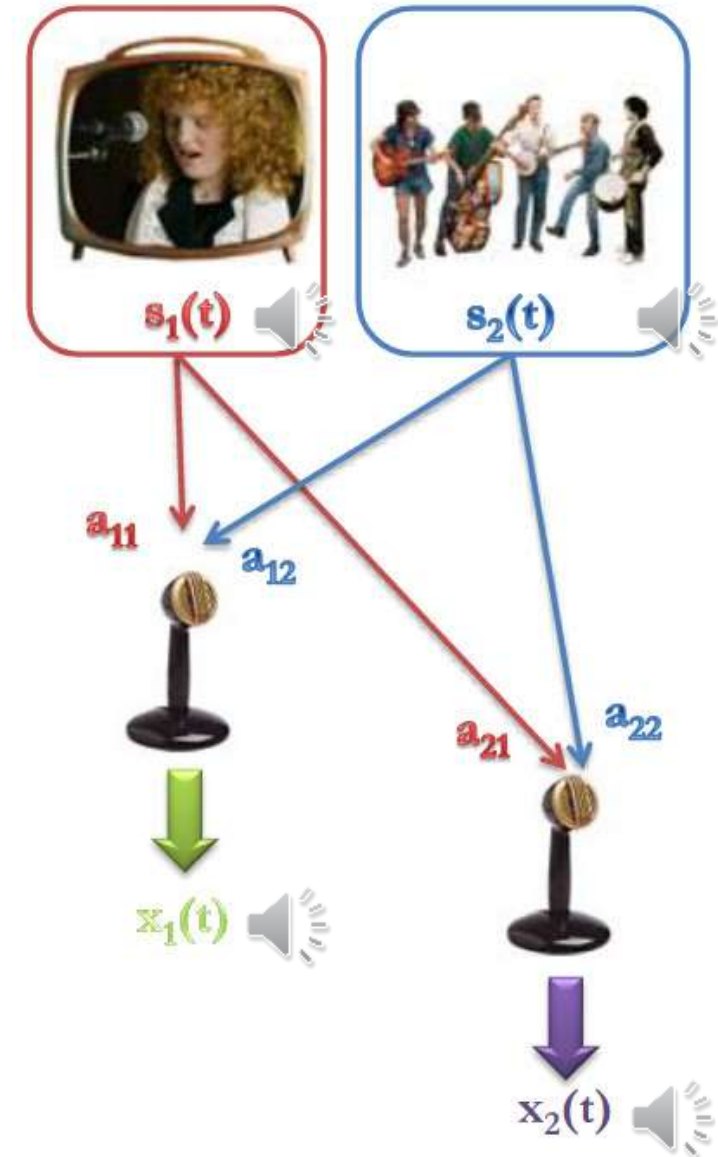


Illustration: http://www.sci.utah.edu/~shireen/pdfs/tutorials/Elhabian_ICA09.pdf

Audio: <http://research.ics.aalto.fi/ica/>

Setup of Traditional ICA

- Traditional ICA formalizes the blind source separation problem as matrix factorization $\mathbf{X} = \mathbf{A}\mathbf{S}$
 - Rows of $\mathbf{X}$ contain the recorded time signals
 - Means are removed to simplify computations later on
 - Rows of $\mathbf{S}$ contain the (unknown) source signals
 - Coefficients a_{ij} of square mixing matrix $\mathbf{A}$ reflect the sensitivity of microphone i for source j
 - Mixing is assumed to be linear and instantaneous
 - $\mathbf{A}$ is required to have full rank
- Estimating $\mathbf{A}$ will provide an unmixing matrix $\mathbf{W} = \mathbf{A}^{-1}$ that allows us to recover the sources as $\mathbf{S} = \mathbf{W}\mathbf{X}$

Assumptions and Ambiguities in Traditional ICA

- Need further constraints to find a meaningful solution to $\mathbf{X} = \mathbf{AS}$
- Key assumptions in ICA:
 1. Sources are statistically independent
 2. Sources are zero-mean non-Gaussian random variables

Intuition: Averaging independent non-Gaussian random variables tends to make the result „more Gaussian“ (cf. central limit theorem)
- Remaining ambiguities:
 - Applying the same permutation to columns of $\mathbf{A}$ and rows of $\mathbf{S}$ cancels out
 - *Consequence:* Ordering of sources is arbitrary and meaningless
 - Scaling columns of $\mathbf{A}$ by non-zero f and rows of $\mathbf{S}$ by $1/f$ cancels out
 - *Consequences:* Assume all sources have unit variance to fix scaling. Signs ($f = \pm 1$) are arbitrary and meaningless.

Reminder: Statistical Independence

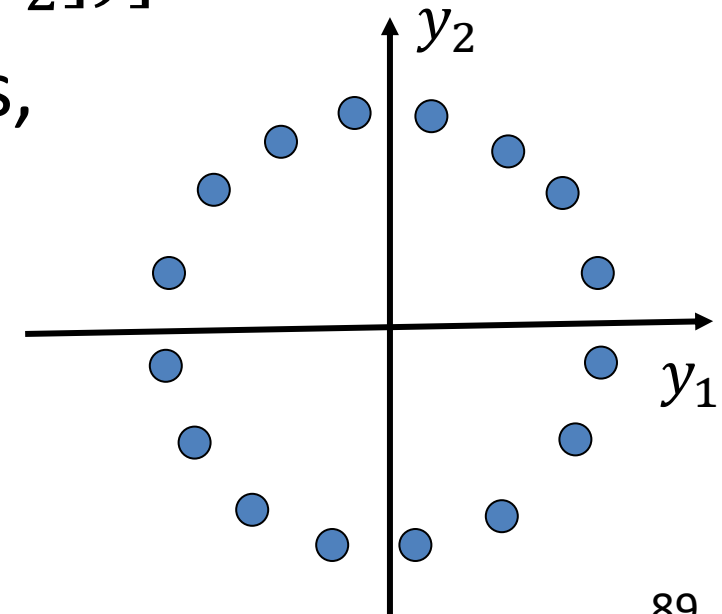
- *Intuition:* Two random variables are **independent** if information about the value of one does not provide any information on the value of the other
- Let $p(y_1, y_2)$ denote the joint probability density function (PDF) of two scalar random variables y_1 and y_2
- The marginal PDFs can be computed as
$$p_1(y_1) = \int p(y_1, y_2) dy_2 \text{ and } p_2(y_2) = \int p(y_1, y_2) dy_1$$
- y_1 and y_2 are independent if and only if the joint PDF factorizes into the marginals: $p(y_1, y_2) = p_1(y_1)p_2(y_2)$

Reminder: Uncorrelatedness

- *Intuition:* Two random variables are **uncorrelated** if there is no linear relationship between them
- y_1 and y_2 are uncorrelated if and only if they have zero covariance

$$\text{cov}(y_1, y_2) = E[(y_1 - E[y_1])(y_2 - E[y_2])] = 0$$

- *Note:* Independence implies uncorrelatedness, but not vice versa (see example)



Solving Traditional ICA

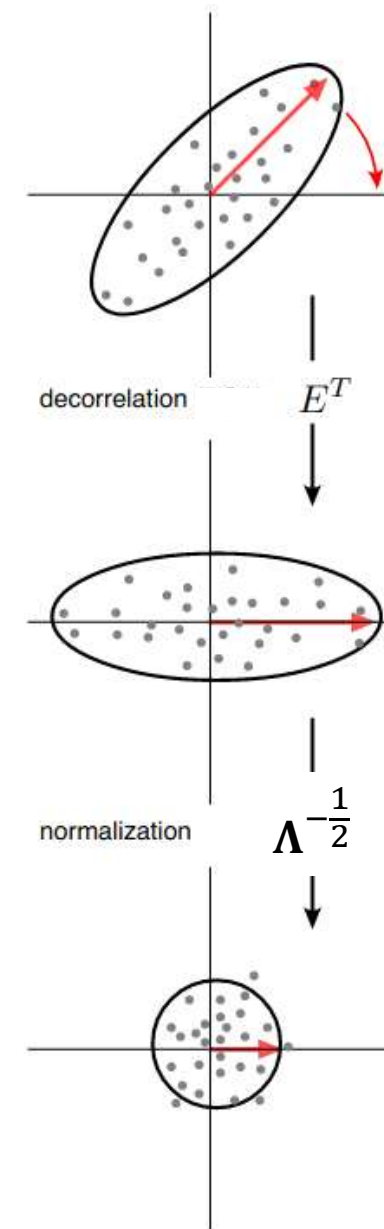
- Under the traditional ICA model, zero-mean observations $\mathbf{x}$ arise as a linear combination of zero-mean sources $\mathbf{s}$:

$$\mathbf{x} = \mathbf{A}\mathbf{s}$$

- Goal*: Find $\mathbf{A}$ via Singular Value Decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$
- Pre-Process**: “Whiten” / “sphere” the observations
 - Remove correlations, normalize variance in all dimensions
 - Given our assumptions on $\mathbf{s}$, $\text{Cov}(\mathbf{s}) = E[\mathbf{s}\mathbf{s}^T] = \mathbf{I}$
 - Therefore: $\text{Cov}(\mathbf{x}) = E[\mathbf{x}\mathbf{x}^T] = E[\mathbf{A}\mathbf{s}\mathbf{s}^T\mathbf{A}^T] = \mathbf{A}\mathbf{A}^T = \mathbf{U}\mathbf{\Sigma}^2\mathbf{U}^T$
 - Whitening is achieved via a spectral decomposition of the sample covariance

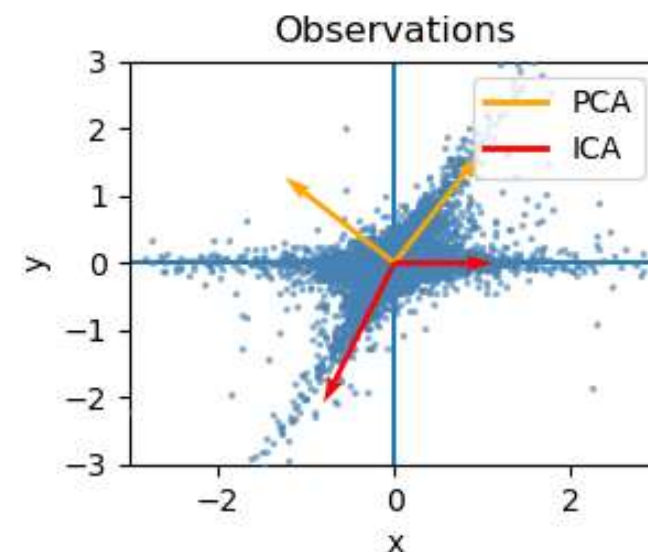
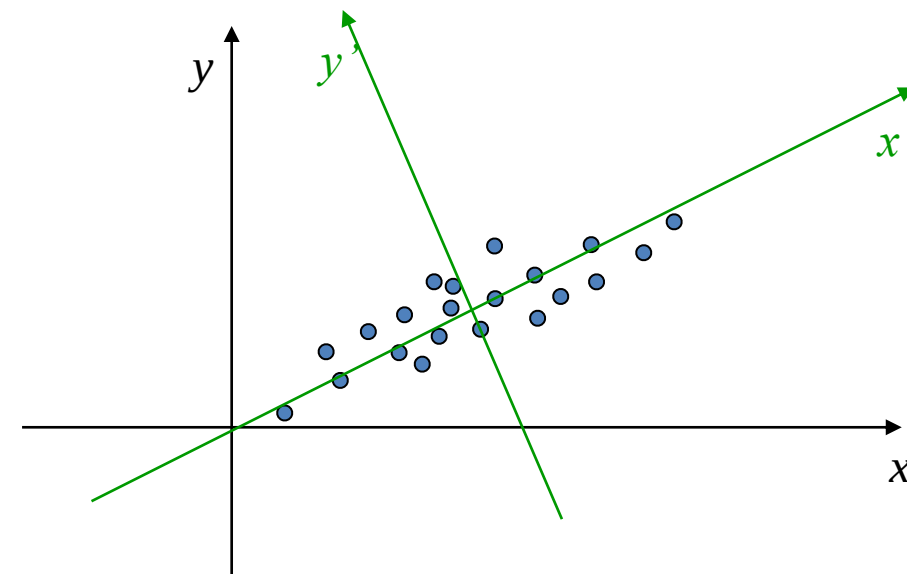
$$\frac{1}{n}\mathbf{X}\mathbf{X}^T = \mathbf{E}\mathbf{\Lambda}\mathbf{E}^T \Rightarrow \mathbf{U} = \mathbf{E}, \mathbf{\Sigma} = \mathbf{\Lambda}^{\frac{1}{2}}$$

Quiz: Can anyone think of a similar technique?



ICA vs. PCA

- Principal Component Analysis (PCA) finds an orthogonal basis that
 - Maximizes the (cumulative) variance along the first axes
 - Removes correlations
- Independent Component Analysis (ICA) decomposes data into
 - independent non-Gaussian sources
 - along potentially non-orthogonal directions



Optimizing for Non-Gaussian Sources

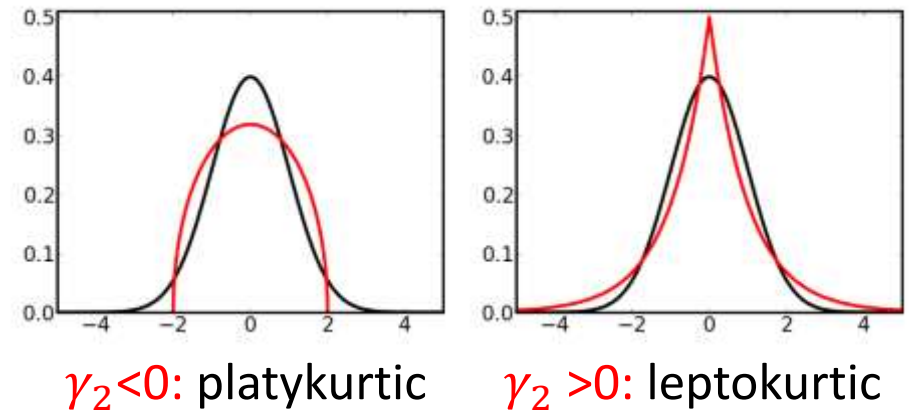
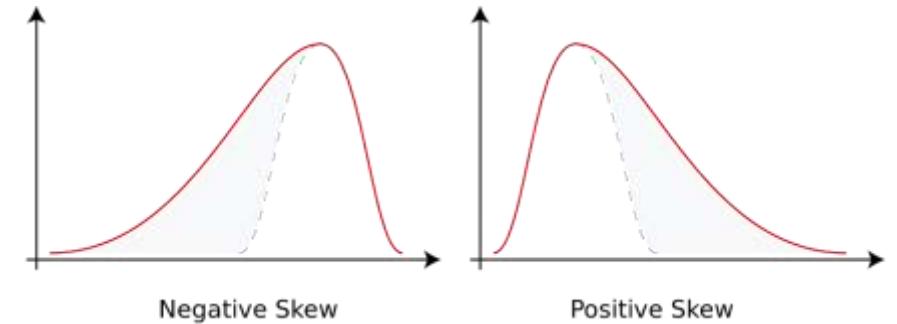
- **So far:** Found $\mathbf{U}$ and $\mathbf{\Sigma}$ for $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ or $\mathbf{W} = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^T$
- **Remaining task:** Find $\mathbf{V}$ that makes the sources independent
 - Has to be based on higher-order dependencies, since we already normalized second-order covariance
- **Approach:**
 - Continue working with „sphered“ data
$$\tilde{\mathbf{x}} = \mathbf{\Lambda}^{-\frac{1}{2}}\mathbf{E}^T\mathbf{x}$$
 - Optimize $\mathbf{V}$ so that sources $\hat{\mathbf{s}} = \mathbf{V}\tilde{\mathbf{x}}$ become as **non-Gaussian** as possible

Quantifying Non-Gaussianity

Popular measures of non-Gaussianity include

- Skewness $\tilde{\mu}_3 = E[\tilde{\mathbf{x}}^3]$
- Excess Kurtosis $\gamma_2 = E[\tilde{\mathbf{x}}^4] - 3$

Unfortunately, computing them from limited data is unstable because large exponents give outliers a high weight



Note: Equations above are simplified based on the facts that $\tilde{\mathbf{x}}$ has zero mean and unit variance

Main Idea of FastICA

- FastICA optimizes one row $\mathbf{v}_i$ of $\mathbf{V}$ as a unit vector that maximizes non-Gaussianity in terms of a contrast function

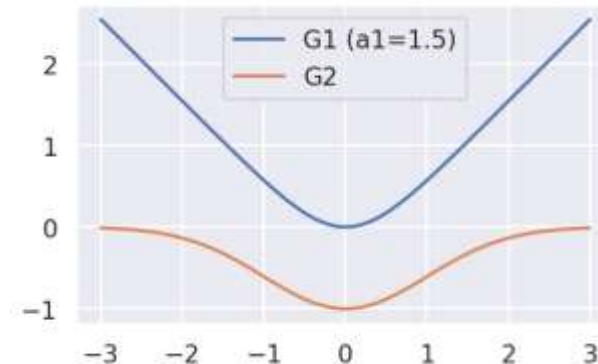
$$J(\mathbf{v}_i^T \tilde{\mathbf{X}}) = \sum_i k_i (\mathbb{E}[G_i(\mathbf{v}_i^T \tilde{\mathbf{X}})] - \mathbb{E}[G_i(\nu)])^2$$

- ν is a standard Gaussian variable
- G_i are nonquadratic, but not too quickly growing functions, e.g.,

- $G_1(u) = \frac{1}{a_1} \log \cosh a_1 u$ with $1 \leq a_1 \leq 2$

- $G_2(u) = -e^{-\frac{u^2}{2}}$

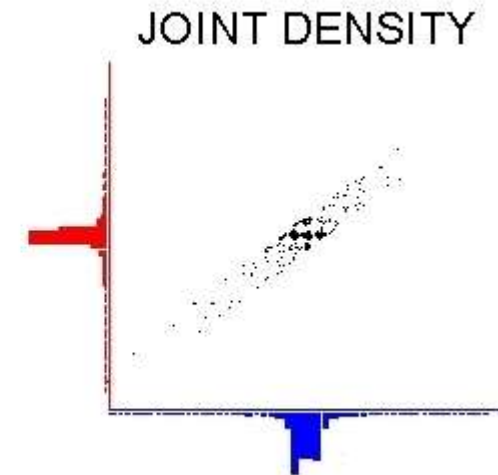
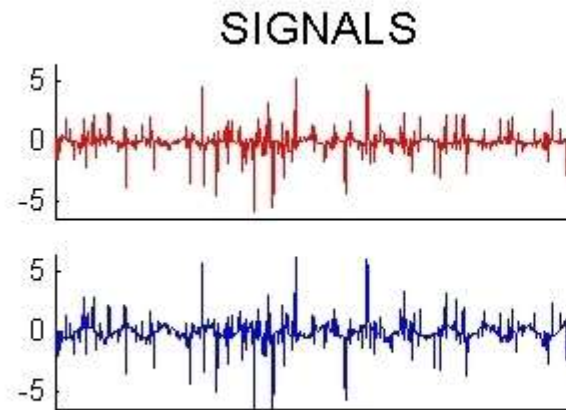
- k_i are positive constants



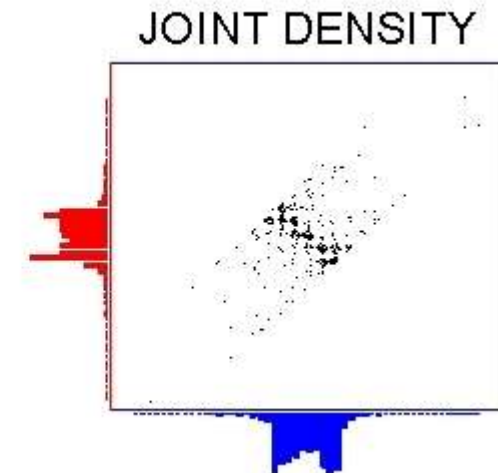
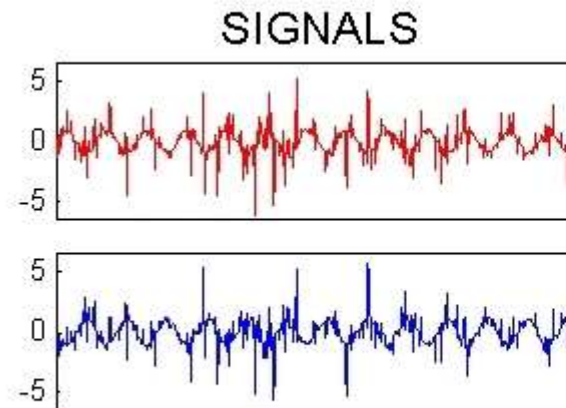
- Set of all $\mathbf{v}_i$ is optimized iteratively, starting from random initializations, preserving orthonormality via Gram-Schmidt

Example: FastICA

1. Initial signals and their joint and marginal densities

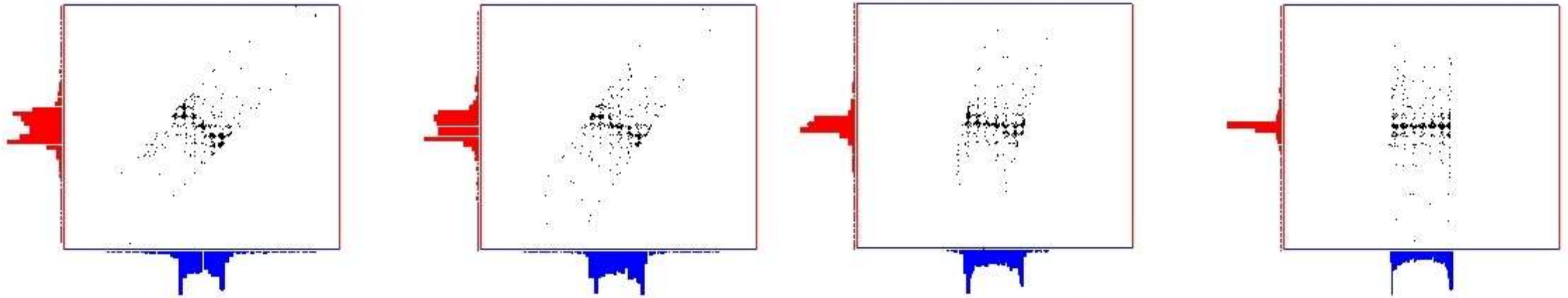


2. After initial whitening

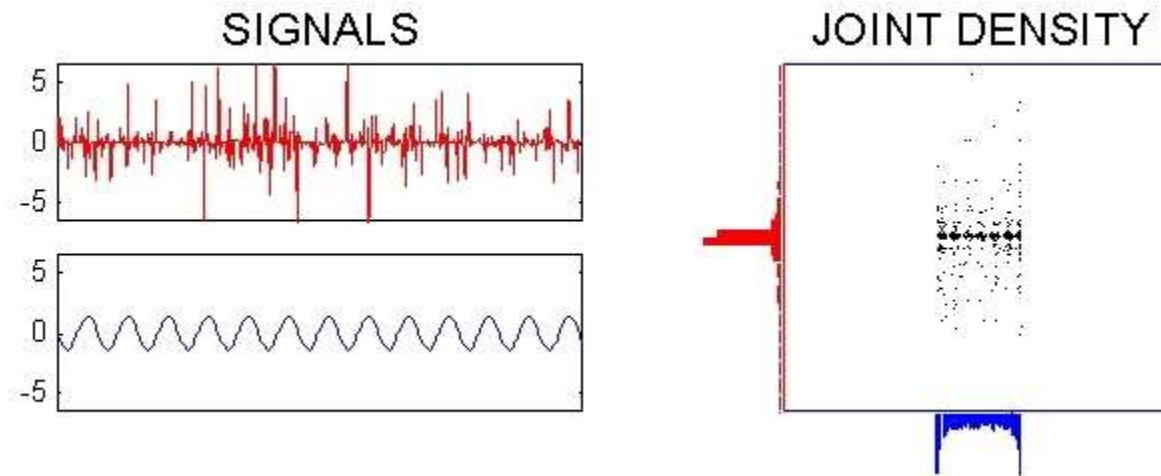


Example Continued: FastICA

3. Four iterations of optimizing $\mathbf{V}$ to achieve non-Gaussianity

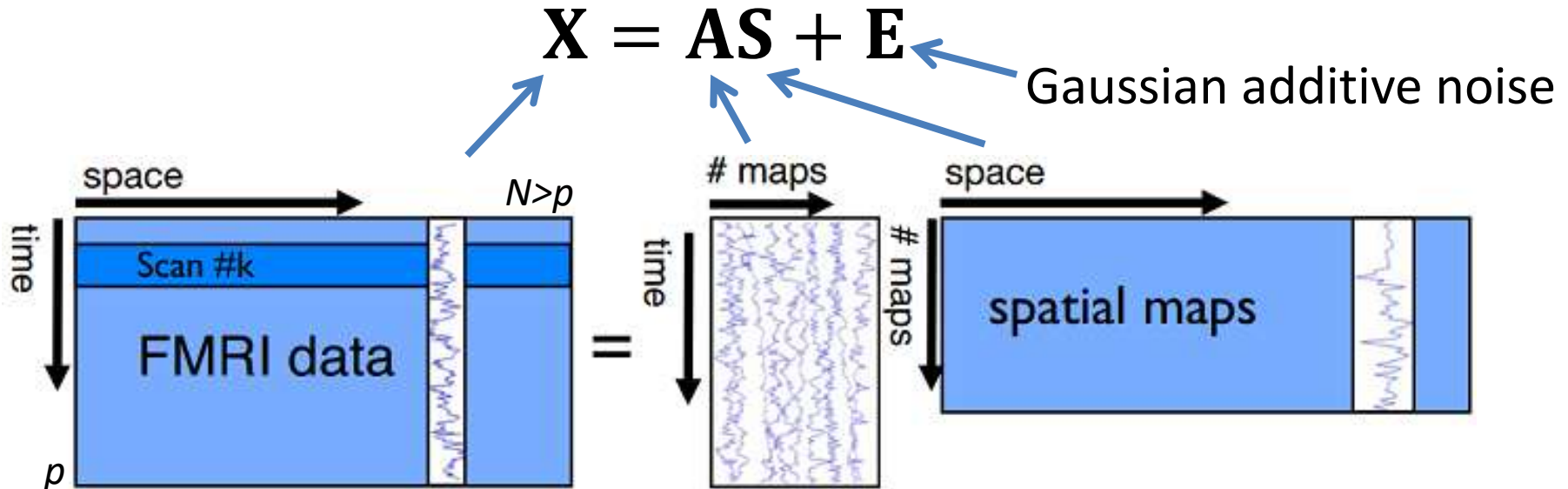


4. Final result



Revisiting ICA for fMRI

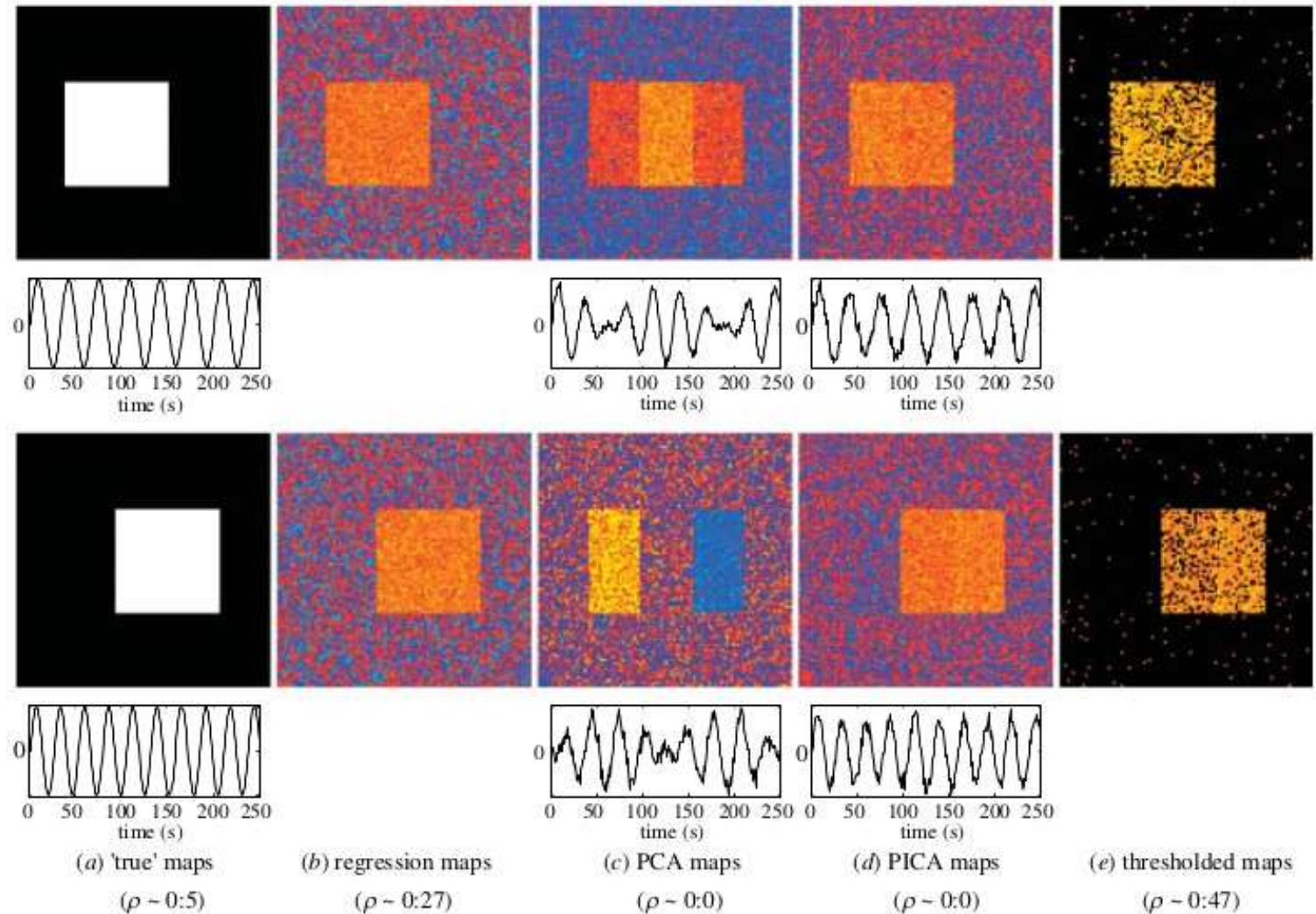
- We introduced ICA for fMRI as:



- Differences to previous example:
 - Time along the **rows** of $\mathbf{X}$, we assume spatially independent sources
 - $\mathbf{A}$ is no longer square, fewer sources than timepoints
 - This leads to a need for an **error term** $\mathbf{E}$ („probabilistic ICA“)

Spatial Independence vs. Spatial Overlaps

- Since spatial resolution far exceeds temporal resolution in fMRI, spatial ICA is more robust
 - **Concern:** We expect functional networks to spatially overlap
 - **Observation:** When estimating sparse sources in noisy data, requiring spatial independence does not preclude spatial overlaps



Setup of Probabilistic ICA

- Probabilistic ICA is based on the generative model

$$\mathbf{x}_i = \boldsymbol{\mu} + \mathbf{A}\mathbf{s}_i + \boldsymbol{\eta}_i \text{ for all voxels } i$$

- $\mathbf{x}_i$ is the timecourse at voxel i , a column vector of $\mathbf{X}$
- $\boldsymbol{\mu}$ contains the spatial averages for all time points
- $\mathbf{s}_i$ is a vector of independent, zero-mean, non-Gaussian random variables (spatial maps), mixed linearly via time courses in $\mathbf{A}$
- $\boldsymbol{\eta}_i \sim N(0, \sigma^2 \mathbf{I})$ is an i.i.d. **Gaussian** noise vector
 - *Note:* fMRI noise typically exhibits voxel-specific covariation Σ_i over time, which is removed in an additional “whitening” step (the previously discussed whitening now happens over space).

Model Selection in Probabilistic ICA

- *Important new step:* Determine a plausible number q of sources
 - When q is too small: Unable to explain true variance in the data
 - When q is too large: Networks break apart, obtain a large number of meaningless maps
- Spectrum (eigenvalues) of sample covariance matrix provides guidance
 - Since $E[\mathbf{xx}^T] = \mathbf{AA}^T$, q equals the rank in the ideal noise-free case
 - Independent additive noise adds $\sigma^2 \mathbf{I}$ to the covariance
 - *Note:* When working with the sample covariance, we cannot expect its spectrum to be exactly isotropic

Implementation Notes on Probabilistic ICA

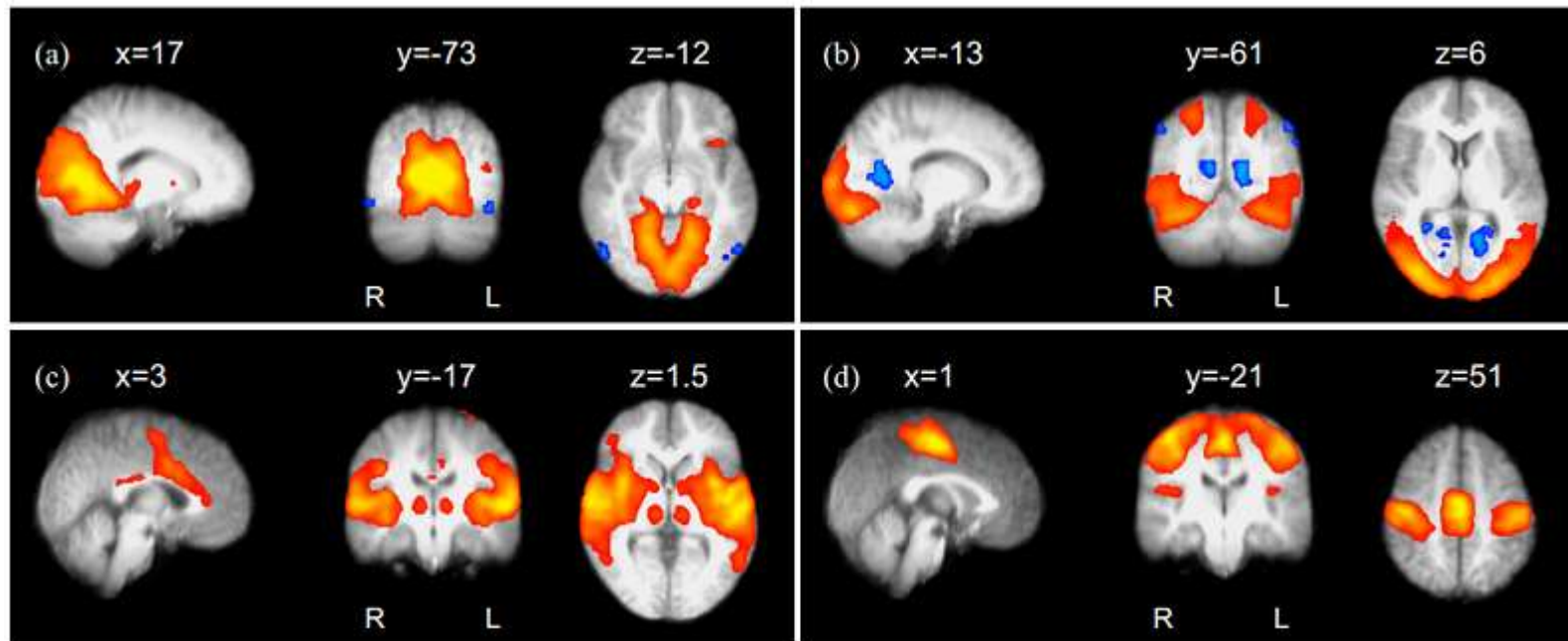
- The **pre-whitening** is now combined with dimensionality reduction:

$$\tilde{\mathbf{X}} = \left(\mathbf{\Lambda}_q - \hat{\sigma}^2 \mathbf{I} \right)^{-\frac{1}{2}} \mathbf{E}_q^T \mathbf{X}$$

- $\mathbf{\Lambda}_q$ and $\mathbf{E}_q$ contain first q eigenvalues and –vectors
- Estimate $\hat{\sigma}^2 = \frac{1}{p-q} \sum_{l=q+1}^p \lambda_l$
- Given the resulting $\tilde{\mathbf{X}}$, we can proceed with FastICA as before to optimize a $q \times q$ rotation $\mathbf{V}$

Resting State Networks Using ICA

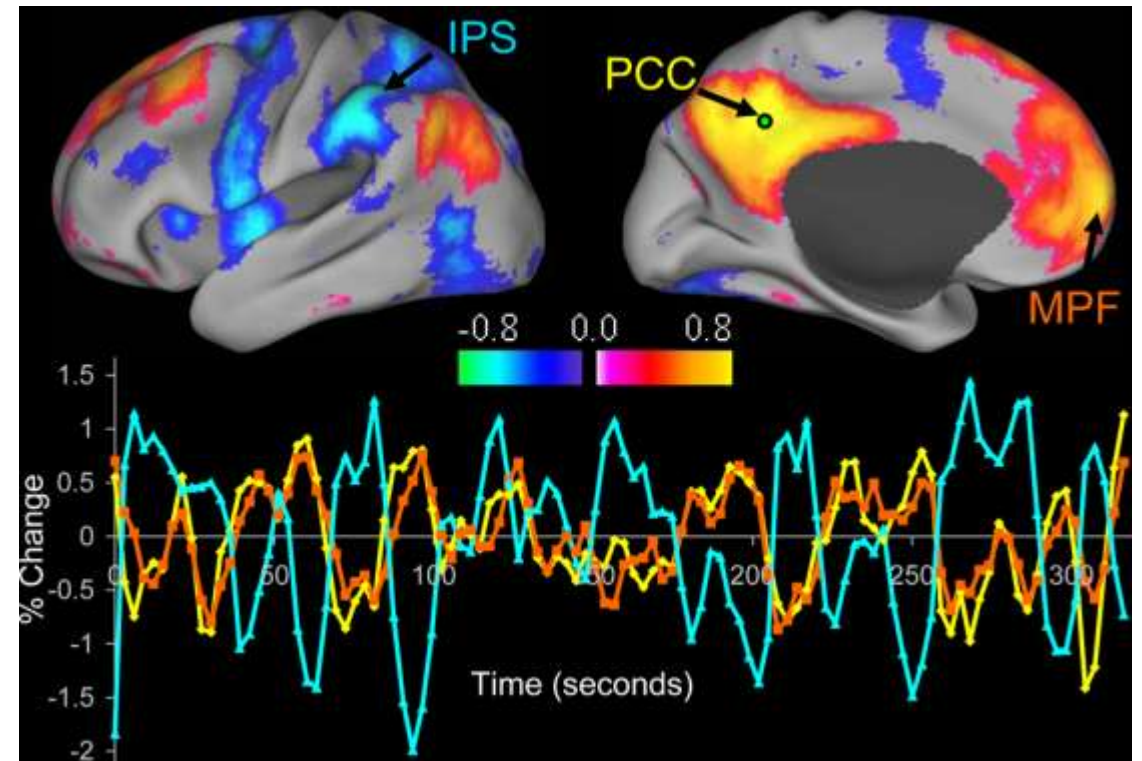
- Certain **resting state networks** consistently form independent components in rs-fMRI data
 - all show activity during rest and during tasks
 - one of the RSNs, the default mode network (DMN), shows a decrease in activity during certain external goal-oriented tasks



Default Mode Network

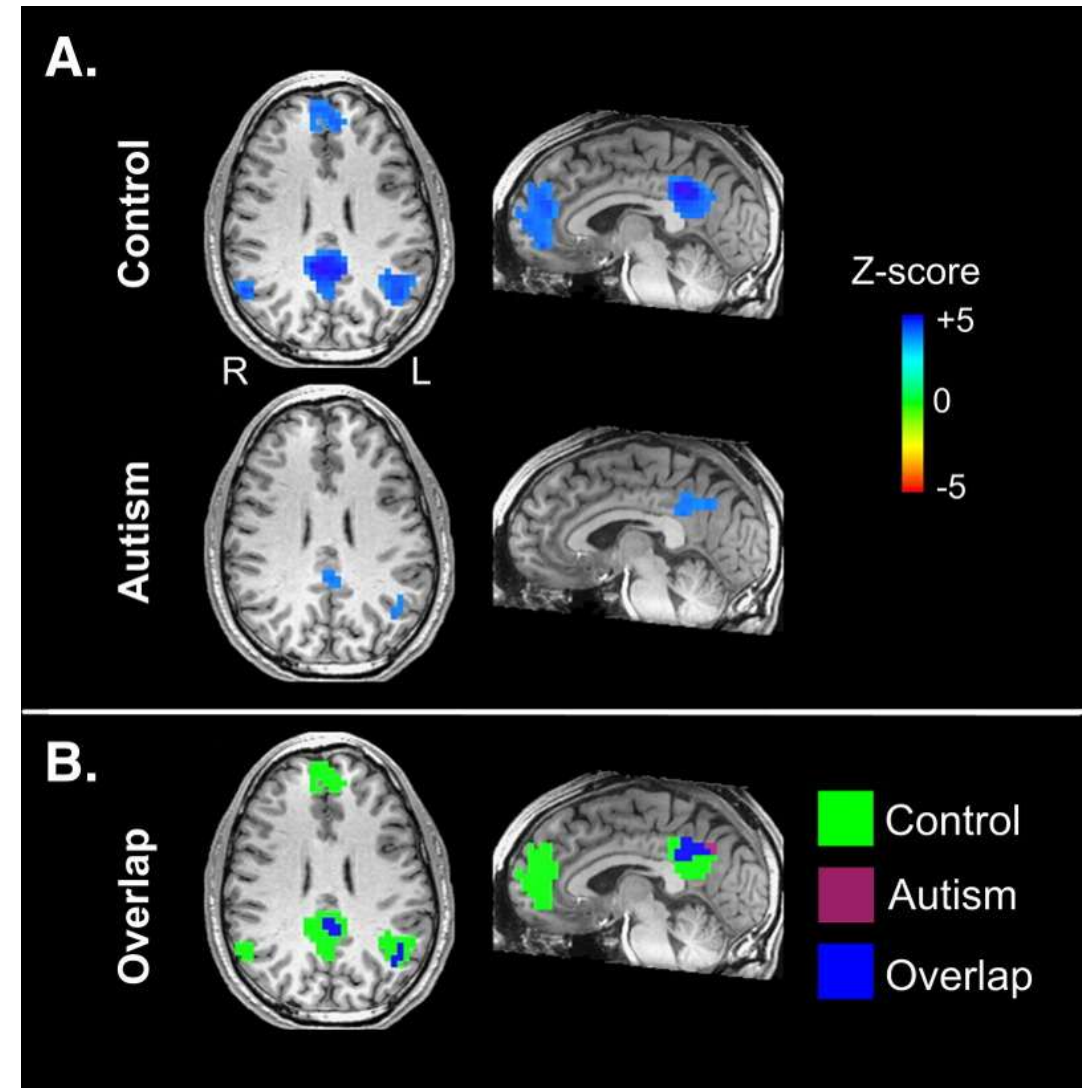
- Regions in the **default mode network** are more active – and more strongly coupled – at rest than during certain goal-oriented tasks
 - Thought to be involved in social and emotional processing, thinking about yourself and others, remembering the past, planning for the future

Intraparietal Sulcus
Posterior Cingulate Cortex
Medial Prefrontal Cortex



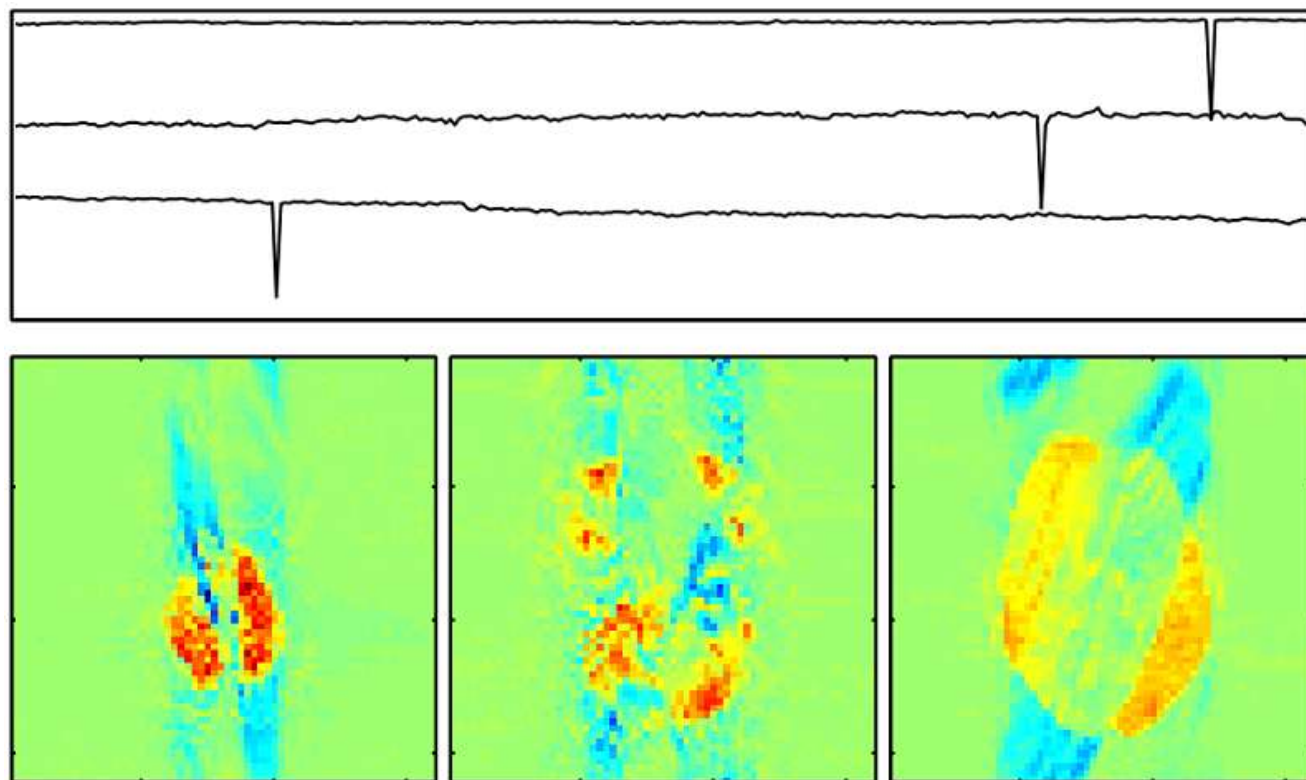
Example: A rs-fMRI Group Study

- [Kennedy et al. 2008] find reduced **resting state fMRI correlations** (“functional connectivity”) within the default mode network in individuals with autism compared to normal controls
 - No comparable changes in other, task and goal directed networks



Artifact Reduction Using ICA

- Many „structured“ imaging artifacts show up as independent components in ICA
 - After identifying them, we can „regress them out“ with a GLM
 - Built into automated tools such as FMRIB’s ICA-based Xnoiseifier (FIX) [Salimi-Khorshidi et al. 2014]
 - *Example: Slice Dropouts*



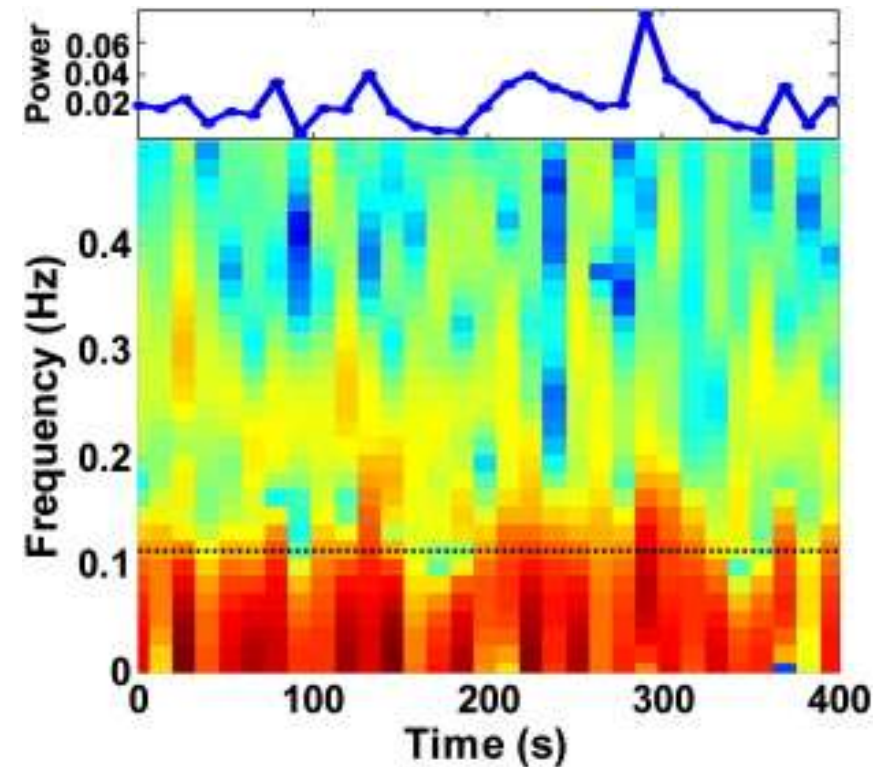
Summary: Resting-State fMRI

- At the **resting state**, subjects relax without sleeping or performing any particular task. Spontaneous low-frequency (0.01-0.10 Hz) oscillations in the BOLD signal are observed
- Correlations are observed between **functionally connected** regions
 - Can be explored by measuring correlation to a seed
- **ICA** is a model-free tool for data cleaning and exploratory analysis
 - Probabilistic ICA avoids „overfitting“ a huge number of components
- Several **resting state networks** are reliably identified; some of them have been shown to be affected by disease

7.7 Clustering-based rs-fMRI Analysis

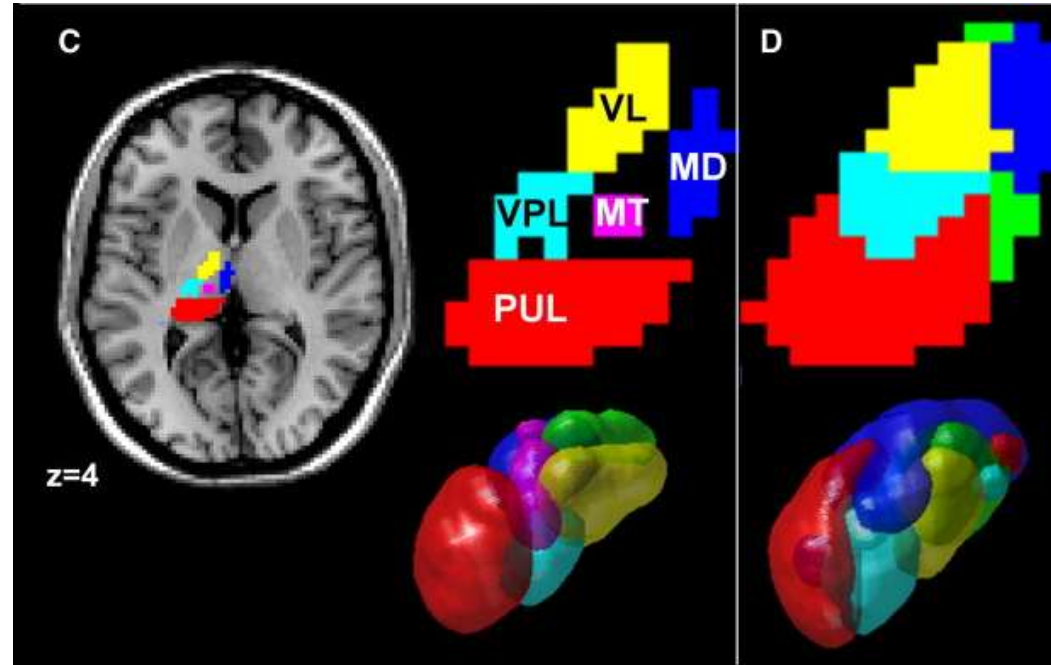
Clustering-based Analysis

- **Idea:** As an alternative tool for exploratory analysis, find clusters of voxels that have a similar time course in the relevant frequency band
- **Possible descriptor (Mezer et al. 2009):** Spectrogram (based on Short-Time Fourier Transform)



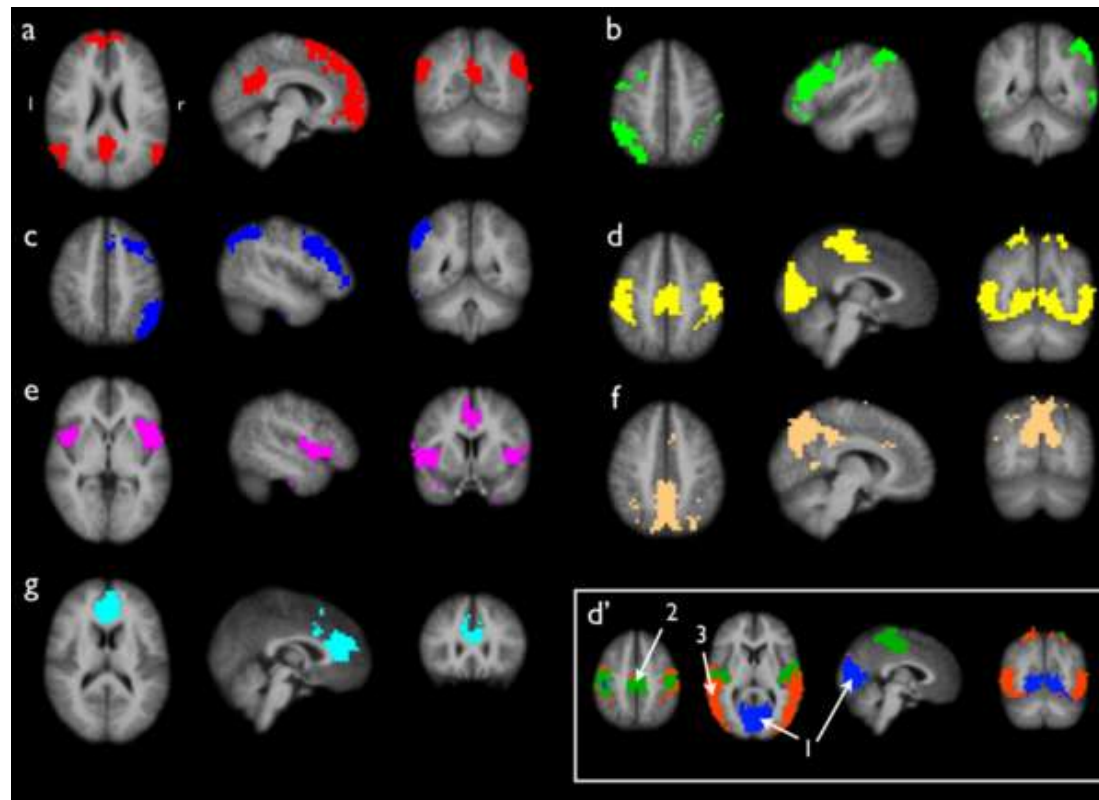
Clustering: Sample Result

- **Mezer et al. 2009** apply *k*-means on truncated and vectorized spectrograms
- *Example result:* Parcellation of the thalamus
 - Reasonable agreement with known anatomy (left)



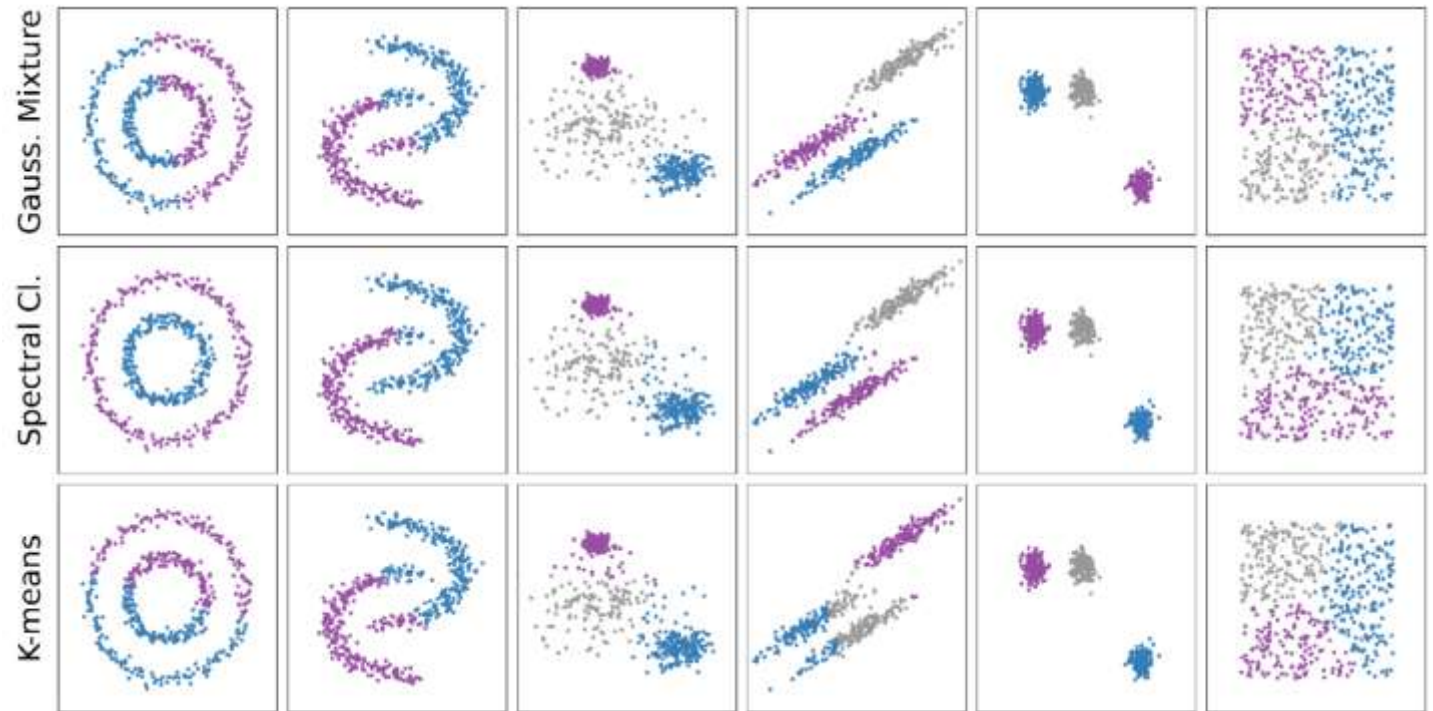
Clustering: Sample Result

- **Van den Heuvel et al. 2008** apply spectral clustering to a graph consisting of gray matter voxels (nodes) and their rs-fMRI correlations (edge weights)



Spectral Clustering as a General Purpose Technique

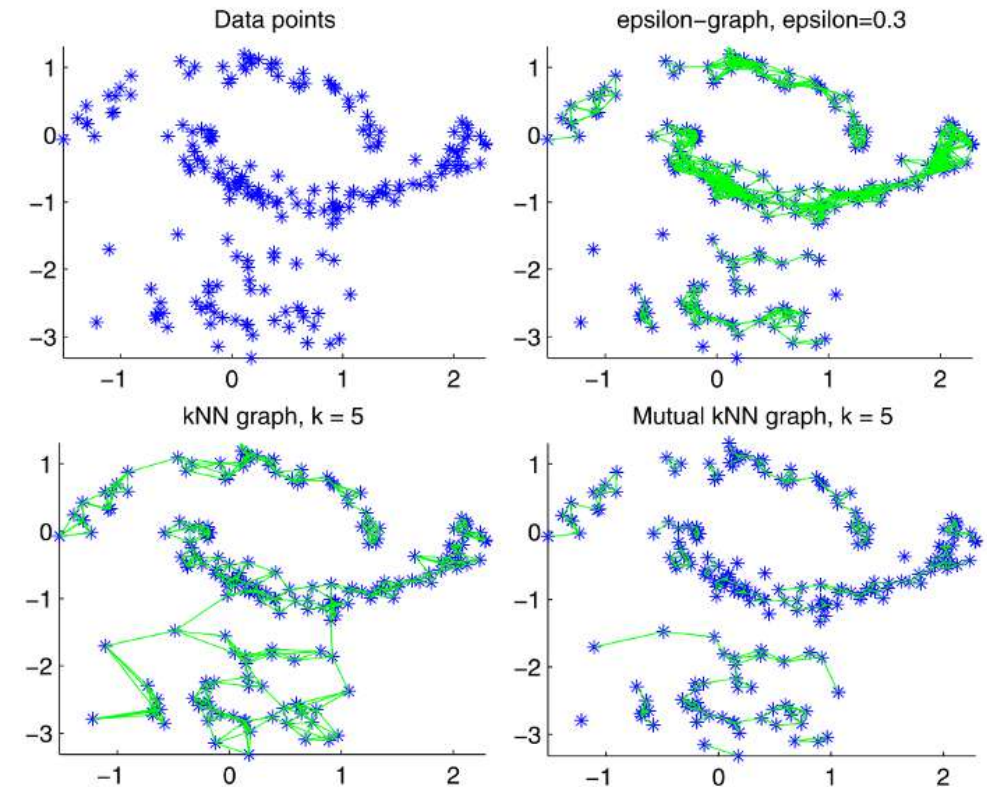
- **Spectral clustering** is a popular general-purpose clustering technique that handles nonlinear cluster shapes
- When given data items without edges, first construct a **neighborhood graph** based on distances / dissimilarities



Neighborhood Graphs

If we are given a distance matrix, e.g., from a set of points, we apply spectral clustering by deriving a **neighborhood graph** (V,E) : Each object is a vertex V , edges connect neighbors

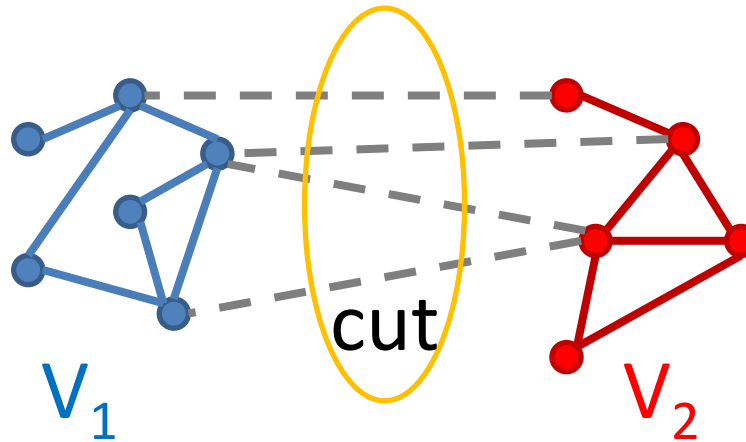
- **ϵ -graph**: Connect to all data points within distance ϵ
- **k NN**: Connect each data point to its k nearest neighbors
- **Mutual k NN**: Connect pairs of points for which both are among each other's k nearest neighbors



Graph Cuts

When **partitioning** an undirected weighted graph (V,E) into two vertex classes V_1 and V_2 , the corresponding “cut” consists of the edges between V_1 and V_2

- $\text{cut}(V_1, V_2)$ denotes the corresponding sum of edge weights



- Simply minimizing the cut usually splits off individual vertices, or very small groups

RatioCut and NCut

- **Ratio cut** avoids splitting off small clusters by normalizing by the number of nodes:

$$\text{RatioCut} = \frac{\text{cut}(V_1, V_2)}{|V_1|} + \frac{\text{cut}(V_1, V_2)}{|V_2|}$$

- **Normalized cut** looks for partition such that
 - Similarities between classes are minimized
 - Similarities within classes are maximized

$$\text{NCut} = \frac{\text{cut}(V_1, V_2)}{\text{vol}(V_1)} + \frac{\text{cut}(V_1, V_2)}{\text{vol}(V_2)}$$

- Uses $\text{vol}(V_i)$ (sum of degrees d_j of all vertices j within V_i) rather than number of nodes in V_i

Finding exact optima of both criteria is NP-hard.

Graph Laplacian

Define:

- $n = |V|$ (number of data points)
- Affinity matrix **W** (symmetric, $n \times n$)
 - w_{ij} contains weight of edge between i and j
- Degree matrix **D** (diagonal, $n \times n$)
 - sum of adjacent edge weights of vertex i : $d_i = \sum_j w_{ij}$
- (Unnormalized) Graph Laplacian: **L** = **D** – **W**

Quadratic Form of Graph Laplacian

Lemma: The quadratic form associated with the (unnormalized) graph Laplacian is

$$f^T L f = \frac{1}{2} \sum_{i,j=1}^n w_{ij} (f_i - f_j)^2$$

Proof:

$$\begin{aligned} f^T L f &= f^T D f - f^T W f = \sum_{i=1}^n d_i f_i^2 - \sum_{i,j=1}^n f_i f_j w_{ij} \\ &= \frac{1}{2} \left(\sum_{i=1}^n d_i f_i^2 - 2 \sum_{i,j=1}^n f_i f_j w_{ij} + \sum_{j=1}^n d_j f_j^2 \right) \\ &= \frac{1}{2} \sum_{i,j=1}^n w_{ij} (f_i - f_j)^2 \end{aligned}$$

Indicator Vector

- Define an **indicator vector** $f \in R^n$ of set $A \subset V$ such that $f_i = \sqrt{|\bar{A}|/|A|}$ if $v_i \in A$ and $f_i = -\sqrt{|A|/|\bar{A}|}$ if $v_i \in \bar{A}$
- f is orthogonal to the constant one vector $\mathbf{1}$

$$\begin{aligned}\sum_{i=1}^n f_i &= \sum_{i \in A} \sqrt{\frac{|\bar{A}|}{|A|}} - \sum_{i \in \bar{A}} \sqrt{\frac{|A|}{|\bar{A}|}} \\ &= |A| \sqrt{\frac{|\bar{A}|}{|A|}} - |\bar{A}| \sqrt{\frac{|A|}{|\bar{A}|}} = 0\end{aligned}$$

- The Euclidean norm of f is $\|f\| = \sqrt{n}$

$$\sum_{i=1}^n f_i^2 = |A| \frac{|\bar{A}|}{|A|} + |\bar{A}| \frac{|A|}{|\bar{A}|} = n$$

RatioCut in Terms of Graph Laplacian

- Given indicator vector f ,

$$\begin{aligned} f^T L f &= \frac{1}{2} \sum_{i,j=1}^n w_{ij} (f_i - f_j)^2 \\ &= \frac{1}{2} \sum_{i \in A, j \in \bar{A}} w_{ij} \left(\sqrt{\frac{|\bar{A}|}{|A|}} + \sqrt{\frac{|A|}{|\bar{A}|}} \right)^2 + \frac{1}{2} \sum_{i \in \bar{A}, j \in A} w_{ij} \left(-\sqrt{\frac{|A|}{|\bar{A}|}} - \sqrt{\frac{|\bar{A}|}{|A|}} \right)^2 \\ &= \text{cut}(A, \bar{A}) \left(\frac{|\bar{A}|}{|A|} + \frac{|A|}{|\bar{A}|} + 2 \right) \\ &= \text{cut}(A, \bar{A}) \left(\frac{|A| + |\bar{A}|}{|A|} + \frac{|A| + |\bar{A}|}{|\bar{A}|} \right) \\ &= |V| \cdot \text{RatioCut}(A, \bar{A}) \end{aligned}$$

RatioCut Re-Written

- Our results allow us to **re-write RatioCut** as a minimization of $f^T L f$
 - minimization over $A \subset V$ with f defined as above
- This equivalent reformulation with discrete f is still NP-hard, but it suggests a **relaxation**
 - permit continuous f
 - maintain constraints $f \perp \mathbf{1}$, $\|f\| = \sqrt{n}$
 - Since L is symmetric, its eigenvectors are the critical points of the Rayleigh quotient, and the eigenvalues the corresponding values

$$R(x) = \frac{x^T L x}{x^T x}$$

Properties of the Graph Laplacian

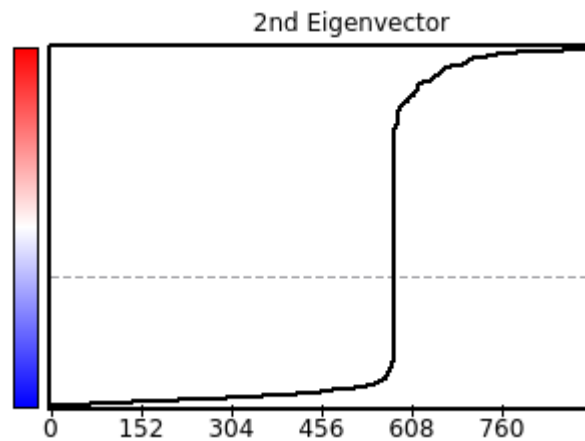
- L is symmetric and positive semi-definite
 - Symmetry: Direct consequence of definition
 - Semi-definiteness: Consequence of $f^T L f = \frac{1}{2} \sum_{i,j=1}^n w_{ij} (f_i - f_j)^2$ and non-negativity of w_{ij}
- The constant one vector **1** is an eigenvector of L with corresponding eigenvalue 0
 - Direct consequence of definition $L=D-W$

Approximate RatioCut

- As a consequence of the observations above, the desired minimum of $f^T L f$ is attained when f is the eigenvector of L corresponding to its **second smallest eigenvalue**
 - “Fiedler vector”, fuzzy cluster indicator
- A hard clustering is achieved by **discretizing the coefficients** of f , e.g., by
 - Thresholding at zero or at $(\max + \min)/2$
 - Applying a clustering method such as K means
 - Sorting the coefficients, computing RatioCut for each possible threshold, taking the optimum
- *Note:* **No approximation guarantees**, but works well in many real-world cases

Example: Approximate RatioCut

- *Example:* ε -graph over points sampled on fissure between lung lobes, edges weighted by similarity in surface normal



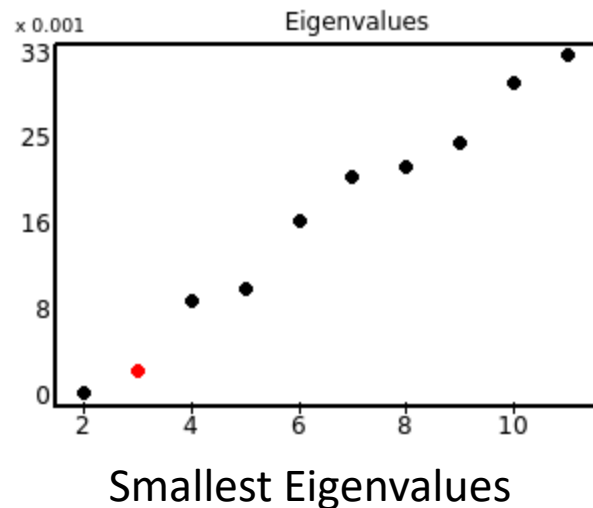
Fiedler vector (sorted)



Color coded on 3D data

Spectral Clustering: Number of Clusters

- Spectral Gap:
 - In case of K connected components, null space of Laplacian is K -dimensional
 - A stable clustering requires that K near-zero eigenvalues should be followed by a “spectral gap” towards a markedly larger $K+1^{\text{st}}$ eigenvalue



3rd Eigenvector

Spectral Clustering: Multi-Cluster Case

Strategies for $K > 2$ clusters:

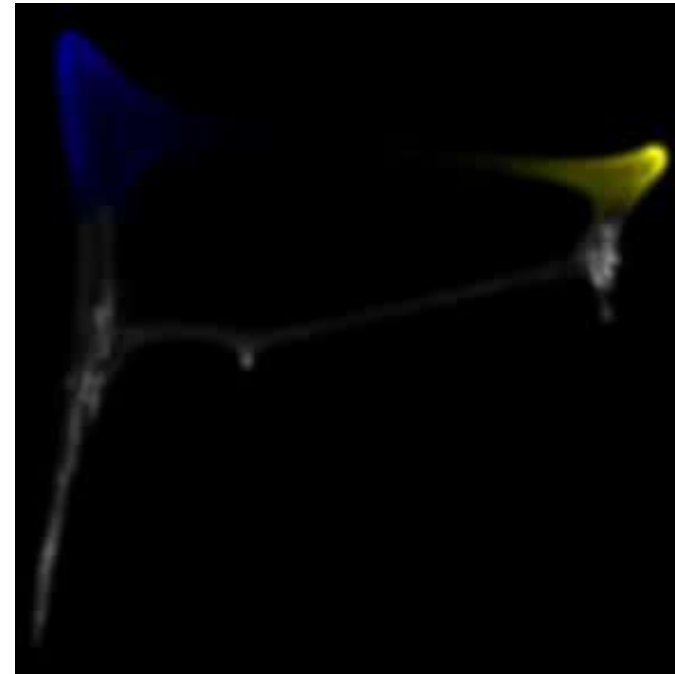
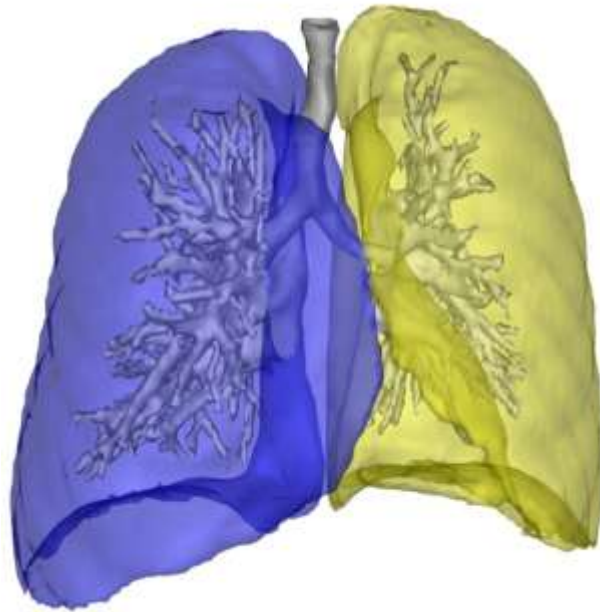
1. Recursively bisect (sub-)clusters [Shi/Malik]
2. Compute K smallest eigenvectors and use them to define K -dimensional coordinates for each point; use a traditional clustering method such as K means on that representation [Ng, Meila/Shi]
 - *Justification:* Approximation of multi-cluster RatioCut, see [von Luxburg 2007]

Algorithm: Spectral Clustering

- **Possible pre-process:** Construction of a neighborhood graph
- **Input:** Undirected weighted graph with affinity matrix $W \in \mathbb{R}^{n \times n}$, degree matrix D , number k of clusters to construct
- **Output:** Partitioning into clusters $A_1, \dots, A_k$
- **Algorithm:**
 - Compute (unnormalized) Laplacian $L = D - W$
 - Compute $U \in \mathbb{R}^{n \times k}$ whose columns are eigenvectors for k smallest eigenvalues of L
 - Let $y_i \in \mathbb{R}^k$ be the i -th row of U
 - Cluster y_i with a standard clustering method (e.g., k means)

Visualizing the Spectral Embedding

The video on the right shows a „grand tour“ of the K -dimensional (here, $K=5$) feature space spanned by five eigenvectors

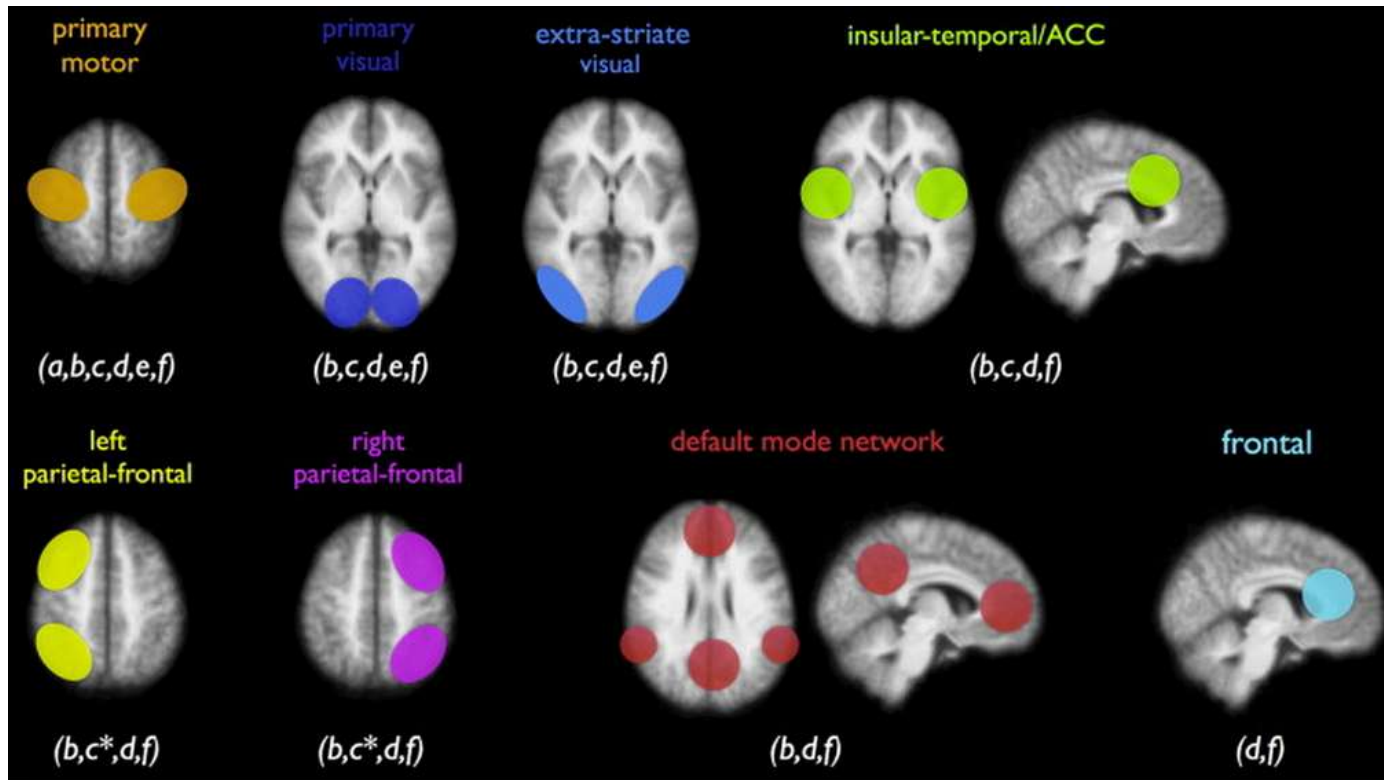


Approximate Normalized Cut

- The **Normalized Cut** can be approximated in a very similar way, with the normalized Laplacian $\tilde{\mathbf{L}} = \mathbf{D}^{-1/2}\mathbf{L}\mathbf{D}^{-1/2}$ replacing its unnormalized counterpart $\mathbf{L} = \mathbf{D} - \mathbf{W}$
 - Can be re-written as generalized eigenproblem
$$\mathbf{L}f = \lambda\mathbf{D}f$$
 - *Details*: [von Luxburg 2007] (see Further Reading)
- **Implementation** (for both RatioCut and NCut):
 - To keep computational effort feasible for a medium to large number of nodes, keep the graph sparse and exploit sparsity in (generalized) eigenvector computation!

Resting State Networks

- Overall, similar resting-state networks (RSNs) have been identified with clustering as with ICA



Van den Heuvel et al. 2010

Summary: Clustering-based Analysis

- **Clustering** provides an alternative model-free exploratory approach to resting state fMRI analysis
- Feature descriptors can be derived from **spectrograms**
- Alternatively, build and partition a **graph representation**
 - Gray matter regions as nodes, correlations define edge weights or presence of edges (after thresholding)
- **Spectral clustering** provides a computationally feasible relaxation of the NP hard RatioCut and Ncut clustering criteria

7.8 Connectomics

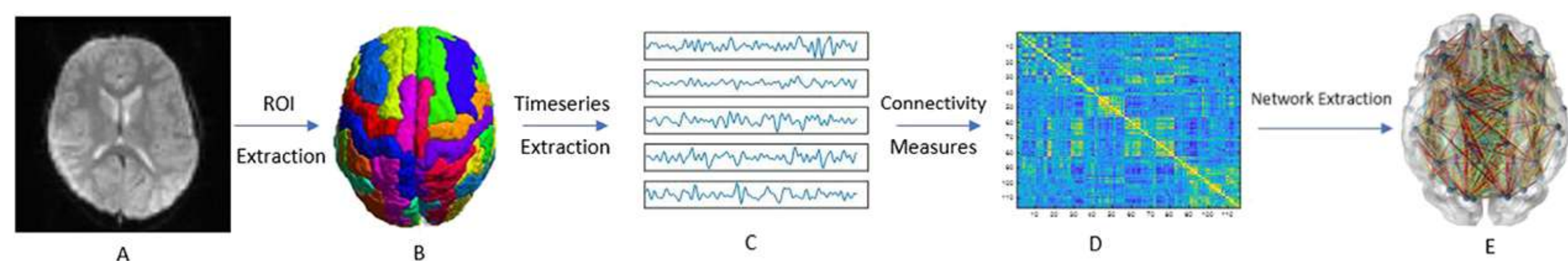
Introduction

- **Graphs** are a natural formalism to describe brain networks
 - Nodes: Individual cells (microscale) or brain regions (macroscale)
 - Edges: Connections between them
- *Goal*: Provide a complete model of the connections in the human brain
 - “**Connectome**” coined in 2005 (Sporns et al. / Hagmann)
 - Functional (this chapter) vs. structural connectivity (next chapter)
- “**Connectomics**” maps and analyzes the properties of brain networks, often using graph theory

Building Functional Networks

Resting-state functional networks are built by

1. Defining nodes via a gray matter parcellation
2. Extracting an average time series for each brain region
3. Computing functional connectivity via correlations
4. Defining edges via super-threshold correlations

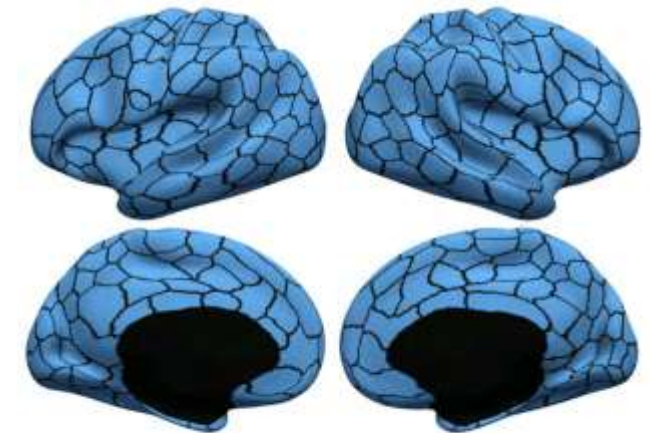


Gray Matter Parcellation

- **Nodes** should represent contiguous brain regions with similar rs-fMRI time courses
- **Parcellation** has a long history (Brodman). fMRI-based approaches include
 - Spectral Clustering [Craddock et al. 2012]
 - Markov Random Fields [Schaefer et al. 2018]
 - Similarity of time courses
 - Spatial proximity
 - “Gradient-weighted” binary potentials
- Many atlases provide multiple resolutions
 - Often: 100-500 regions, sometimes fewer/more



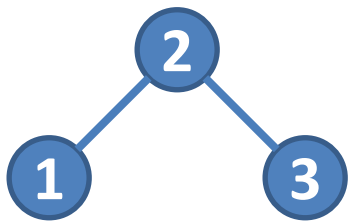
Craddock



gwMRF

Direct vs. Indirect Connections

- **Problem:** Computing full correlations does not distinguish between direct and indirect connections
- *Example:* If regions 1 and 3 depend on region 2, but do not have a direct connection, their time series will still be correlated because they are indirectly connected via 2



-	1	0
1	-	1
0	1	-

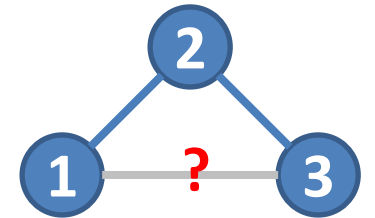
True
Connectivity

-	1	1
1	-	1
1	1	-

Estimated
Connectivity

Partial Correlations

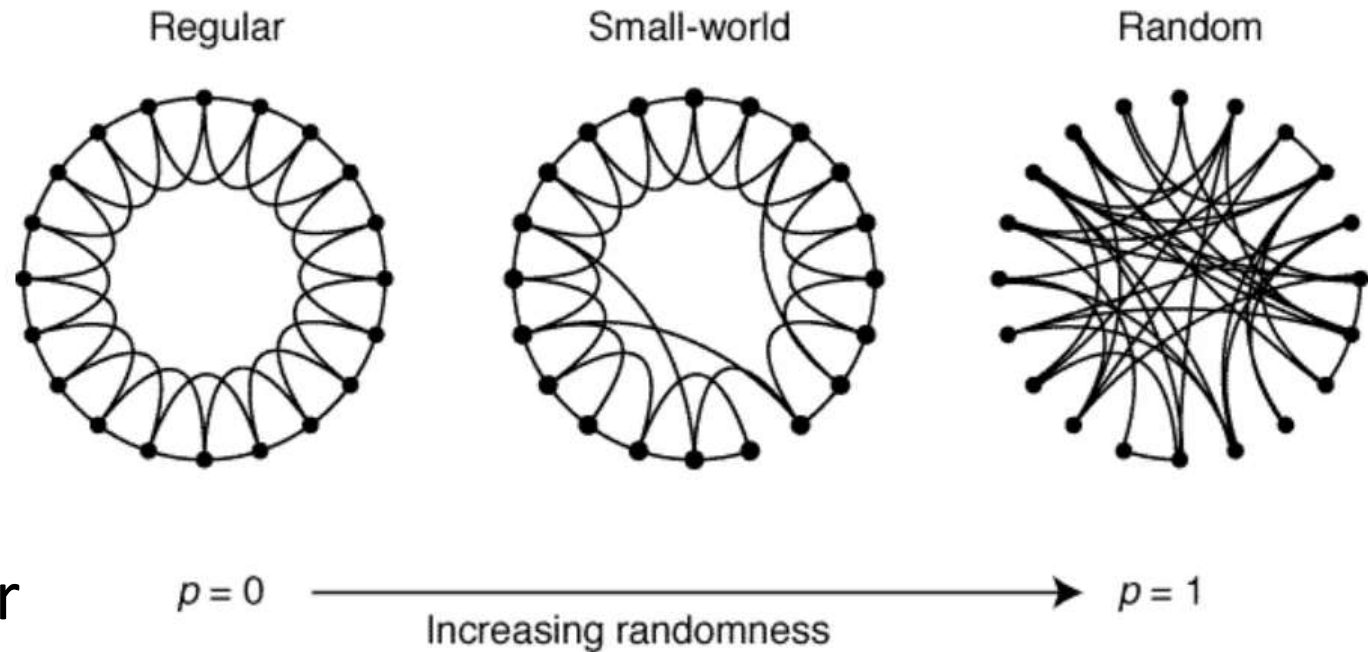
- **Partial correlations** are a popular strategy for removing indirect connections
 - *Idea*: We compute $\rho_{13 \cdot 2}$, the partial correlation coefficient between 1 and 3 controlling for 2, from their residual time series, after regressing out the time course from 2
- Given a larger number of nodes, we need to regress *all other nodes* out of the pair in question
 - Partial correlation $\rho_{ij \cdot R}$ between i and j , controlling for all remaining nodes R , can be obtained from the precision matrix $P = \Sigma^{-1}$



$$\rho_{ij \cdot R} = - \frac{p_{ij}}{\sqrt{p_{ii} p_{jj}}}$$

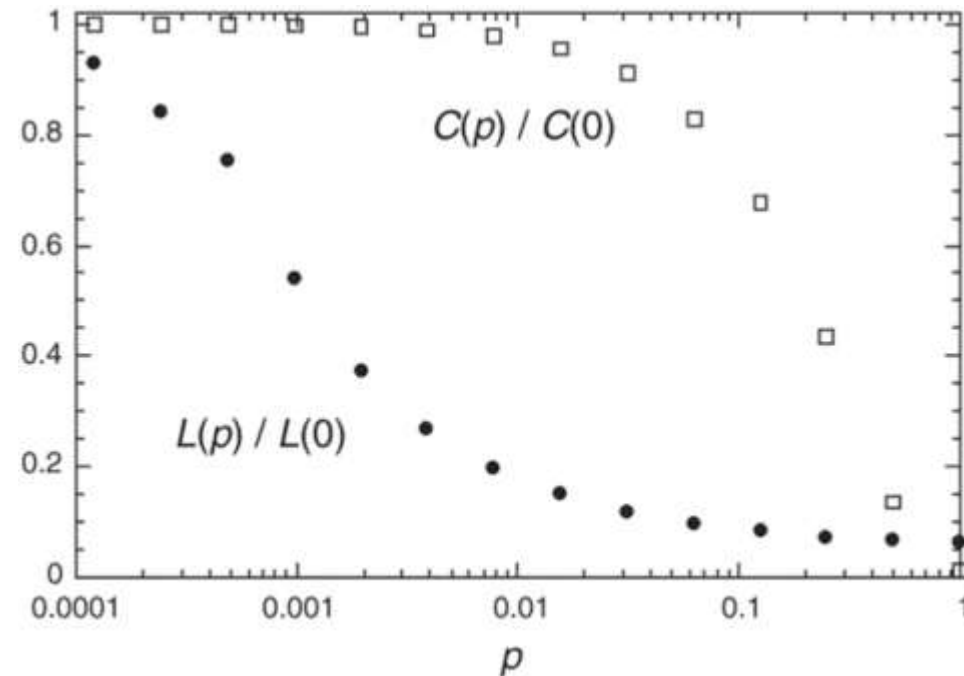
Idea: Small-World Networks

- **Regular graphs** (each vertex connects to a fixed number of neighbors) cluster, but shortest paths are long on average
- **Random graphs** (edges are drawn randomly) do not cluster, but shortest paths are much shorter
- [Watts/Strogatz 1998]: Many real-world graphs are “**small-world**” graphs, in between these cases
 - Vertices connect efficiently
 - Vertices still cluster together



Small-World Networks: Formalization

- **Characteristic path length L** is the shortest path length, averaged over all pairs of vertices
 - “Six degrees of separation”
- **Clustering coefficient C** is the ratio of the number of edges that exist between the k neighbors of a node, and the maximum number of $k(k - 1)/2$ edges that could exist between them, averaged over all vertices
- **Small-world networks** have similar L as a random graph, but similar C as a regular one



The Brain as a Small-World Network

- Examples by Watts/Strogatz include the cellular-level connectome of *C. elegans*

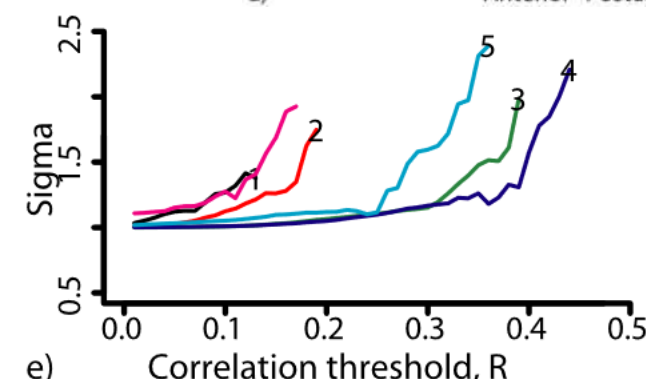
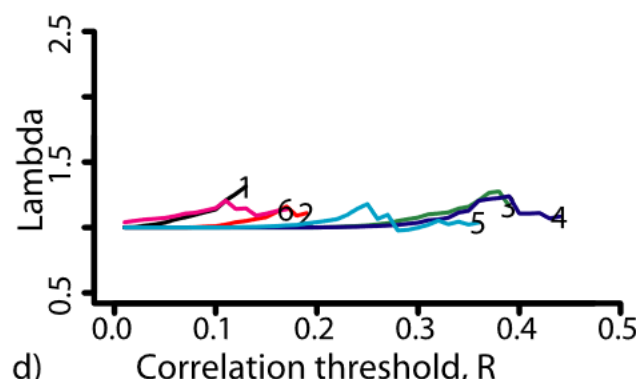
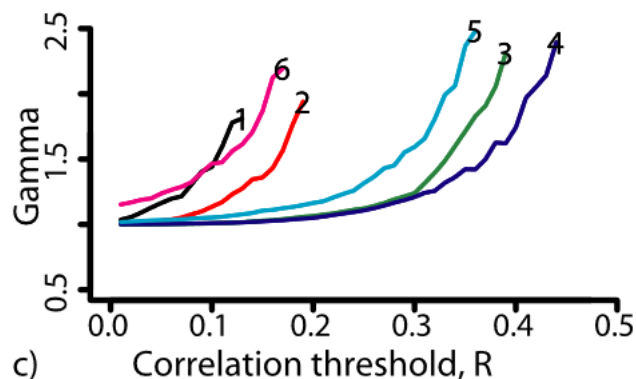
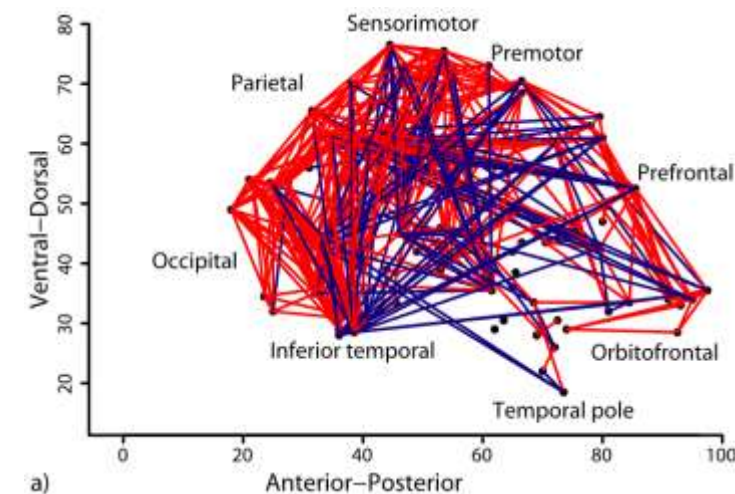
Table 1 Empirical examples of small-world networks

	L_{actual}	L_{random}	C_{actual}	C_{random}
Film actors	3.65	2.99	0.79	0.00027
Power grid	18.7	12.4	0.080	0.005
<i>C. elegans</i>	2.65	2.25	0.28	0.05

- [Achard et al. 2006] (and many others) argue that functional networks from rs-fMRI have small-world properties

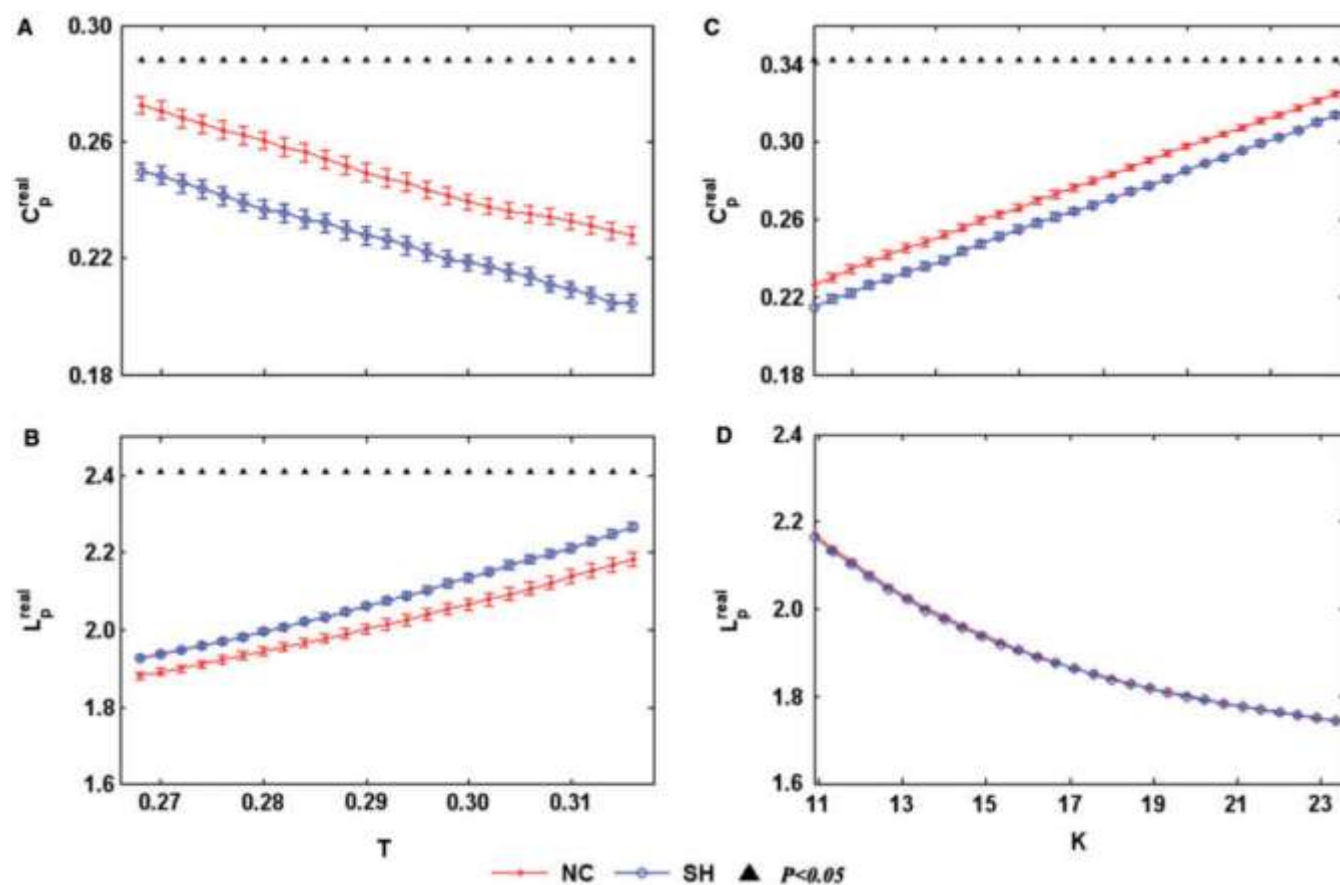
- Analyzed six frequency bands (4: 0.03-0.06 Hz)

- $\gamma = \frac{C}{C_{\text{rand}}}$, $\lambda = \frac{L}{L_{\text{rand}}}$, “small-worldness” $\sigma = \frac{\gamma}{\lambda}$



Altered Connectivity Patterns in Disease

- Several studies report that global patterns in functional connectivity are affected by mental disorders
- *Example:* [Liu et al. 2008] find reduced clustering coefficients, and increased path lengths, in **patients with schizophrenia** compared to **matched normal controls**
 - For a range of thresholds T and average degrees K

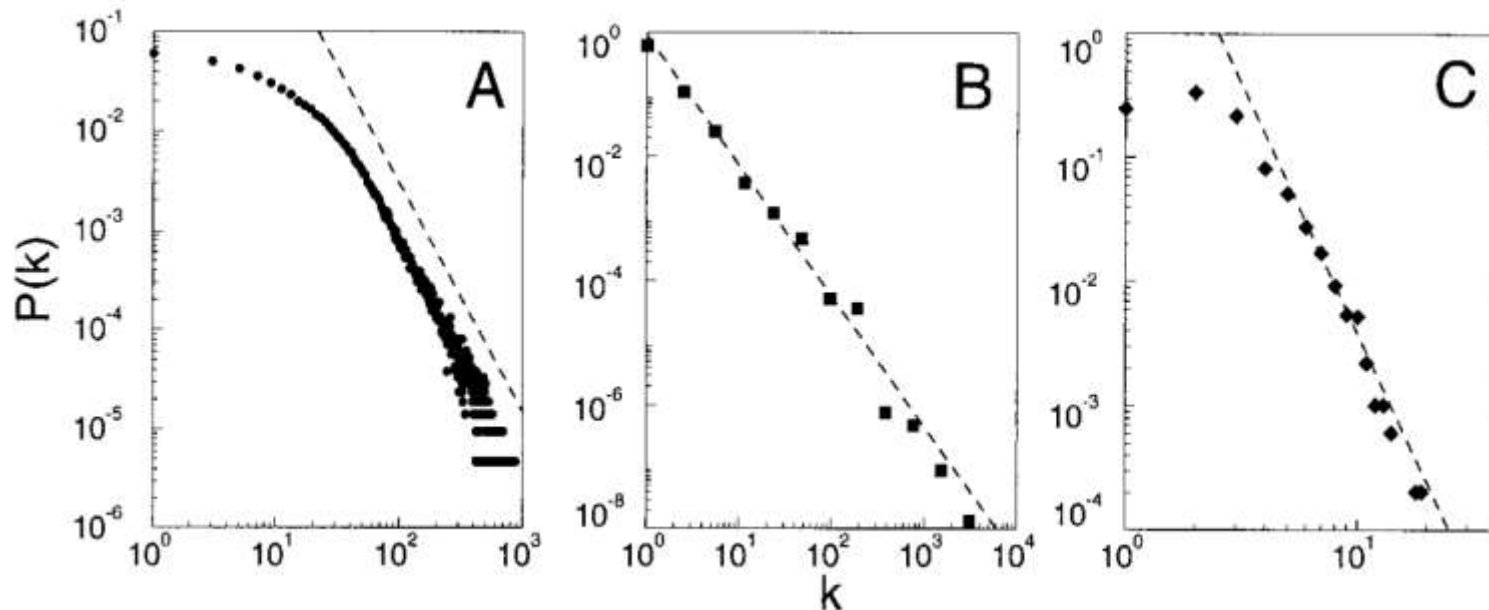


Idea: Scale-Free Networks

- In **random graphs**, most vertices have a similar degree
- “Scale-free” networks have a **power-law** degree distribution

$$P(k) \propto k^{-\gamma}$$

- Arises when networks expand, and new vertices attach preferentially to well-connected ones [Barabási/Albert 1999]



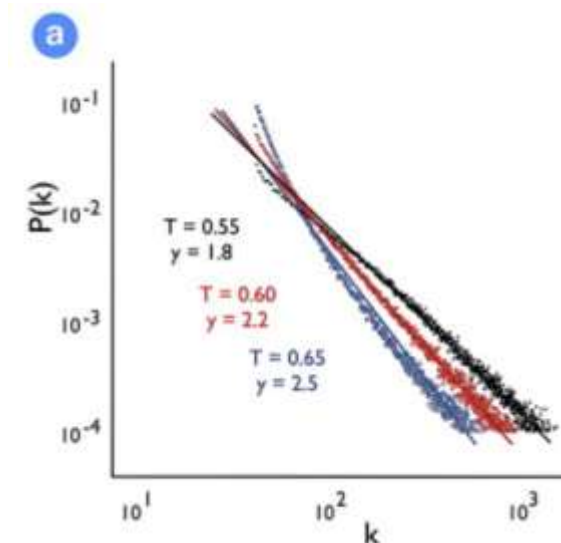
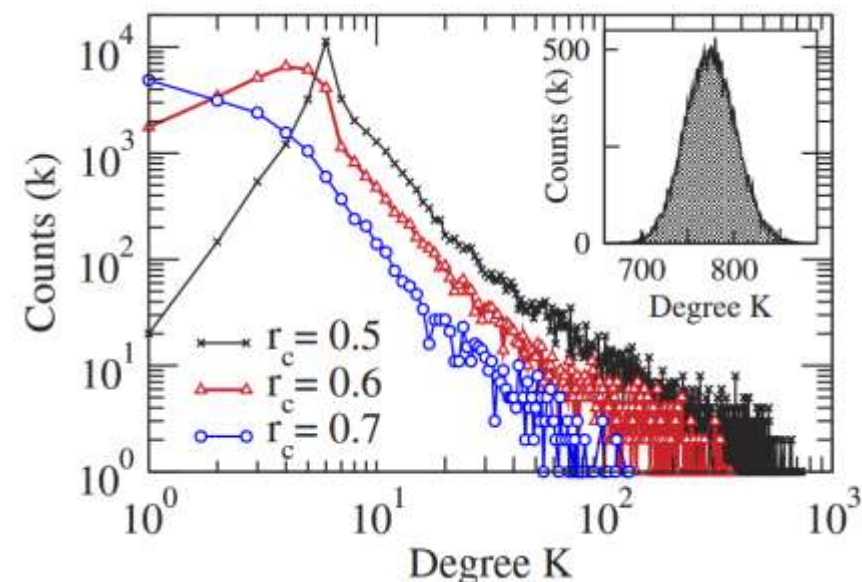
A: Actor collaboration ($\gamma = 2.3$)

B: WWW ($\gamma = 2.1$)

C: Power grid ($\gamma = 4$)

Is the Brain a Scale-Free Network?

- [Eguíluz et al. 2005] report scale-free degree distribution in task fMRI
- [Achard et al. 2006] find their rs-fMRI networks are small-world, *not* scale-free
 - Brain network is more resilient to removal of hubs than a scale-free network
- [van den Heuvel et al. 2008] report small-world *and* scale-free networks in a voxel-based rs-fMRI analysis
 - Argue that intra-regional connectivity should be included in the analysis
- [Broido/Clauset 2019] suggest more generally: “Scale-free networks are rare”



Summary: Connectomics

- **Graph-based models** of functional connectivity are based on
 - Parcellating gray matter into regions of similar time courses
 - Computing functional connectivity via (partial) correlations
 - Defining edges via thresholding (often, multiple ones are considered)
- **Functional networks** can be used to investigate
 - “Small worldness” (local clustering behavior, global efficiency)
 - Node degrees (scale invariance, hubs)
- Some **mental disorders** appear to disrupt global patterns of functional connectivity

Further Reading: fMRI Analysis

- K.J. Friston et al. (Eds): *Statistical Parametric Mapping: The Analysis of Functional Brain Images*. Elsevier, 2007
- Nicole A. Lazar: *The Statistical Analysis of Functional MRI Data*. Springer, 2008
- Russell A. Poldrack, Jeanette A. Mumford, Thomas E. Nichols: *Handbook of Functional MRI Data Analysis*. Cambridge University Press, 2011
- Thomas Schultz: *Neuroimage Acquisition and Analysis*. In: *Neurotechnology: methods, advances and applications*, pp. 37-67, IET 2020

Further Reading: ICA, Spectral Clustering

- A. Hyvärinen, J. Karhunen, E. Oja: *Independent Component Analysis*. John Wiley & Sons, 2001
- CF Beckmann, S Smith: *Probabilistic independent component analysis for functional magnetic resonance imaging*. IEEE Trans. Med. Imaging 23(2):137-52, 2004
- Ulrike von Luxburg: *A Tutorial on Spectral Clustering*. Stat Comput 17:395-416, 2007

Further Reading: Connectomics

- D.J. Watts, S.H. Strogatz: *Collective dynamics of 'small-world' networks*. Nature 393:440-442, 1998
- A.L. Barabási, R. Albert: *Emergence of scaling in random networks*. Science 286:509-512, 1999
- E. Bullmore, O. Sporns: *Complex brain networks: graph theoretical analysis of structural and functional systems*. Nature Reviews Neuroscience 10:186-198, 2009
- S.M. Smith et al.: *Network modelling methods for fMRI*. NeuroImage 54:875-891, 2011

Update / Announcement

- The MR scanner tour on Monday, February 2, will be moved to the **DZNE** and start at **13:00 sharp**
 - Please arrive a few minutes earlier
 - Venusberg-Campus 1/99
 - Bus stop Uniklinikum Süd
- We are planning to see an ultra high field (7T), regular (3T) and ultra low field (0.064T) scanner
- Please sign up via eCampus (bottom of course page) if you plan to attend



Reminder Regarding Exercises

- Admission to the exam requires 50% of the points on the assignments **plus presenting at least once**
- At this point, many more students have registered for the exam than presented
- The only remaining opportunities are
 - Thursday, January 15 (Exercise 5)
 - Monday, January 26 (Exercise 6)

Exam-like Question

1. Does the BOLD effect increase or decrease MR signal intensity when neural activity increases? Briefly explain why.
2. Name four operations that are frequently performed while pre-processing functional MRI data.
3. Point out three errors in the following pseudocode for the spectral clustering of a graph with degree matrix D and affinity matrix W into two clusters:
 - Compute Laplacian $L = DW$
 - Compute the eigenvector $\mathbf{v}$ corresponding to the smallest eigenvalue of L
 - Let θ be the median of the v_i
 - Assign node i to cluster 1 if $v_i \geq \theta$, to cluster 2 if $v_i < \theta$